

CMPE 256 – Large Scale Analytics

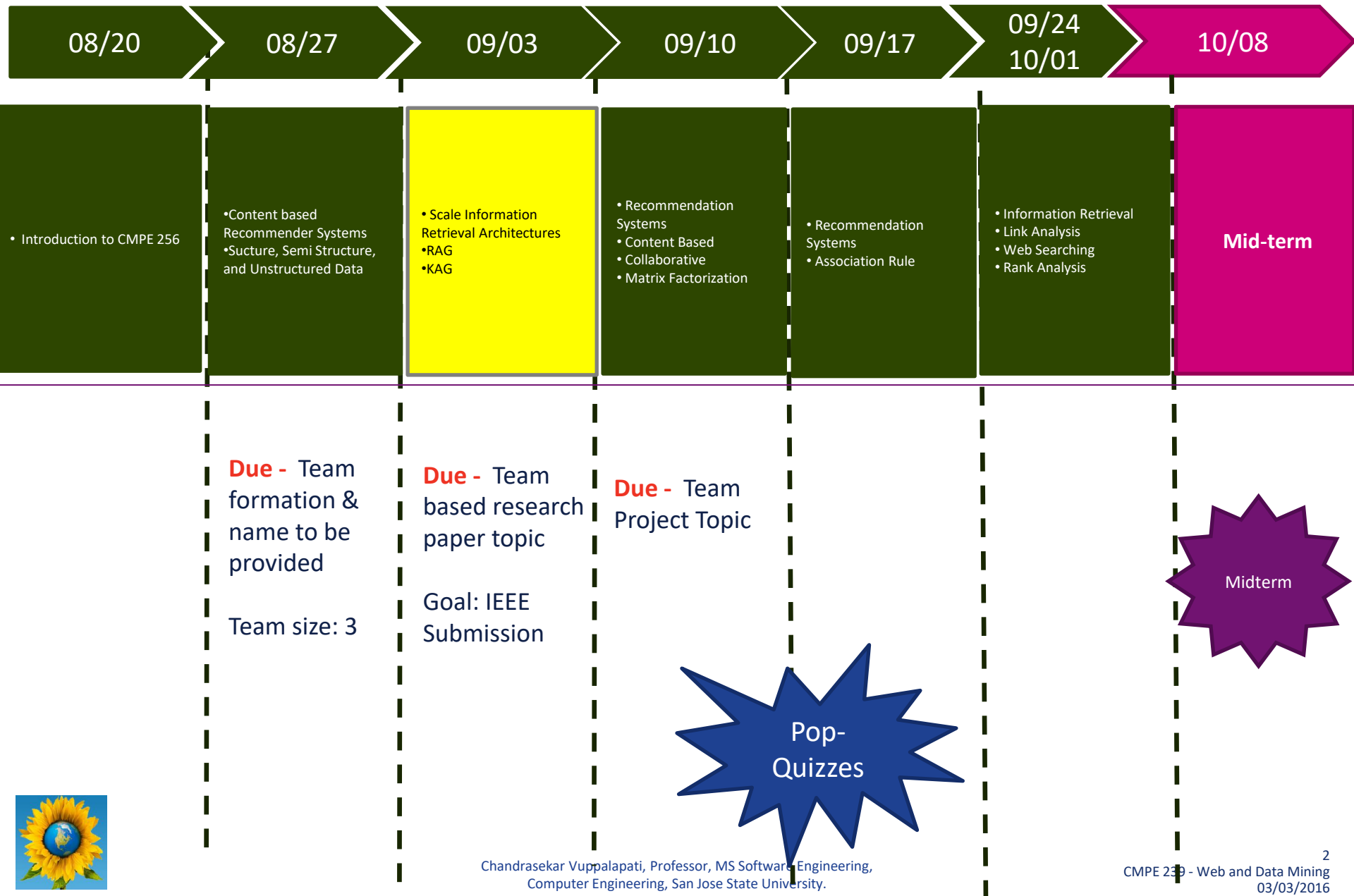


Chandrasekar Vuppalapati

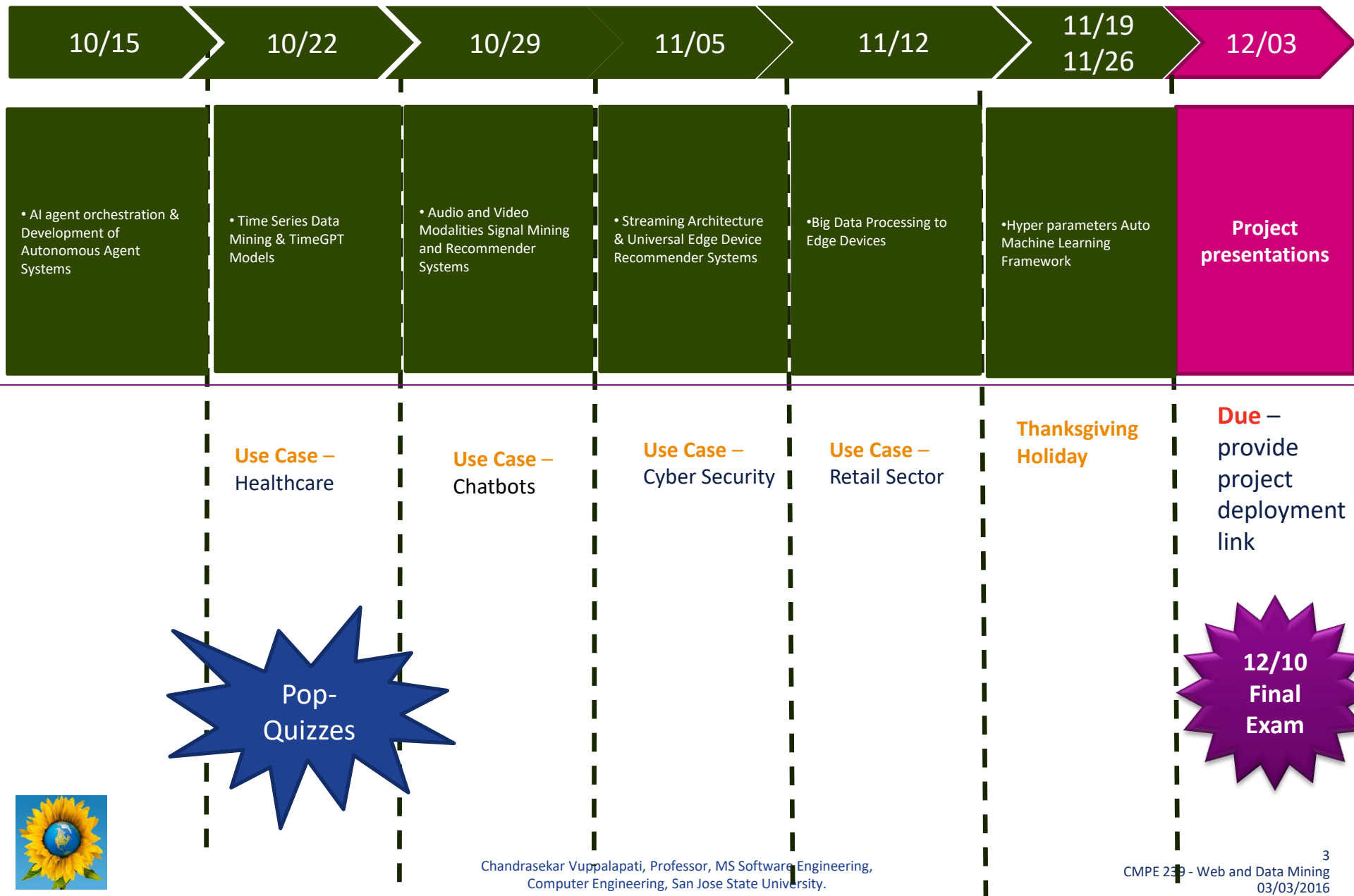
Professor, MS Software Engineering – Data Science,
Department of Computer Engineering, San Jose State
University.



Class Roadmap



Class Roadmap



Agenda

1. Clustering



$$\frac{\partial}{\partial \theta} \int_{\mathbb{R}_n} T(x) f(x, \theta) dx = \int_{\mathbb{R}_n} \frac{\partial}{\partial \theta} T(x) f(x, \theta) dx$$

$$\frac{\partial}{\partial a} \ln f_{a, \sigma^2}(\xi_1) = \frac{(\xi_1 - a)}{\sigma^2} f_{a, \sigma^2}(\xi_1) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{(\xi_1 - a)^2}{2\sigma^2}\right\}$$

$$\int_{\mathbb{R}_n} T(x) \cdot \frac{\partial}{\partial \theta} f(x, \theta) dx = M\left(T(\xi) \cdot \frac{\partial}{\partial \theta} \ln L(\xi, \theta)\right) = \int_{\mathbb{R}_n} \frac{\partial}{\partial \theta} T(x) f(x, \theta) dx$$

$$\int_{\mathbb{R}_n} T(x) \cdot \left(\frac{\partial}{\partial \theta} \ln L(x, \theta)\right) \cdot f(x, \theta) dx = \int_{\mathbb{R}_n} T(x) \cdot \left(\frac{\frac{\partial}{\partial \theta} f(x, \theta)}{f(x, \theta)}\right) \cdot f(x, \theta) dx$$

$$\frac{\partial}{\partial \theta} M T(\xi) = \frac{\partial}{\partial \theta} \int_{\mathbb{R}_n} T(x) f(x, \theta) dx = \int_{\mathbb{R}_n} \frac{\partial}{\partial \theta} T(x) f(x, \theta) dx$$

$$\left\{ \frac{(\xi_1 - a)^2}{\sigma^2} \right\} \frac{\partial}{\partial a} \ln f_{a, \sigma^2}(\xi_1) = \frac{(\xi_1 - a)}{\sigma^2}$$

Rise of the replicants

Technology Rapid advances in artificial intelligence now threaten the jobs of educated white-collar workers. *By Richard Waters*

If Daniel Nadler is right, a generation of college graduates with well-paid positions as junior researchers and analysts in the banking industry should be worried about their jobs. Very worried.

Mr Nadler's start-up, staffed with ex-Google engineers and backed partly by money from Google's venture capital arm, is trying to put them out of work.

Its algorithms assess how different securities are likely to react after the release of a market-moving piece of information, such as a monthly employment report. That is the kind of work usually done by well-educated junior analysts, who pull data from terminals, fill in spreadsheets and crunch numbers. "There are several hundred thousand people employed in that capacity. We do it with machines," says Mr Nadler. "We're not competing with other [tech] providers. We're competing with people."

Warren – the name given to the system, in homage to investor Warren Buffett – is part of a new army of "smart" machines that are threatening to invade office life. These computers do not just collect and process information; they draw inferences, answer questions and recommend actions, too.

The threat to jobs stretches beyond the white-collar world. Advances in artificial intelligence (AI) also make possible more versatile robots capable of taking over many types of manual work. "It's going to decimate jobs at the low end," predicts Jerry Kaplan, a Silicon Valley entrepreneur who teaches a class about AI at Stanford University. Like others working in the field, he says he is surprised by the speed at which the new technologies are moving out of the research labs.

"People don't understand it, they don't get what it's going to mean," adds Mr Kaplan, who says he was "radicalised" by a sudden awareness of the job-destroying capabilities of the new technology. "I feel like one of the early guys warning about global warming."



Companies involved in digital advertising, for instance, engage in big number crunching to improve their campaigns, making constant data-driven judgments about which sites to use, at which times of day and in front of which types of audience.

"It's a real drag to have that job – it's a constant, crushing load," says George John, chief executive of Rocket Fuel, a digital advertising company whose technology instead automates the process.

Robotics companies trying to replace various types of manual labour make similar claims for their products, notes Mr Kaplan. A weeding machine smart enough to pick its way through a field without harming the crop, for instance, is advertised by its manufacturer, Blue River, as an advance on the "back-breaking job" it is replacing.

However, for the humans who risk being put out of work, Mr Kaplan adds, it may be little consolation to be told that their jobs were undesirable in the first place.

A second defence of the smart machines is that the slow speed of human labour causes bottlenecks in an increasingly digital production chain. Even an army of analysts with calculators and spreadsheets would be unable to process all the data being churned out by some of today's systems.

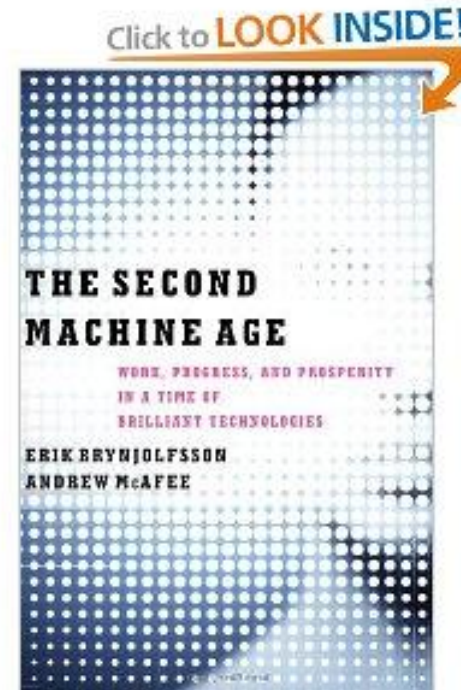
Kris Hammond, chief scientist of Narrative Science, says that many companies "have spent a tremendous amount on collecting data, and they might have analysts who only produce one or two reports a day".

Companies with large numbers of branch managers or franchisees, insurance companies with big sales forces and wealth management concerns with thousands of customers are among the businesses his company is targeting for its automated report-writing software.

This points to a technology dividend that often follows widespread automa-

Rise of the replicants

- Relentless fall in cost of computing
- Availability of vast bodies of digital data



Projects

Team Research and Project

ChandraSekar

← → ↺ 🏠

https://sjsu.instructure.com/courses/1137910/pages/team-research-and-project-details-page

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SP15: CMPE-239 Sec 02 - Web and Data Mining Spring 2015

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Team Research and Project Details Page

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- Crowd-sourced Big Data Mobile Diagnostics Web Dashboard Design
- Crowd-sourced Big Data Healthcare Dashboard
- Crowd-sourced Big Data Application for Disease Propagation
- Crowd-sourced Traffic Optimization using Big Data Application & Web UI Design
- Crowd-sourced Medical Emergency Vehicle Navigation System Using Big Data & Mobile Application - Dashboard Design

Team Name	Team Member	Research Topic	Project Topic	Data Sources and any additional details
One	Saideepak Budaraju	Cookieless User Activity Tracking and Analytics	Diet recommendation using Fitbit data.	
	Leona Cherla			
	Amey More			
	Aniket Savanand			
Heineken	Ameya Manjrekar, Chao Nie, Rajas Hegiste, Yaopeng Wu	Sentiment Analysis on StackOverflow	Recommendation System on StackOverflow	https://archive.org/details/stackexchange
Data Miners	Gagan Gautam	Real time streaming with Spark		

Supervised vs. unsupervised learning

- ▶ **Supervised learning:** discover patterns in the data that relate data attributes with a target (class) attribute.
 - ▶ These patterns are then utilized to predict the values of the target attribute in future data instances.
- ▶ **Unsupervised learning:** The data have no target attribute.
 - ▶ We want to explore the data to find some intrinsic structures in them.



Slide adapted from slides of B. Liu for “Web Data Mining”



Clustering

The process of grouping a set of physical or abstract objects into classes of *similar* objects is called clustering.

Clustering is the process of examining a collection of “points,” and grouping the points into “clusters” according to some distance measure. The goal is that points in the same cluster have a small distance from one another, while points in different clusters are at a large distance from one another.

A cluster is a collection of data objects that are *similar* to one another within the same cluster and are *dissimilar* to the objects in other clusters. A cluster of data objects can be treated collectively as one group and so may be considered as a form of data compression. Although classification is an effective means for distinguishing groups or classes of objects, it requires the often costly collection and labeling of a large set of training tuples or patterns, which the classifier uses to model each group.

Cluster analysis has been widely used in numerous applications, including market research, pattern recognition, data analysis, and image processing. In business, clustering can help marketers discover distinct groups in their customer bases and characterize customer groups based on purchasing patterns. In biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionality, and gain insight into structures inherent in populations.



Clustering

Clustering is also called data segmentation in some applications because clustering partitions large data sets into groups according to their *similarity*. Clustering can also be used for outlier detection, where outliers (values that are “far away” from any cluster) may be more interesting than common cases.

As a branch of statistics, cluster analysis has been extensively studied for many years, focusing mainly on *distance-based cluster analysis*. Cluster analysis tools based on *k*-means, *k*-medoids, and several other methods have also been built into many statistical analysis software packages or systems, such as S-Plus, SPSS, and SAS.

In machine learning, clustering is an example of unsupervised learning. Unlike classification, clustering and unsupervised learning do not rely on predefined classes and class-labeled training examples. For this reason, clustering is a form of learning by observation, rather than *learning by examples*. In data mining, efforts have focused on finding methods for efficient and effective cluster analysis in *large databases*. Active themes of research focus on the *scalability* of clustering methods, the effectiveness of methods for clustering *complex shapes and types of data*, *high-dimensional* clustering techniques, and methods for clustering *mixed numerical and categorical data* in large databases.



Clustering

- ▶ Clustering is a technique for finding **similarity groups** in data, called **clusters**.
- ▶ Clustering is often called an **unsupervised learning** task as no class values denoting an *a priori* grouping of the data instances are given, which is the case in supervised learning.
- ▶ Due to historical reasons, clustering is often considered synonymous with unsupervised learning.
 - ▶ In fact, association rule mining is also unsupervised

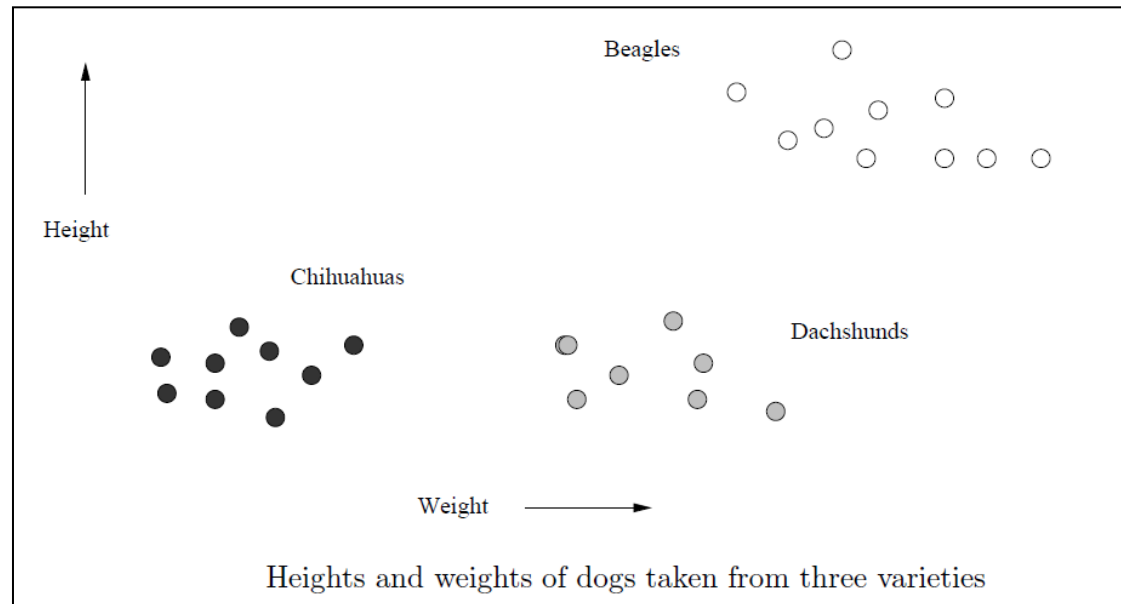


Slide adapted from slides of B. Liu for "Web Data Mining"



Dataset – Points, Space, and Distance

A dataset suitable for clustering is a collection of points, which are objects belonging to some space. In its most general sense, a space is just a universal set of points, from which the points in the dataset are drawn. However, we should be mindful of the common case of a Euclidean space which has a number of important properties useful for clustering. In particular, a Euclidean space's points are vectors of real numbers. The length of the vector is the number of dimensions of the space. The components of the vector are commonly called coordinates of the represented points. All spaces for which we can perform a clustering have a distance measure, giving a distance between any two points in the space.



Use Cases

Use the data you've gathered to develop consumer clusters—groups who share distinct preferences related to what, where, and when you sell. To keep things simple, use the minimum number of clusters that enable you to localize the most promising elements of your offering. Example: **Best Buy** identified four consumer clusters: 1) busy suburban moms who are the chief buyers for their households and want quick, personalized help navigating the world of technology, 2) young tech junkies who want cutting-edge entertainment and gaming gear, 3) affluent, time-pressed professionals seeking high-end equipment and personalized service, 4) budget minded family men desiring technology to enhance home life.



Source: Localization The Revolution in Consumer Markets

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

Use Cases

Locating Your Next Strategic Opportunity

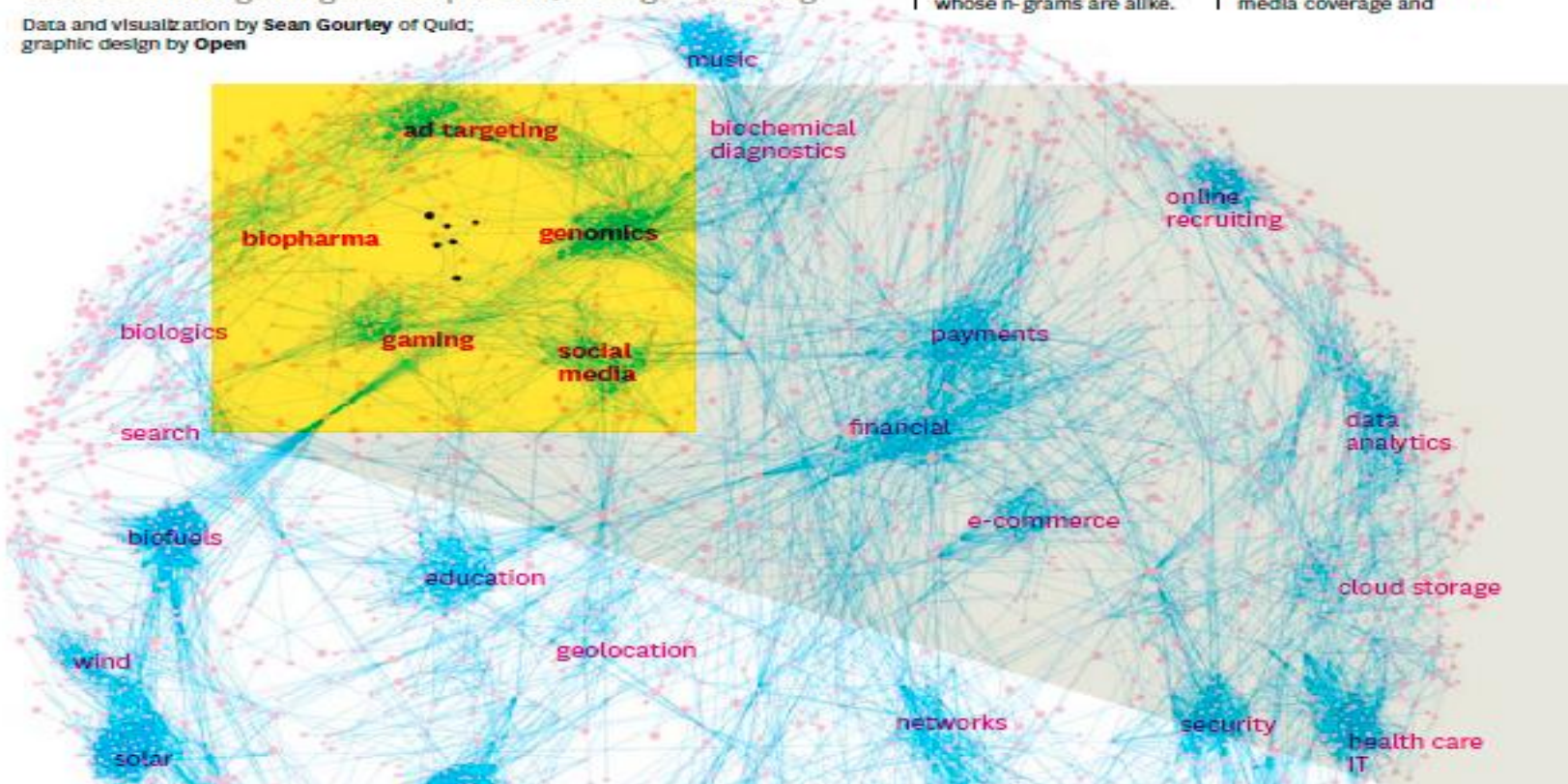
Where and how do strategists find growth opportunities? Sometimes by literally drawing a map, using a technique called semantic-clustering analysis. Such maps can reveal not only which sectors are thick with competition but where in the market white spaces are open for the taking. For example, while it may seem odd to find opportunity in the nexus between gaming and biopharma, seeing is believing.

Data and visualization by Sean Gourley of Quid;
graphic design by Open

To map semantic clusters, Quid software first identifies hundreds of key phrases associated with individual companies and organizations, or their "n-grams." Applying algorithms and other analytical tools, the technology parses text in millions of corporate documents, from patent filings, to press releases, to Twitter posts. The software then creates a map with lines connecting companies whose n-grams are alike.

The lines act like gravitational pull: The more lines there are between companies, the more tightly together those companies are drawn. Similar firms become clustered into industry sectors.

The result is a multi-dimensional industry map like the one below. It represents 4,000 technology enterprises—from venture-backed start-ups to established public companies—that received media coverage and



Use Cases (in-class)

Movies & Clustering



Use Cases

Car Dealers & Clustering Example



Types of Data in Cluster

Suppose that a data set to be clustered contains n objects, which may represent **persons, houses, documents, countries, and so on**. Main memory-based clustering algorithms typically operate on either of the following two data structures.

Data matrix (or object-by-variable structure): This represents n objects, such as persons, with p variables (also called measurements or attributes), such as age, height, weight, gender, and so on. The structure is in the form of a relational table, or n -by- p matrix (n objects p variables):

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$



Types of Data in Cluster

Dissimilarity matrix (or *object-by-object structure*): ***This stores a collection of proximities that are available for all pairs of n objects.*** It is often represented by an n -by- n table:

$$\begin{bmatrix} 0 & & & & \\ d(2, 1) & 0 & & & \\ d(3, 1) & d(3, 2) & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ d(n, 1) & d(n, 2) & \dots & \dots & 0 \end{bmatrix}$$

where $d(i, j)$ is the measured difference or dissimilarity between objects i and j . In general, $d(i, j)$ is a nonnegative number that is close to 0 when objects i and j are highly similar or “near” each other, and becomes larger the more they differ.



Distance Calculation

the dissimilarity (or similarity) between the objects described by interval-scaled variables is typically computed based on the distance between each pair of objects. The most popular distance measure is **Euclidean distance**, which is defined as

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{in} - x_{jn})^2},$$

where $i = (x_{i1}, x_{i2}, \dots, x_{in})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jn})$ are two n -dimensional data objects.

Another well-known metric is **Manhattan (or city block) distance**, defined as

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{in} - x_{jn}|.$$

Both the Euclidean distance and Manhattan distance satisfy the following mathematic requirements of a distance function:



Distance Calculation

Both the Euclidean distance and Manhattan distance satisfy the following mathematic requirements of a distance function:

1. $d(i, j) \geq 0$: Distance is a nonnegative number.
2. $d(i, i) = 0$: The distance of an object to itself is 0.
3. $d(i, j) = d(j, i)$: Distance is a symmetric function.
4. $d(i, j) \leq d(i, h) + d(h, j)$: Going directly from object i to object j in space is no more than making a detour over any other object h (*triangular inequality*).



Distance Calculation

Minkowski distance is a generalization of both *Euclidean distance* and *Manhattan distance*. It is defined as

$$d(i, j) = (|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \cdots + |x_{in} - x_{jn}|^p)^{1/p},$$

where p is a positive integer. Such a distance is also *called Lp norm*, in some literature. It represents the Manhattan distance when $p = 1$ (i.e., $L1$ norm) and Euclidean distance when $p = 2$ (i.e., $L2$ norm). *If each variable is assigned a weight according to its perceived importance, the weighted Euclidean distance can be computed as*

$$d(i, j) = \sqrt{w_1|x_{i1} - x_{j1}|^2 + w_2|x_{i2} - x_{j2}|^2 + \cdots + w_m|x_{in} - x_{jn}|^2}.$$

Weighting can also be applied to the Manhattan and Minkowski distances.



Distance Metrics

- ▶ Euclidean Distance (L_2 norm)

$$d([x_1, \dots, x_n], [y_1, \dots, y_n]) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

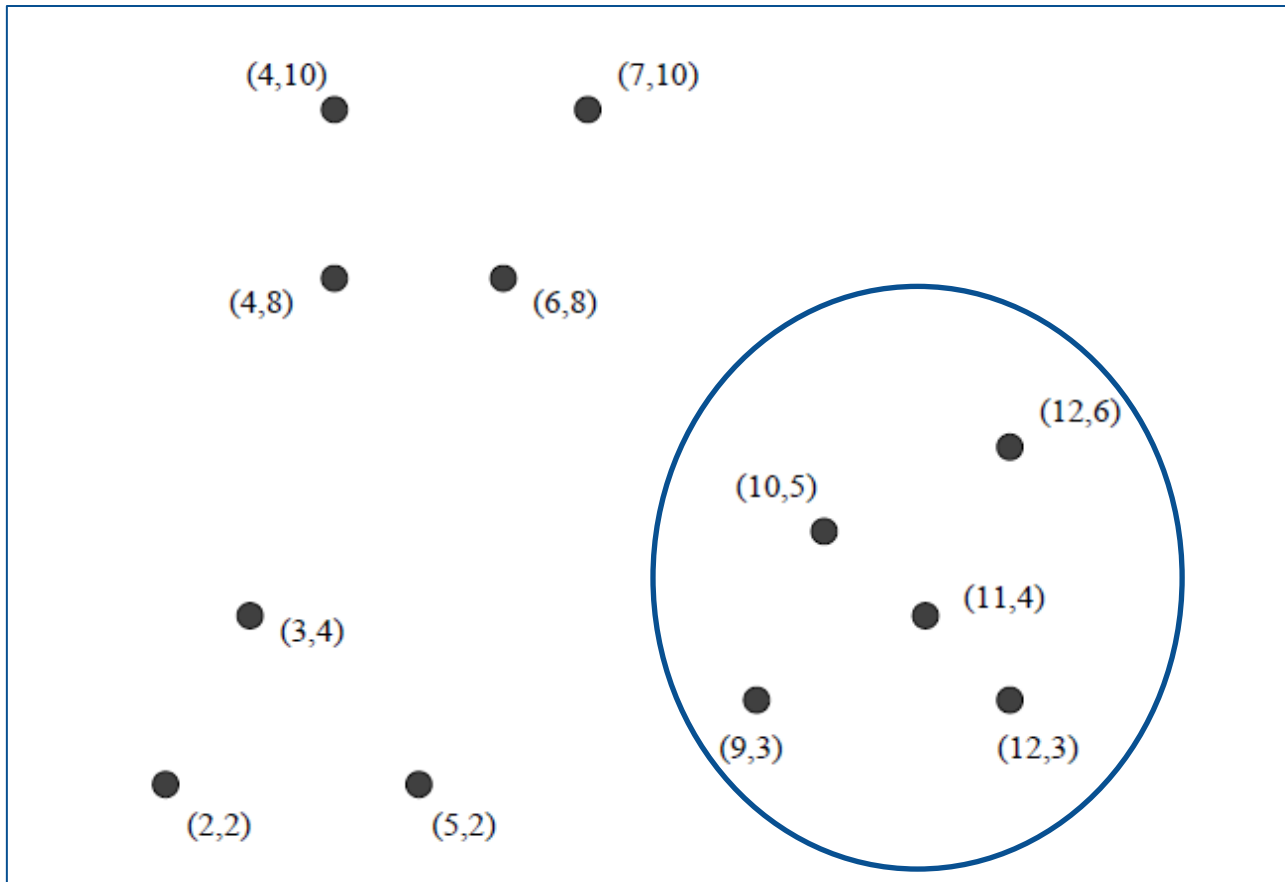
- ▶ L_r norm

$$d([x_1, \dots, x_n], [y_1, \dots, y_n]) = \left(\sum_{i=1}^n |x_i - y_i|^r \right)^{1/r}$$

- ▶ Manhattan Distance (L_1 norm)

$$d([x_1, \dots, x_n], [y_1, \dots, y_n]) = \left(\sum_{i=1}^n |x_i - y_i| \right)$$





Standardized Process

Interval-scaled variables are continuous measurements of a roughly linear scale. Typical examples include weight and height, latitude and longitude coordinates (e.g., when clustering houses), and weather temperature.

The measurement unit used can affect the clustering analysis. For example, changing measurement units from meters to inches for height, or from kilograms to pounds for weight, may lead to a very different clustering structure. In general, expressing a variable in smaller units will lead to a larger range for that variable, and thus a larger effect on the resulting clustering structure. To help avoid dependence on the choice of measurement units, the data should be standardized. **Standardizing** measurements attempts to give all variables an equal weight. This is particularly useful when given no prior knowledge of the data.

To standardize: **To standardize measurements, one choice is to convert the original measurements to unitless variables**



Standardized Process

1. Calculate the mean absolute deviation, s_f :

$$s_f = \frac{1}{n}(|x_{1f} - m_f| + |x_{2f} - m_f| + \cdots + |x_{nf} - m_f|), \quad (7.3)$$

where $x_{1f}, \dots, x_{nf}$ are n measurements of f , and m_f is the *mean* value of f , that is, $m_f = \frac{1}{n}(x_{1f} + x_{2f} + \cdots + x_{nf})$.

2. Calculate the standardized measurement, or z-score:

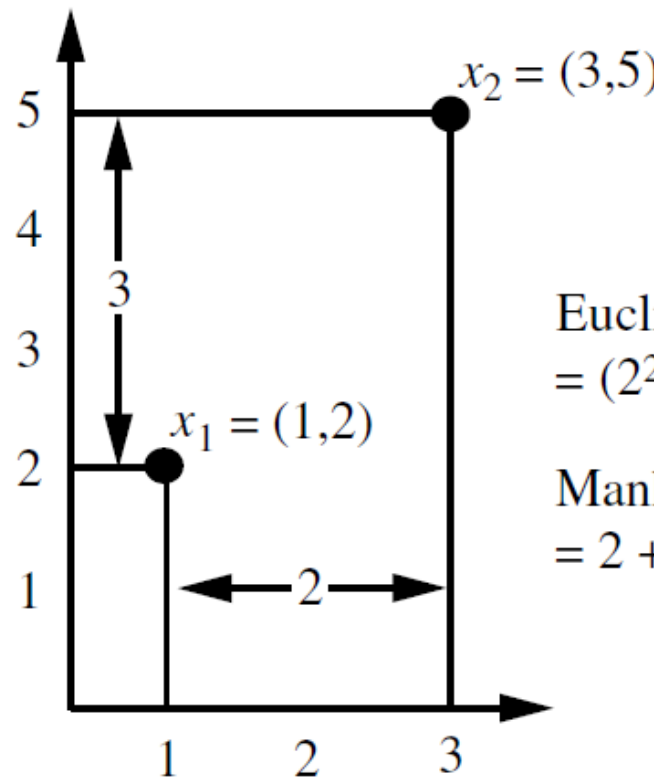
$$z_{if} = \frac{x_{if} - m_f}{s_f}. \quad (7.4)$$

The mean absolute deviation, s_f , is more robust to outliers than the standard deviation, σ_f . When computing the mean absolute deviation, the deviations from the mean (i.e., $|x_{if} - m_f|$) are not squared; hence, the effect of outliers is somewhat reduced. There are more robust measures of dispersion, such as the *median absolute deviation*. However, the advantage of using the mean absolute deviation is that the z-scores of outliers do not become too small; hence, the outliers remain detectable.



In-Class Example

Euclidean distance and Manhattan distance. Let $\mathbf{x}_1 = (1, 2)$ and $\mathbf{x}_2 = (3, 5)$ represent two objects. The Euclidean distance between the two and the Manhattan distance between the two :



Euclidean distance
 $= (2^2 + 3^2)^{1/2} = 3.61$

Manhattan distance
 $= 2 + 3 = 5$



Binary Variables

Let us see how to compute the dissimilarity between objects described by either **symmetric** or **asymmetric binary variables**.

A binary variable has only two states: 0 or 1, where 0 means that the variable is absent, and 1 means that it is present. Given the variable *smoker* describing a patient, for instance, 1 indicates that the patient smokes, while 0 indicates that the patient does not. **Treating binary variables as if they are interval-scaled can lead to misleading clustering results. Therefore, methods specific to binary data are necessary for computing dissimilarities.**

A contingency table for binary variables.

		object j		
		1	0	sum
object i	1	q	r	$q + r$
	0	s	t	$s + t$
sum		$q + s$	$r + t$	p



Binary Variables

Approach involves computing a dissimilarity matrix from the given binary data. If all binary variables are thought of as having the same weight, we have the 2-by-2 contingency table of Table, where

- q is the number of variables that equal 1 for both objects i and j ,
- r is the number of variables that equal 1 for object i but that are 0 for object j ,
- s is the number of variables that equal 0 for object i but equal 1 for object j , and
- t is the number of variables that equal 0 for both objects i and j .

Variable	i	j
q	1	1
r	1	0
s	0	1
t	0	0

The total number of variables is p , where $p = q+r+s+t$.



Binary Variables

“What is the difference between symmetric and asymmetric binary variables?”

A binary variable is **symmetric** if both of its states are equally valuable and carry the same weight; that is, ***there is no preference on which outcome should be coded as 0 or 1***. One such example could be the attribute *gender* having the states *male* and *female*.



Important

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

symmetric binary
dissimilarity

Variable	i	j
q	1	1
r	1	0
s	0	1
t	0	0



Binary Variables

A binary variable is **asymmetric** if the outcomes of the states are not equally important, such as the *positive* and *negative* outcomes of a disease test. By convention, **we shall code the most important outcome, which is usually the rarest one, by 1 (e.g., HIV positive) and the other by 0 (e.g., HIV negative).** Given two asymmetric binary variables, the agreement of two 1s (a positive match) is then considered more significant than that of two 0s (a negative match).



$$d(i, j) = \frac{r + s}{q + r + s}.$$

asymmetric binary
dissimilarity

Variable	i	j
q	1	1
r	1	0
s	0	1
T	0	0



Binary Variables

Complementarily, we can measure the distance between two binary variables based on the notion of *similarity* instead of *dissimilarity*. For example, the asymmetric binary similarity between the objects i and j , or $sim(i, j)$, can be computed as,

$$sim(i, j) = \frac{q}{q + r + s} = 1 - d(i, j).$$

The coefficient $sim(i, j)$ is called the **Jaccard coefficient**, which is popularly referenced in the literature



Important



Device QA Validation

A medical device manufacturer conducting final product validation before releasing the product into the Market. As part of validation phase, the device QA team conducted integration testing. Here are the outcome of the test results by team members:

Name	Title	Number of Defects	FW Algorithm test	Telemetry Test	Therapy Test	Charging Test	Device Backup test
Anitha	Sr. Engineer	25	P	P	N	N	P
Tien	Sr. Engineer	30	N	P	P	P	N
Tami	Sr. Engineer	20	N	P	P	P	P

Evaluate test members? (Hint: compute asymmetric binary dissimilarity)



Device QA Validation

Name	Title	Number of Defects	FW Algorithm test	Telemetry Test	Therapy Test	Charging Test	Device Backup test
Anitha	Sr. Engineer	25	1	1	0	0	1
Tien	Sr. Engineer	30	0	1	1	1	0
Tami	Sr. Engineer	20	0	1	1	1	1

$$d(i, j) = \frac{r + s}{q + r + s}.$$

$d(\text{Anitha}, \text{Tien}) = 2 + 2 / 1 + 2 + 2 = 4/5 = 0.8$

$d(\text{Anitha}, \text{Tami}) = 1 + 2 / 2 + 1 + 2 = 3/5 = 0.6$



In-Class Example (Graded)

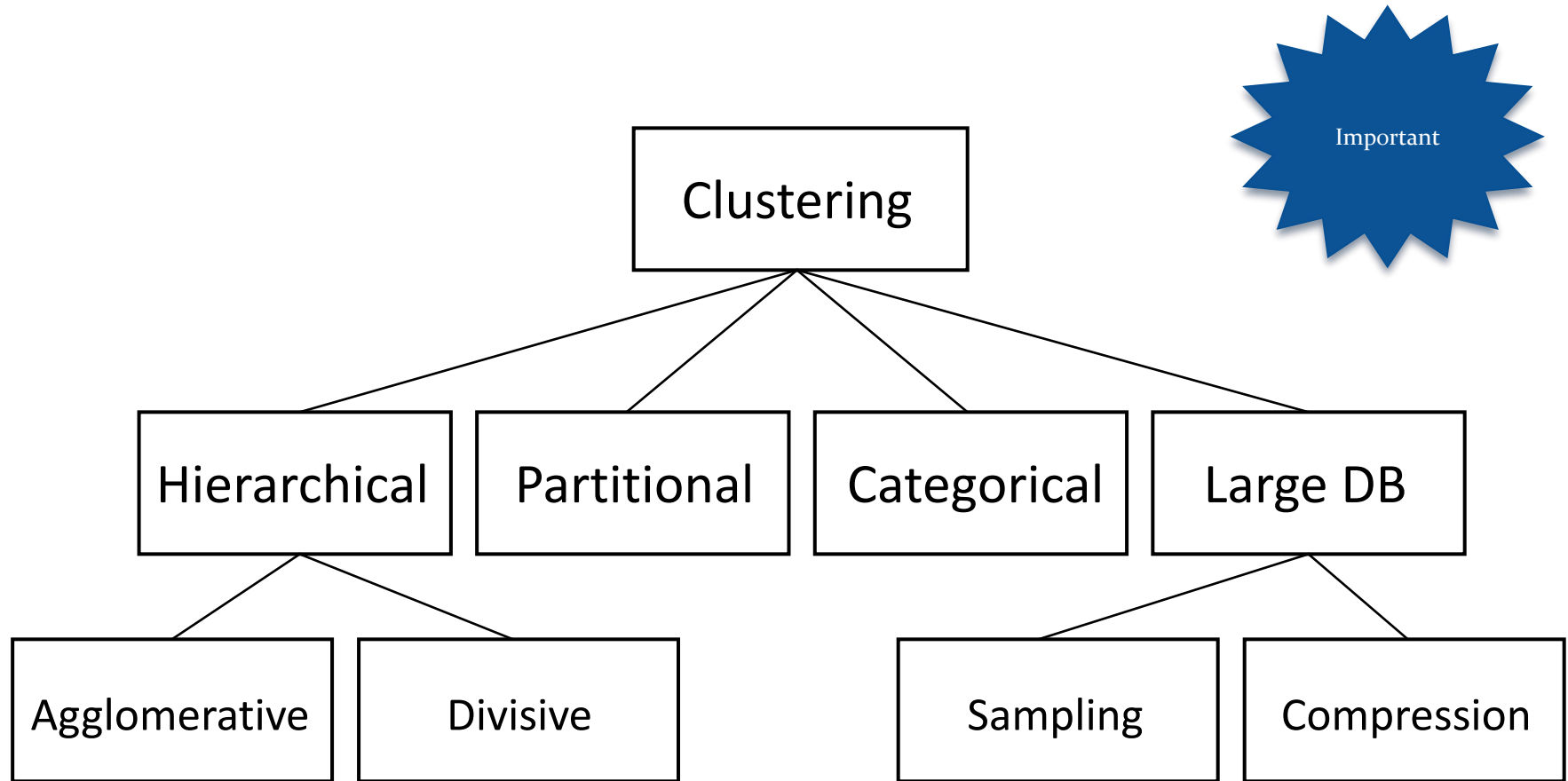
Dissimilarity between binary variables. Suppose that a patient record table contains the attributes *name*, *gender*, *fever*, *cough*, *test-1*, *test-2*, *test-3*, and *test-4*, where *name* is an object identifier, *gender* is a symmetric attribute, and the remaining attributes are asymmetric binary.

A relational table where patients are described by binary attributes.

<i>name</i>	<i>gender</i>	<i>fever</i>	<i>cough</i>	<i>test-1</i>	<i>test-2</i>	<i>test-3</i>	<i>test-4</i>
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	Y	N	N	N	N
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮



Clustering Approaches



Clustering - Background



Two of the most common data clustering algorithms are hierarchical clustering and non-hierarchical clustering. ***In hierarchical clustering, successive clusters are discovered using previously established clusters, while clusters are established all at once using non-hierarchical algorithms.***

Hierarchical algorithms can be further divided into divisive algorithms and agglomerative algorithms. Divisive clustering begins with all data elements in one cluster, which is later successively divided into smaller ones. On the other hand, agglomerative clustering begins with single-element clusters and works by successively merging them into larger ones.

As for non-hierarchical clustering, some of the common algorithms are K-means and fuzzy C-means. Other less conventional clustering algorithms include ant-based clustering, which is inspired by the collective behaviour in decentralized, self-organized systems. In addition to the clustering algorithm, a function or measure for determining the similarity, distance or relatedness between elements is essential. ***In conventional clustering, feature-based measures such as cosine similarity, Jaccard distance or Dice similarity (Salton and McGill, 1983) are employed to support the use of hierarchical and non-hierarchical clustering.***



Clustering - Background



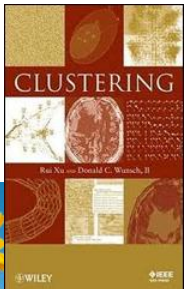
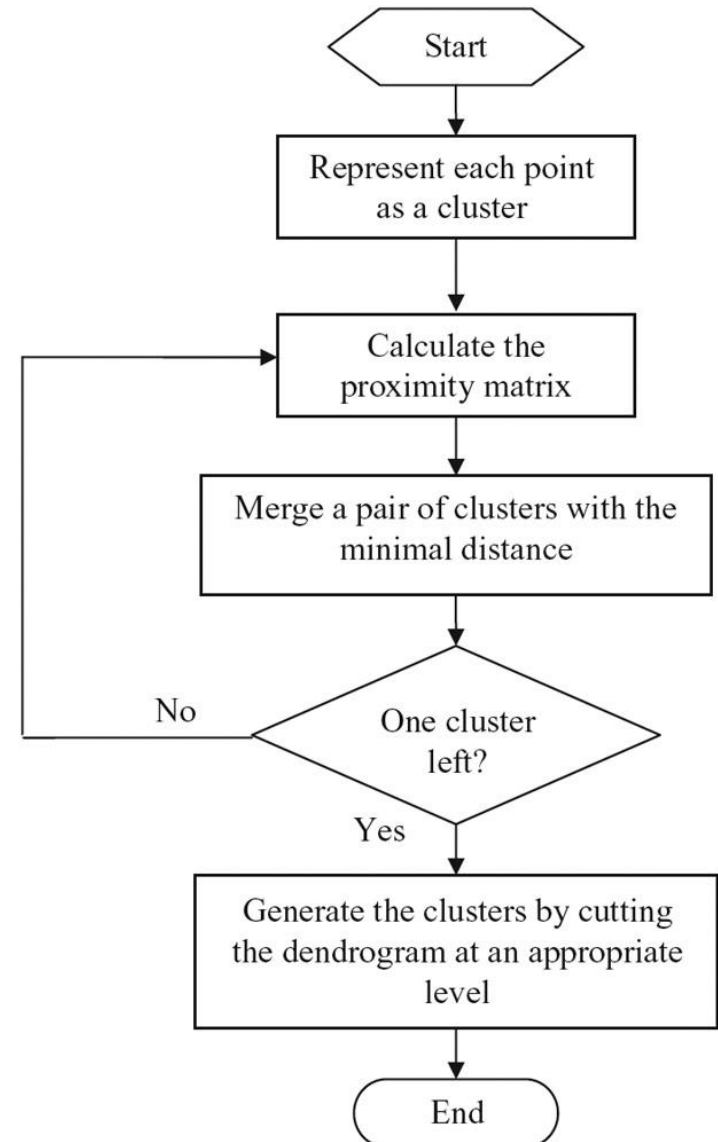
In regard to the types of semantic relations, certain researchers consider the terms "similarity" and "relatedness" to be interchangeable. In the strictest sense, semantic similarity deals only with synonymy, while semantic relatedness incorporates a wider range of concepts such as antonymy, hyponymy and meronymy. For example, the words "buy" and "purchase" belong to the same semantic class and can be used interchangeably within the same context. In that sense, the two words are semantically similar. On the other hand, the words "car" and "tyre" cannot be considered as similar. Nonetheless, we can definitely accept that "car" is related to "tyre" in some way (e.g. through meronymy) more than "car" is to "vegetarian". Due to the absence of feature extraction and selection during the measurement of semantic relation using featureless measures, the distinction between similarity and relatedness can be rather obscure. As such, having discussed the differences, we will remain to use the terms semantic similarity and semantic relatedness interchangeably.



Flowchart - hierarchical clustering algorithm

Flowchart of the agglomerative hierarchical clustering algorithm.

Agglomerative clustering considers each data point as a cluster in the beginning. Two clusters are then merged in each step until all objects are forced into the same group.



Cluster Summary Parameters

$$\textit{centroid} = C_m = \frac{\sum_{i=1}^N (t_{mi})}{N}$$

$$\textit{radius} = R_m = \sqrt{\frac{\sum_{i=1}^N (t_{mi} - C_m)^2}{N}}$$

$$\textit{diameter} = D_m = \sqrt{\frac{\sum_{i=1}^N \sum_{j=1}^N (t_{mi} - t_{mj})^2}{(N)(N-1)}}$$



Hierarchical clustering

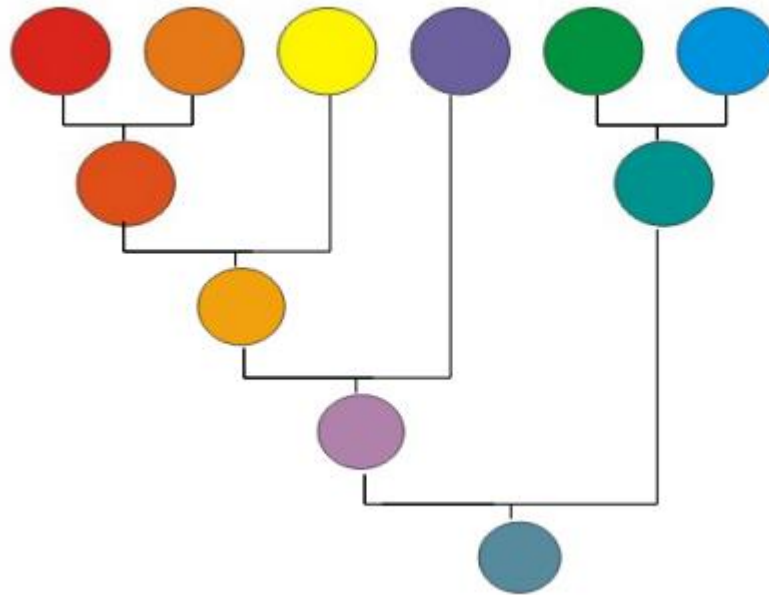
Hierarchical clustering is an agglomerative (top down) clustering method. As its name suggests, the idea of this method is to build a hierarchy of clusters, showing relations between the individual members and ***merging clusters of data based on similarity.***

In the first step of clustering, the algorithm will look for the two most similar data points and merge them to create a new "pseudo-datapoint", which represents the average of the two merged datapoints. Each iterative step takes the next two closest datapoints (or pseudo-datapoints) and merges them. This process is generally continued until there is one large cluster containing all the original datapoints. Hierarchical clustering results in a "tree", showing the relationship of all of the original points.



Colors Example - Agglomerative clustering

The example below demonstrates the agglomerative clustering using colors. In the first stages, the red and orange are combined to make reddish-orange. Green and blue are similar to each other and are merged to form aqua. Next, the reddish-orange and yellow are the closest colors, and create a light orange. Now we have three remaining colors: light orange, the original purple and aqua. **The hierarchical clustering algorithm is only interested in finding the closest relationship at each stage regardless of the degree of similarity.** So in this case, the light orange and purple are more similar than the any other combination. We are now left with the orange-purple mixture and aqua. They are now the most similar samples left and so form a final cluster. The resulting tree shows the overall similarity of the original samples (this is the hierarchy).



Hierarchical clustering

- Hierarchical clustering takes as input a set of points
- It creates a tree in which the points are leaves and the internal nodes reveal the similarity structure of the points.
 - The tree is often called a “dendogram.”
- The method is summarized below:

Place all points into their own clusters
While there is more than one cluster, do
Merge the closest pair of clusters



- The behavior of the algorithm depends on how “closest pair of clusters” is defined



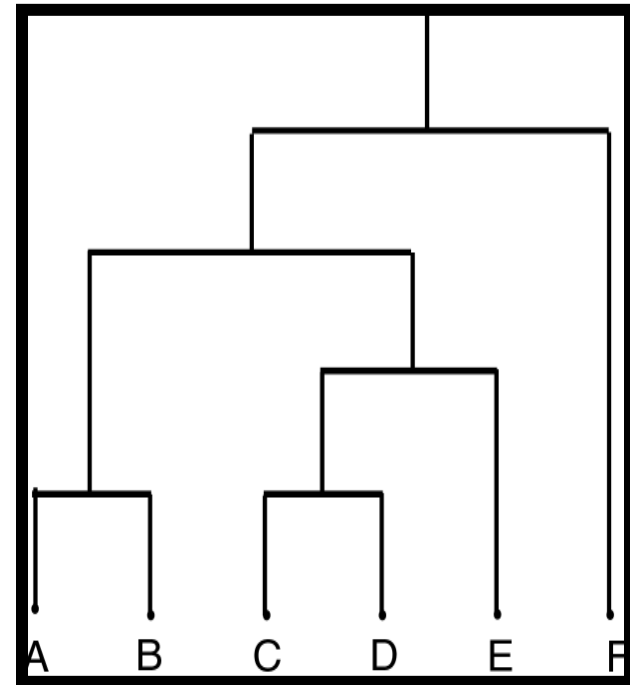
Hierarchical clustering

- Clusters are created in levels actually creating sets of clusters at each level.
- **Agglomerative**
 - Initially each item in its own cluster
 - Iteratively clusters are merged together
 - Bottom Up
- **Divisive**
 - Initially all items in one cluster
 - Large clusters are successively divided
 - Top Down

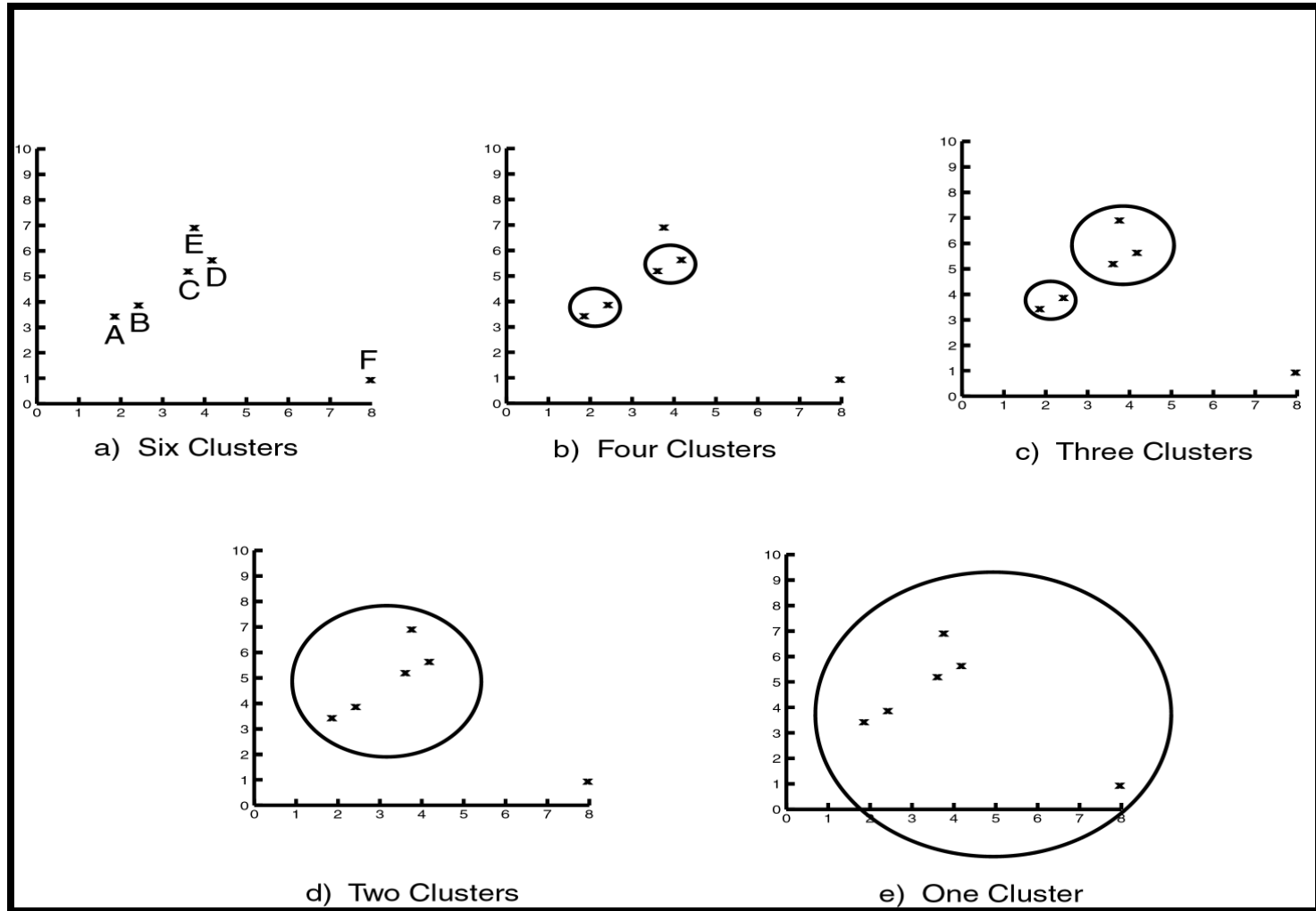


Dendrogram

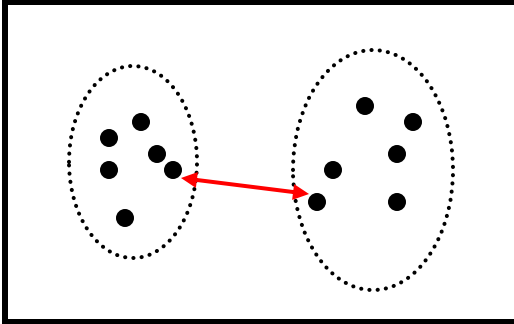
- ***Dendrogram***: a tree data structure which illustrates hierarchical clustering techniques.
- Each level shows clusters for that level.
 - Leaf – individual clusters
 - Root – one cluster
- A cluster at level i is the union of its children clusters at level $i+1$.



Levels of Clustering

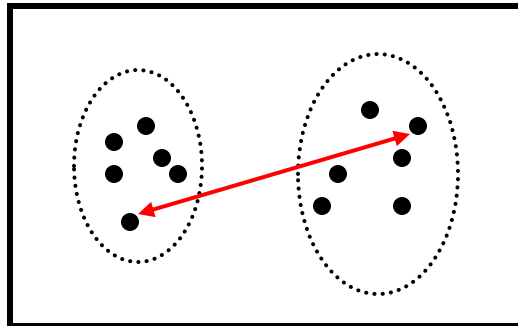
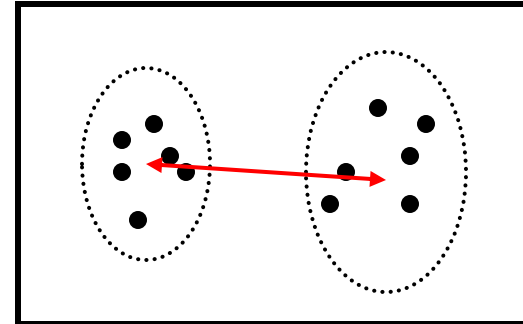


Hierarchical clustering - Merging Clusters



Single Link: Distance between two clusters is the distance between the closest points. Also called “neighbor joining.”

Average Link: Distance between clusters is distance between the cluster centroids.



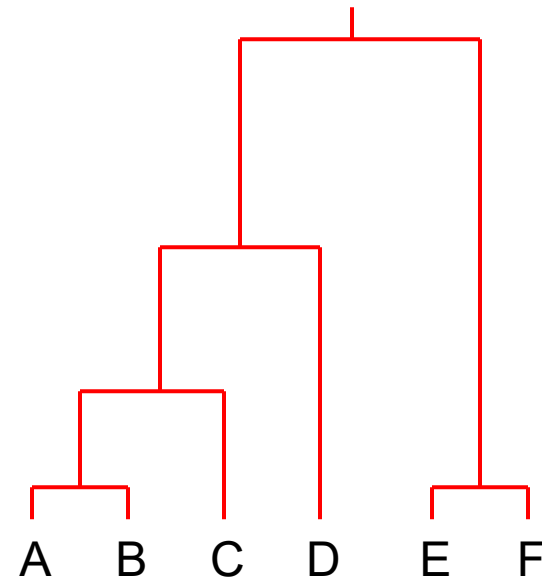
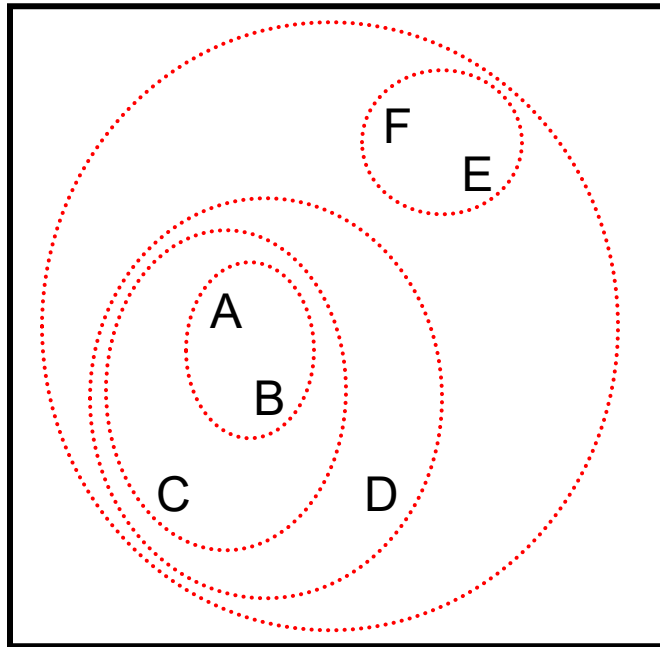
Complete Link: Distance between clusters is distance between farthest pair of points.

Important



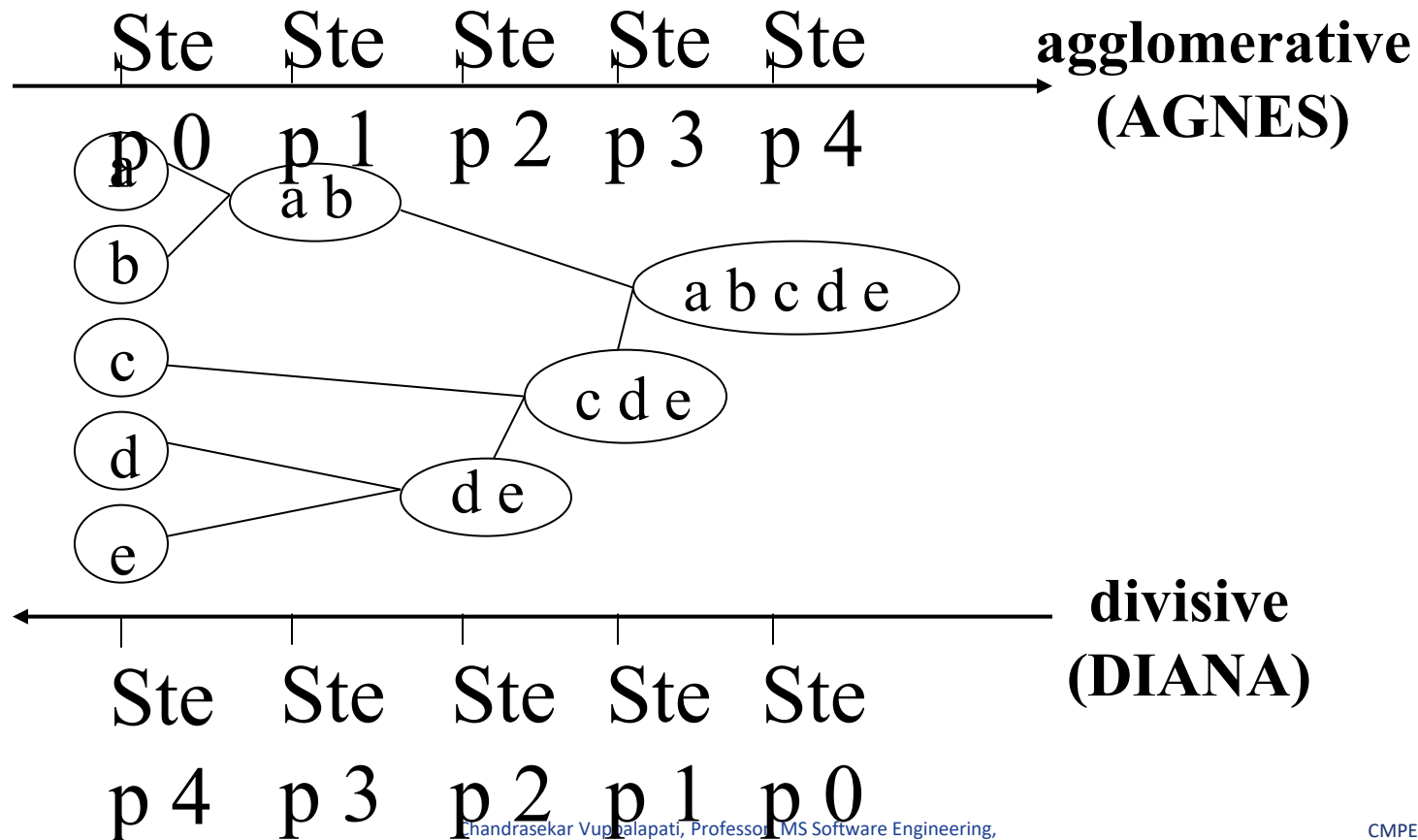
Hierarchical Clustering: Example

This example illustrates single-link clustering in Euclidean space on 6 points.

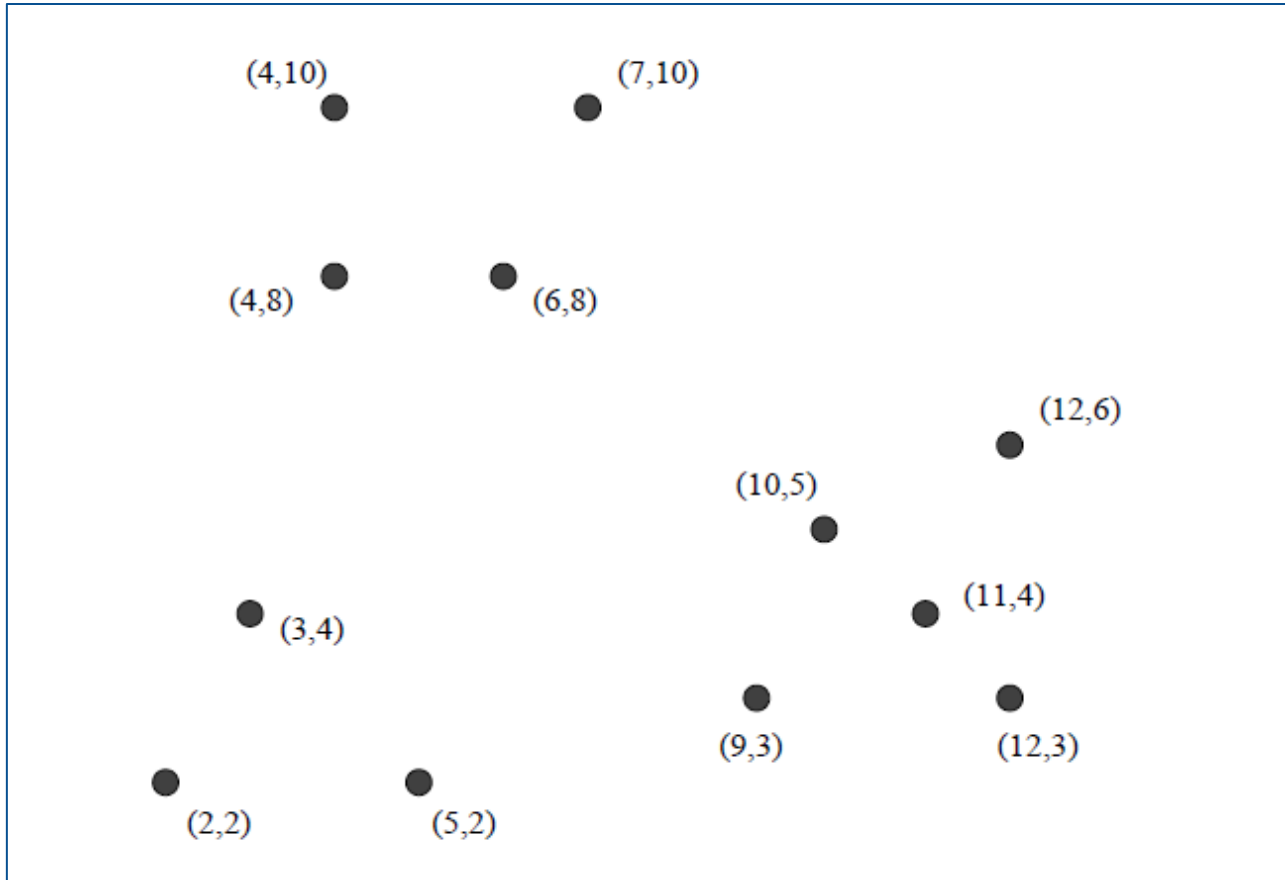


Hierarchical Clustering

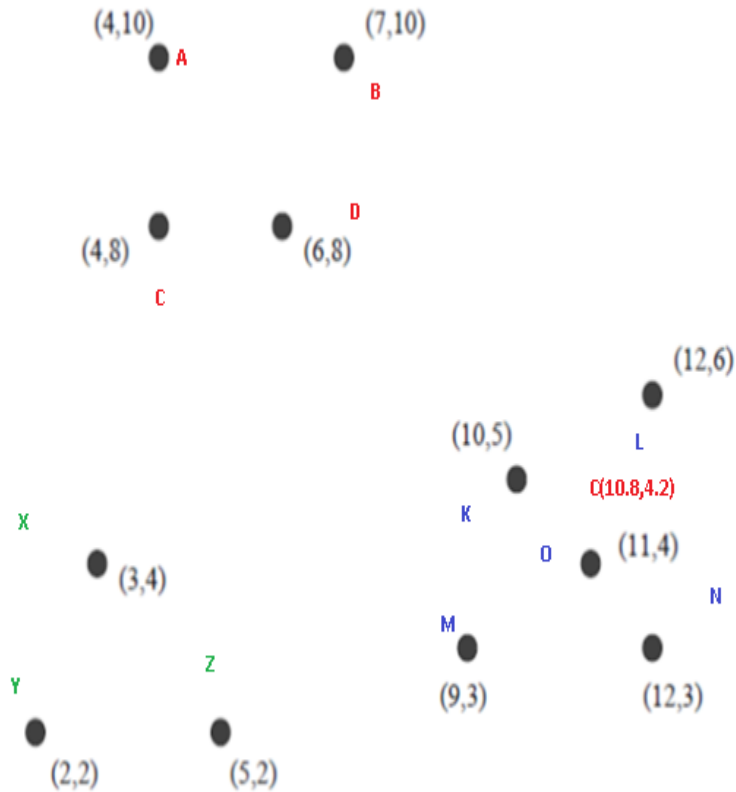
- Use distance matrix as clustering criteria. This method does not require the number of clusters k as an input, but needs a termination condition



Hierarchical Clustering - Example



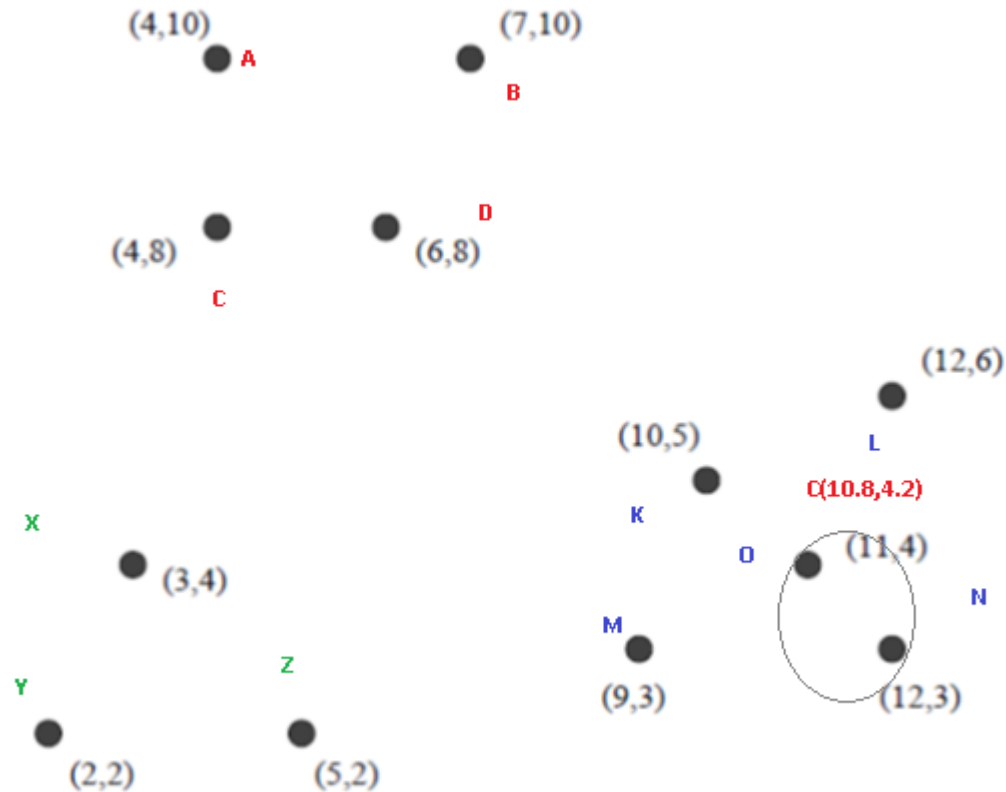
Hierarchical Clustering - Example



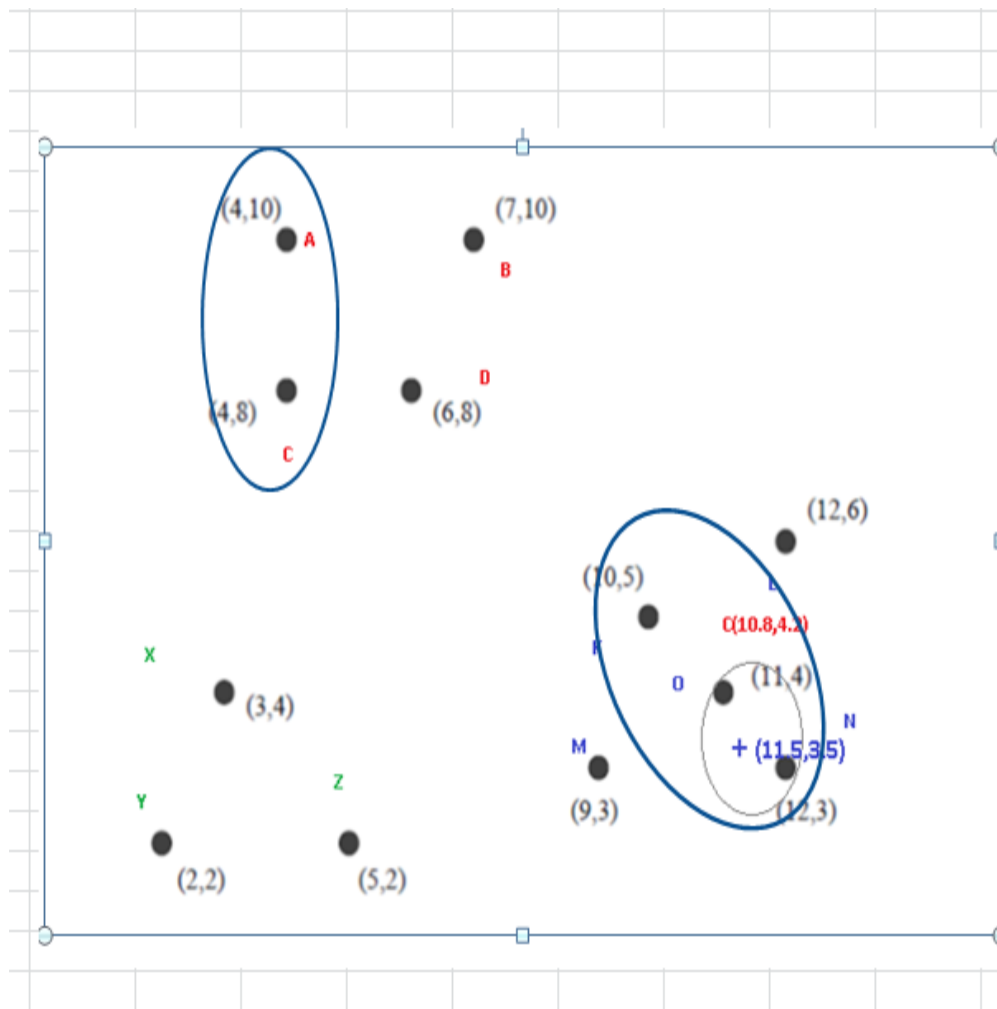
				10.8
AB	3			4.2
BD	SQRT(5)			
AC	2			
CD	2			
KL	SQRT(5)	Centroid	10.8, 4.2	
LO	SQRT(5)			
KO	SQRT(2)			
ON	SQRT(2)			
OM	SQRT(5)			
MN	3			
XY	SQRT(5)			
XZ	SQRT(5)			
YZ	3			



Hierarchical Clustering - Example



Hierarchical Clustering - Example

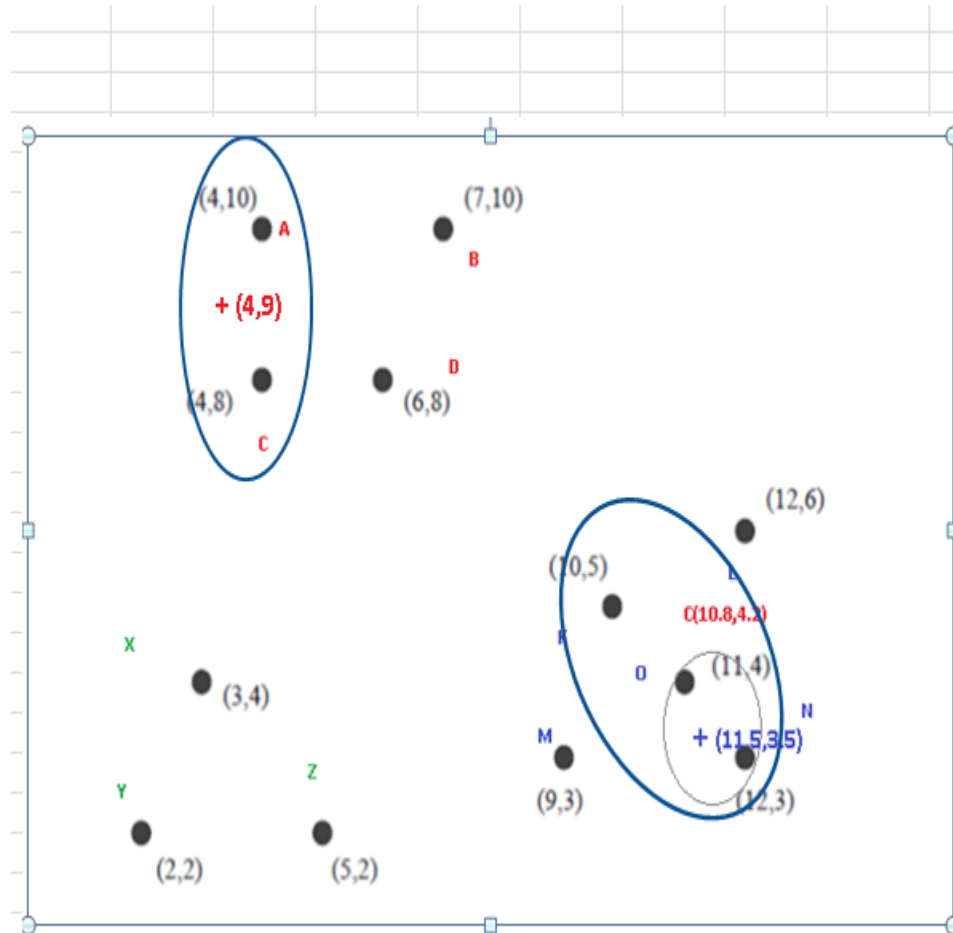


AB	3	
BD	$\text{SQRT}(5)$	2.236068
AC	2	
CD	2	
KL	$\text{SQRT}(5)$	Centroid 10.8,4.2
L+	2.54951	
K+	2.44949	

XY	$\text{SQRT}(5)$
XZ	$\text{SQRT}(5)$
YZ	3



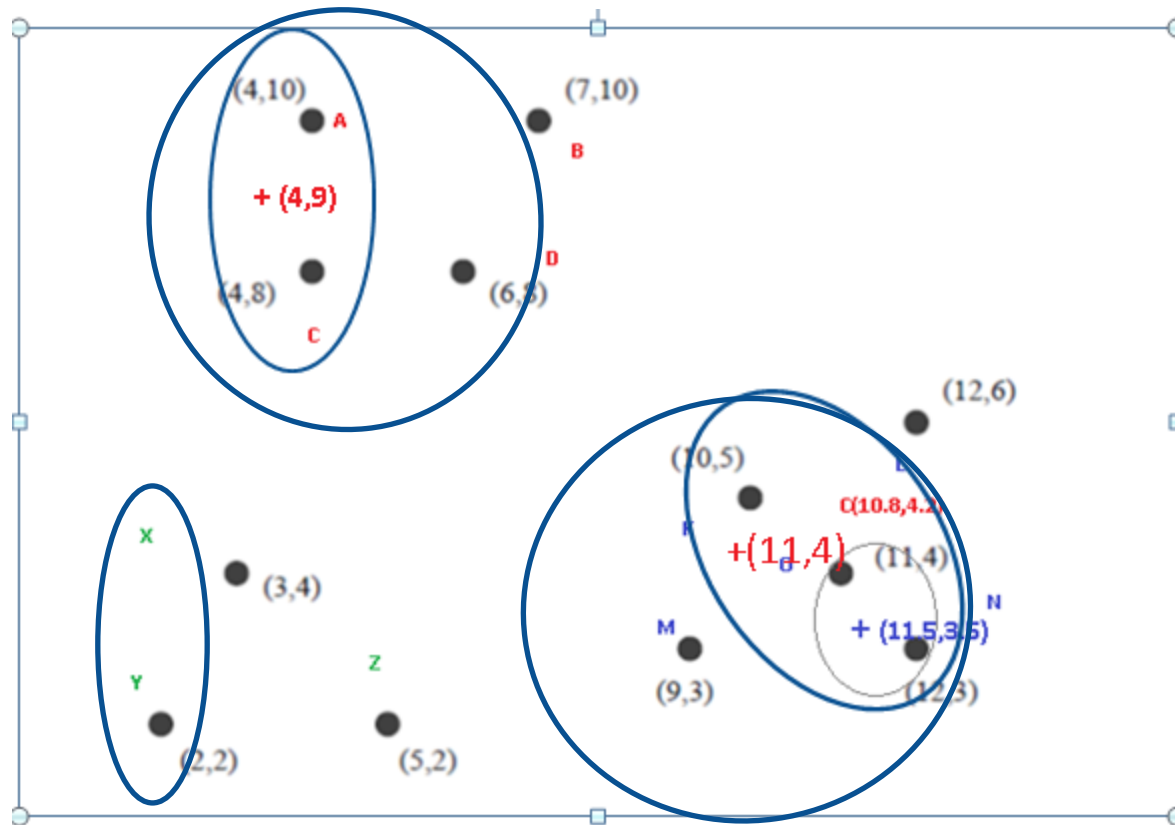
Hierarchical Clustering - Example



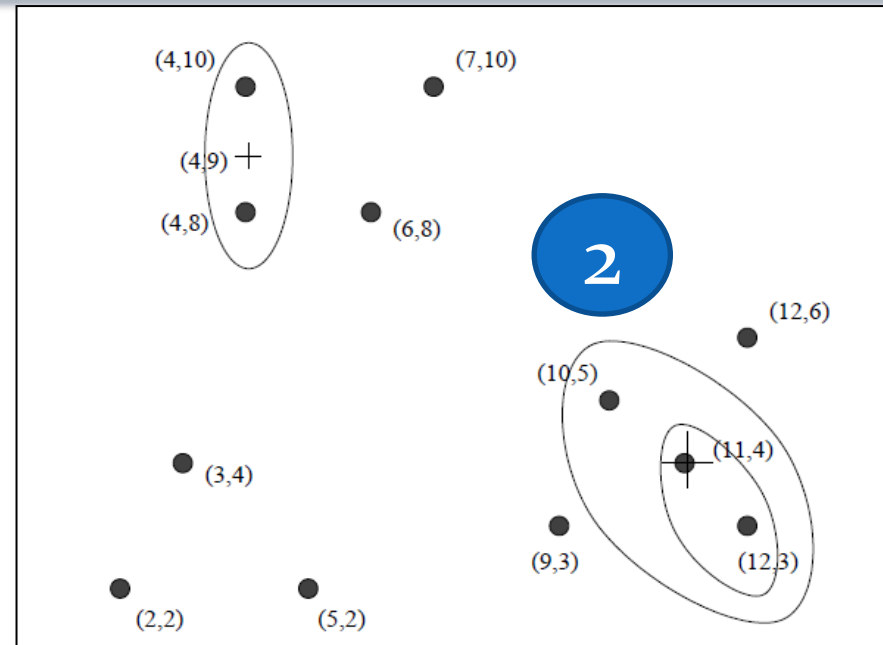
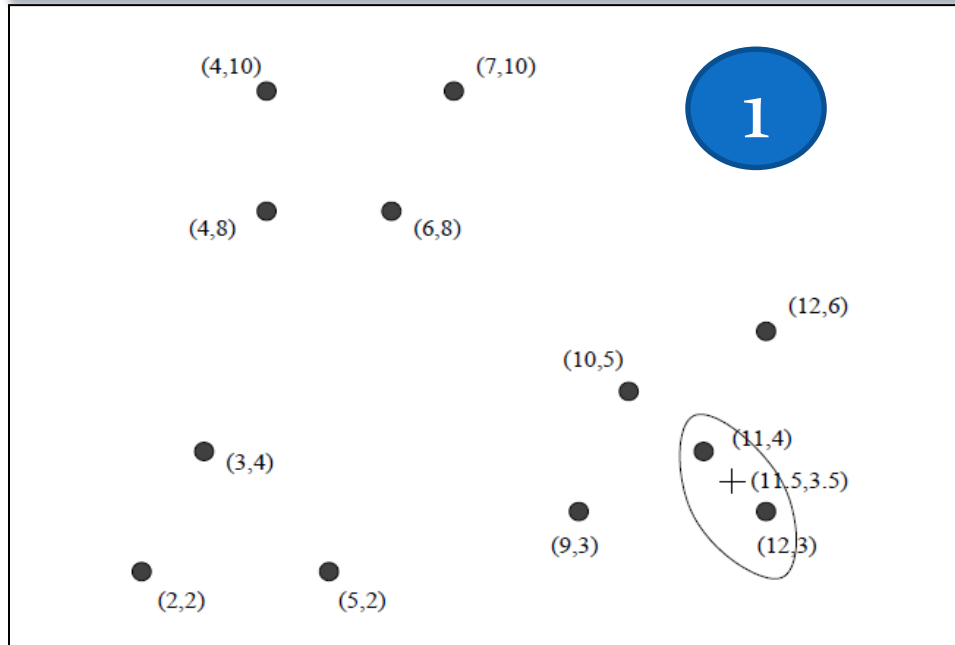
B+	SQRT(10)		
D+	SQRT(5)	2.236068	
BD	SQRT(10)		
KL	SQRT(5)		Centroid 10.8,4.2
L+	2.54951		
K+	2.44949		
XY	SQRT(5)		
XZ	SQRT(5)		
YZ	3		



Hierarchical Clustering - Example

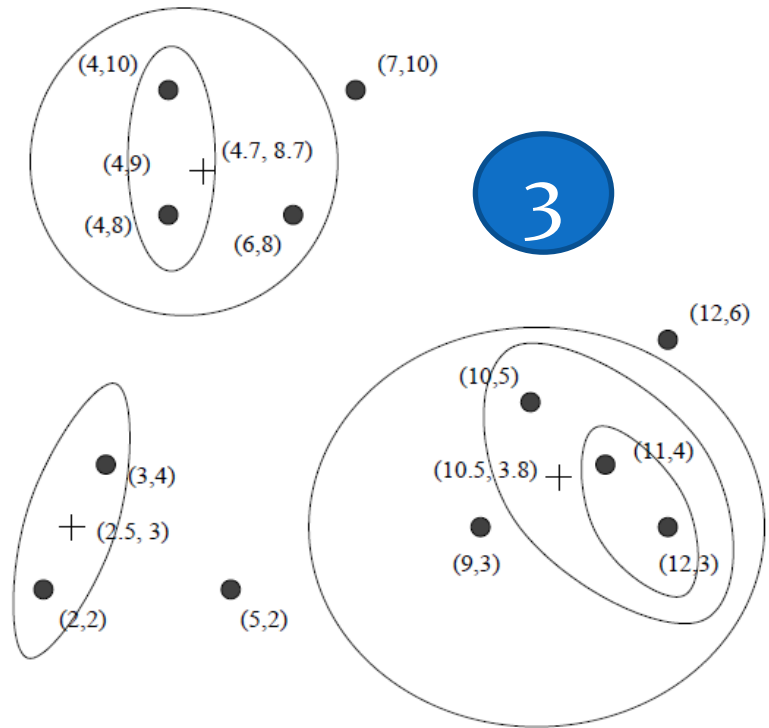


Clustering Solution

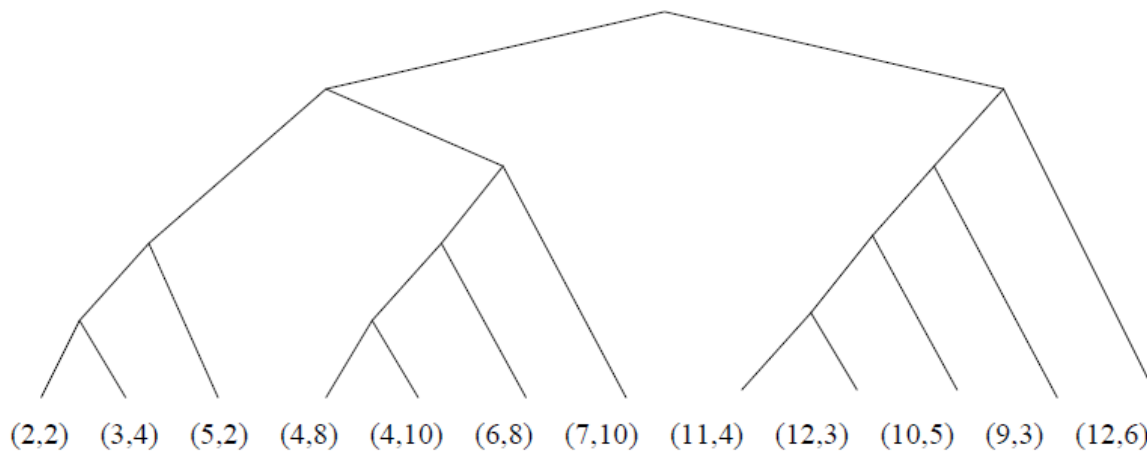


Clustering Solution

3



4



When do you stop?

There are several approaches we might use to stopping the clustering process.

- We could be told, or have a belief, about how many clusters there are in the data. For example, if we are told that the data about dogs is taken from Chihuahuas, Dachshunds, and Beagles, then we know to stop when there are three clusters left.
- We could stop combining when at some point the best combination of existing clusters produces a cluster that is inadequate.
- We could continue clustering until there is only one cluster. However, it is meaningless to return a single cluster consisting of all the points. Rather, we return the tree representing the way in which all the points were combined. This form of answer makes good sense in some applications, such as one in which the points are genomes of different species, and the distance measure reflects the difference in the genome. Then, the tree represents the evolution of these species, that is, the likely order in which two species branched from a common ancestor.



Radius of Cluster

The radius of a cluster is the maximum distance between all the points and the centroid. Combine the two clusters whose resulting cluster has the lowest radius. A slight modification is to combine the clusters whose result has the lowest average distance between a point and the centroid. Another modification is to use the sum of the squares of the distances between the points and the centroid. In some algorithms, we shall find these variant definitions of “radius” referred to as “the radius.”



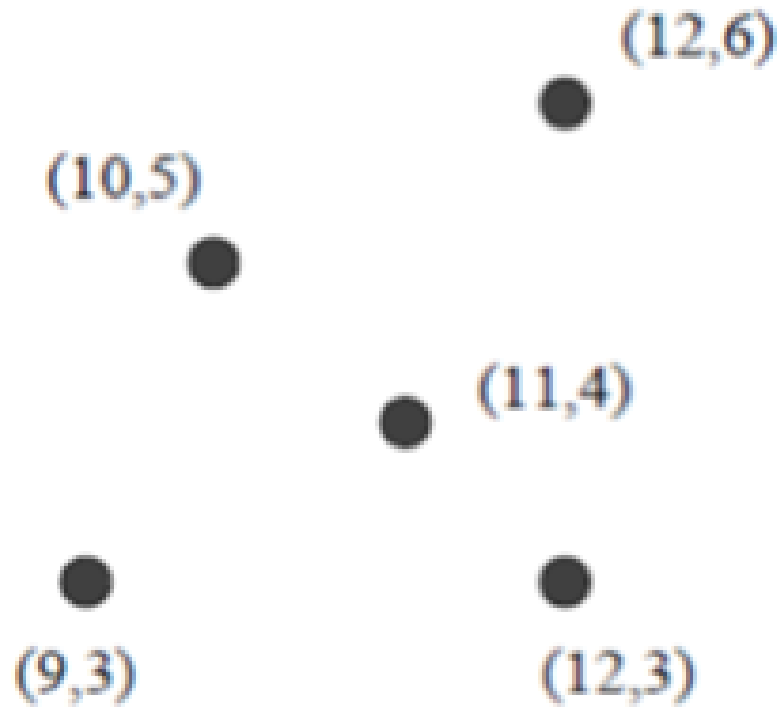
Diameter of Cluster

The diameter of a cluster is the maximum distance between any two points of the cluster. Note that the **radius and diameter of a cluster are not related directly, as they are in a circle**, but there is a tendency for them to be proportional. We may choose to merge those clusters whose resulting cluster has the smallest diameter. Variants of this rule, analogous to the rule for radius, are possible.

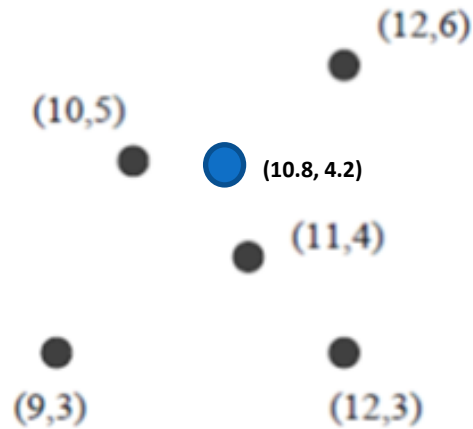


Diameter of Cluster

Compute Radius and Diameter?



Diameter of Cluster



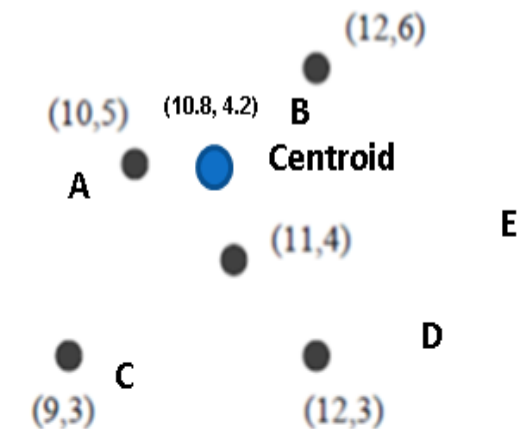
Centroid

$$\begin{aligned} X &= (10+9+12+11+12) / 5 \\ &= 54/5 \\ &= 10.8 \end{aligned}$$

$$\begin{aligned} Y &= (5+3+3+4+6) / 5 \\ &= 21/5 \\ &= 4.2 \end{aligned}$$



Radius & Diameter



Centroid

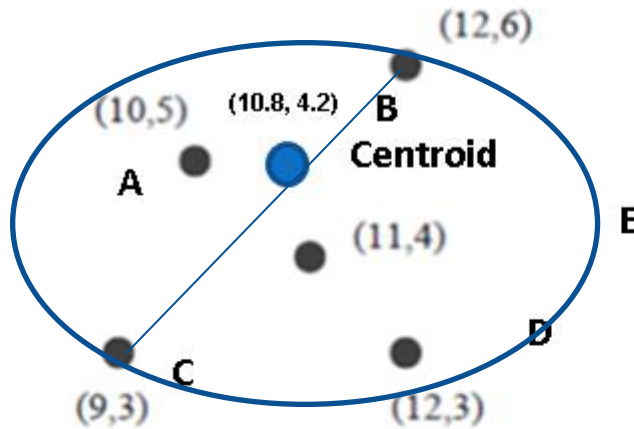
$$\begin{aligned} X &= (10+9+12+11+12) / 5 \\ &= 54/5 \\ &= 10.8 \end{aligned} \quad \begin{aligned} Y &= (5+3+3+4+6) / 5 \\ &= 21/5 \\ &= 4.2 \end{aligned}$$

A-Centroid	$0.8 * \text{SQRT}(2)$	1.131371
B-Centroid	2.163330765	
C-Centroid	2.163330765	
D-Centroid	$1.2 * \text{SQRT}(2)$	1.697056
E-Centroid	$0.2 * \text{SQRT}(2)$	0.282843

There is a tie for the two furthest points from the centroid: (9,3) and (12,6), both at distance $\sqrt{4.68} = 2.16$. Thus, the radius is 2.16. For the diameter, we find the two points in the cluster having the greatest distance. These are again (9,3) and (12,6). Their distance is $\sqrt{18} = 4.24$, so that is the diameter. Notice that the diameter is not exactly twice the radius, although it is close in this case. The reason is that the centroid is not on the line between (9,3) and (12,6).



Radius & Diameter



Radius: 2.16
Diameter: 4.24

Centroid

$$\begin{aligned} X &= (10+9+12+11+12) / 5 \\ &= 54/5 \\ &= 10.8 \end{aligned}$$

$$\begin{aligned} Y &= (5+3+3+4+6) / 5 \\ &= 21/5 \\ &= 4.2 \end{aligned}$$

Important



SCIKIT Reference – Hierarchical Clustering

Hierarchical clustering is a general family of clustering algorithms that build nested clusters by merging or splitting them successively. This hierarchy of clusters is represented as a tree (or dendrogram). The root of the tree is the unique cluster that gathers all the samples, the leaves being the clusters with only one sample. See the Wikipedia page for more details.

The `AgglomerativeClustering` object performs a hierarchical clustering using a bottom up approach: each observation starts in its own cluster, and clusters are successively merged together. The linkage criteria determines the metric used for the merge strategy:

Ward minimizes the sum of squared differences within all clusters. It is a variance-minimizing approach and in this sense is similar to the k-means objective function but tackled with an agglomerative hierarchical approach.

Maximum or complete linkage minimizes the maximum distance between observations of pairs of clusters.

Average linkage minimizes the average of the distances between all observations of pairs of clusters.

Single linkage minimizes the distance between the closest observations of pairs of clusters.

<http://scikit-learn.org/stable/modules/clustering.html>



Demo

Anaconda Prompt

```
(base) C:\Hanumayamma\Clustering>python cmpe256_inclass_plot_points.py  
  
single link cluster  
  
complete link cluster  
  
average link cluster  
  
(base) C:\Hanumayamma\Clustering> _
```

<http://scikit-learn.org/stable/modules/clustering.html>



Code

```
import numpy as np

X = np.array([[7,10],
              [4,10],
              [4,8],
              [6,8],
              [3,4],
              [2,2],
              [9,3],
              [12,3],
              [5,2],
              [10,5],
              [12,6],
              [11,4],])

import matplotlib.pyplot as plt

labels = range(1, 13)
plt.figure(figsize=(10, 7))
plt.subplots_adjust(bottom=0.1)
plt.scatter(X[:,0],X[:,1], label='True Position')

for label, x, y in zip(labels, X[:, 0], X[:, 1]):
    plt.annotate(
        label,
        xy=(x, y), xytext=(-3, 3),
        textcoords='offset points', ha='right', va='bottom')
plt.show()
```



<http://scikit-learn.org/stable/modules/clustering.html>

Code

```
from scipy.cluster.hierarchy import dendrogram, linkage
from matplotlib import pyplot as plt
```

```
print ("\n single link cluster \n")
linked = linkage(X, 'single')
```

```
labelList = range(1, 13)
```

```
plt.figure(figsize=(10, 7))
dendrogram(linked,
            orientation='top',
            labels=labelList,
            distance_sort='descending',
            show_leaf_counts=True)
plt.show()
```

```
print ("\n complete link cluster \n")
```

```
linked_complete = linkage(X, 'complete')
```

```
labelList = range(1, 13)
```

```
plt.figure(figsize=(10, 7))
dendrogram(linked_complete,
            orientation='top',
            labels=labelList,
            distance_sort='descending',
            show_leaf_counts=True)
plt.show()
```

<http://scikit-learn.org/stable/modules/clustering.html>



Code

```
print ("\n average link cluster \n")

linked_average = linkage(X, 'average')

labelList = range(1, 13)

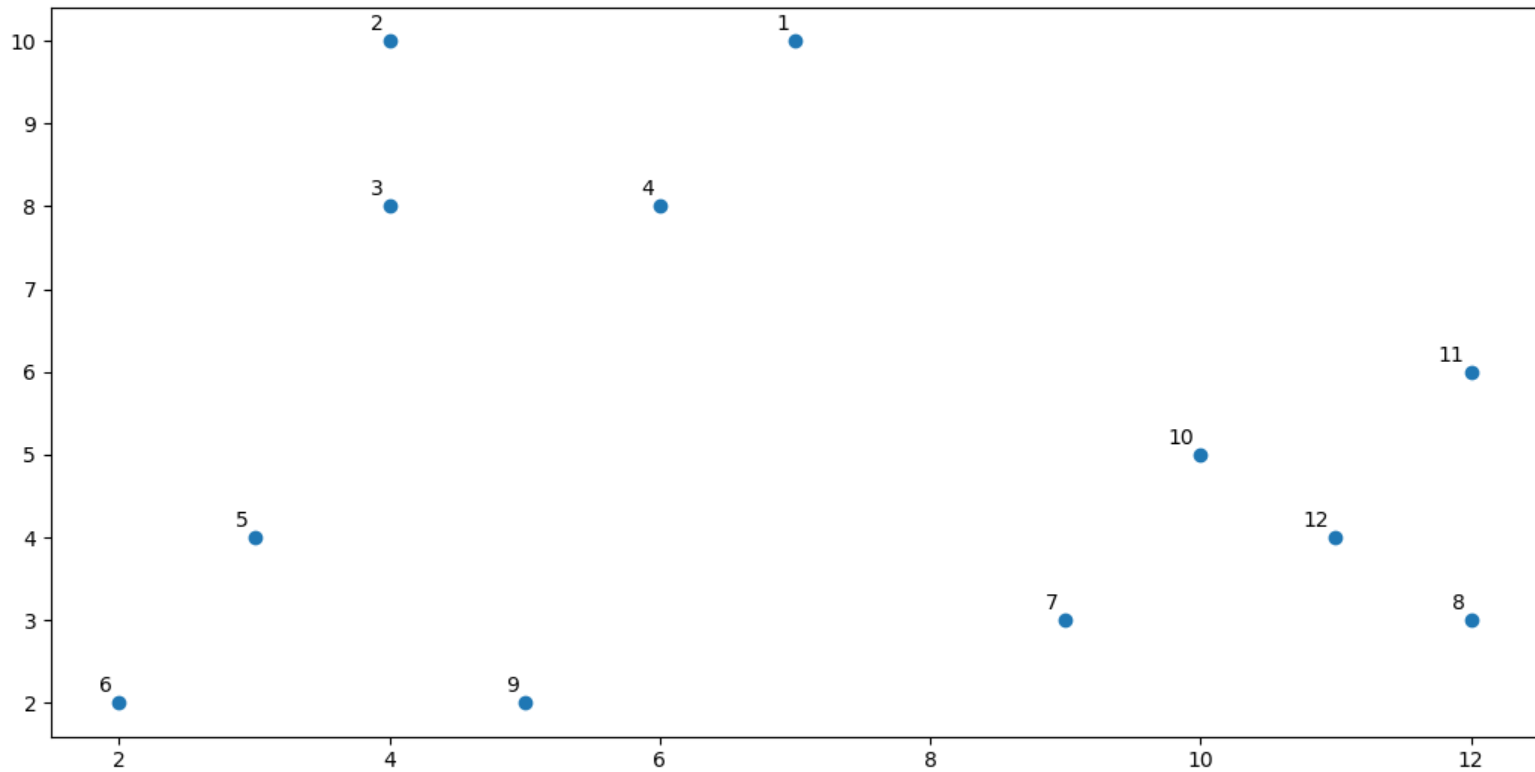
plt.figure(figsize=(10, 7))
dendrogram(linked_complete,
            orientation='top',
            labels=labelList,
            distance_sort='descending',
            show_leaf_counts=True)
plt.show()
```

<http://scikit-learn.org/stable/modules/clustering.html>



Plot

Figure 1

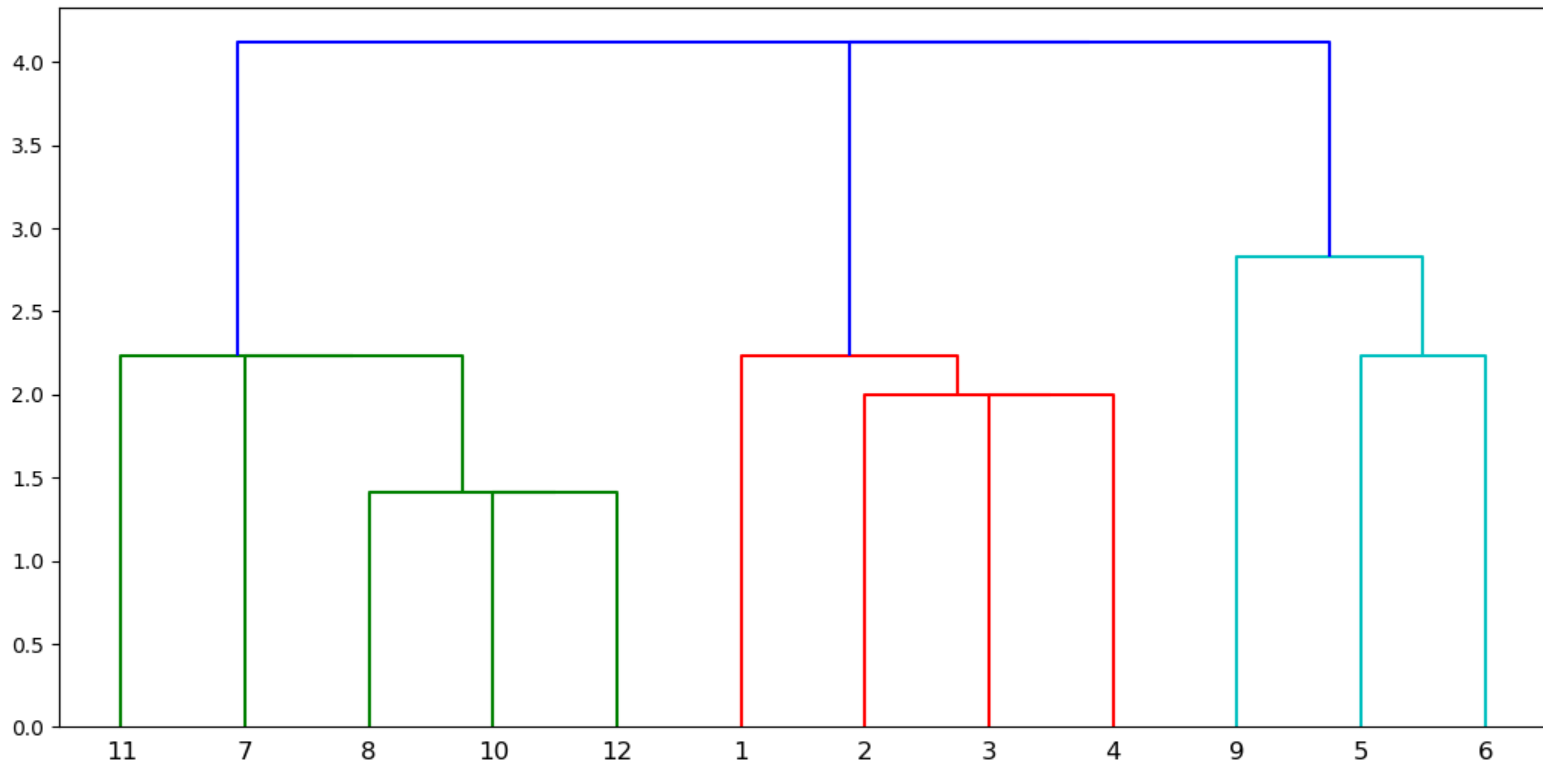


<http://scikit-learn.org/stable/modules/clustering.html>



Single Link Dendrogram

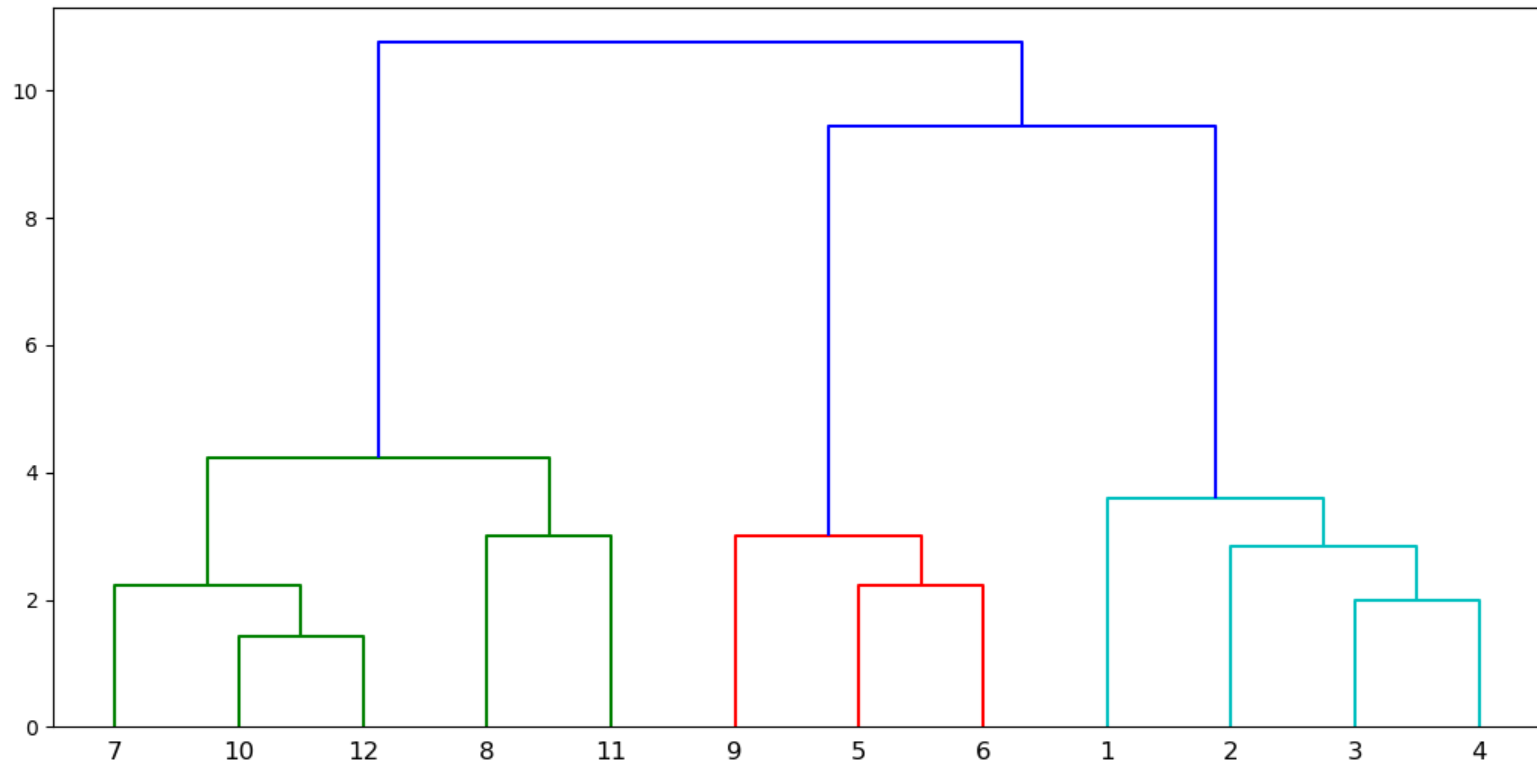
Figure 1



<http://scikit-learn.org/stable/modules/clustering.html>

Complete Link Dendrogram

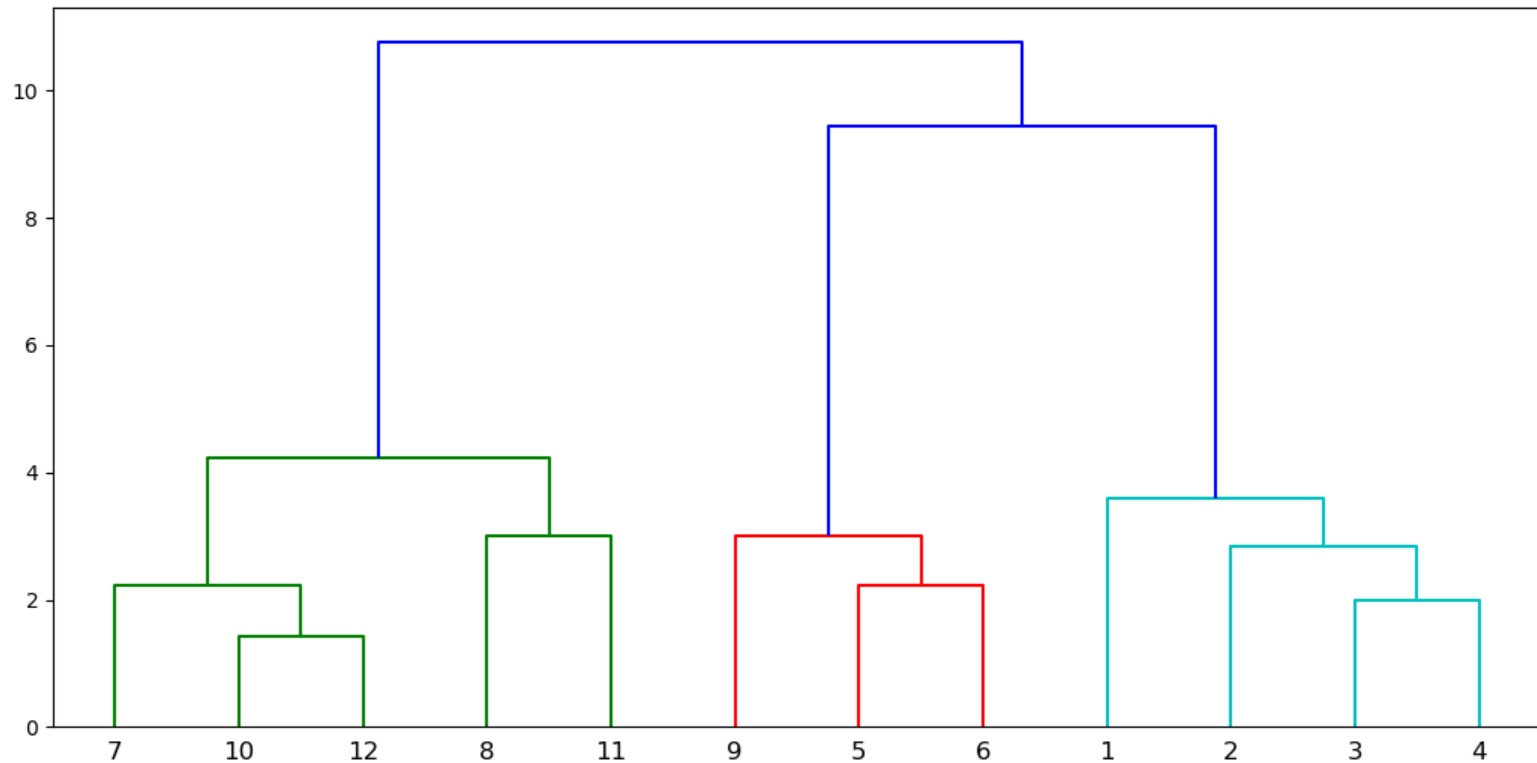
Figure 1



<http://scikit-learn.org/stable/modules/clustering.html>

Average Link Dendrogram

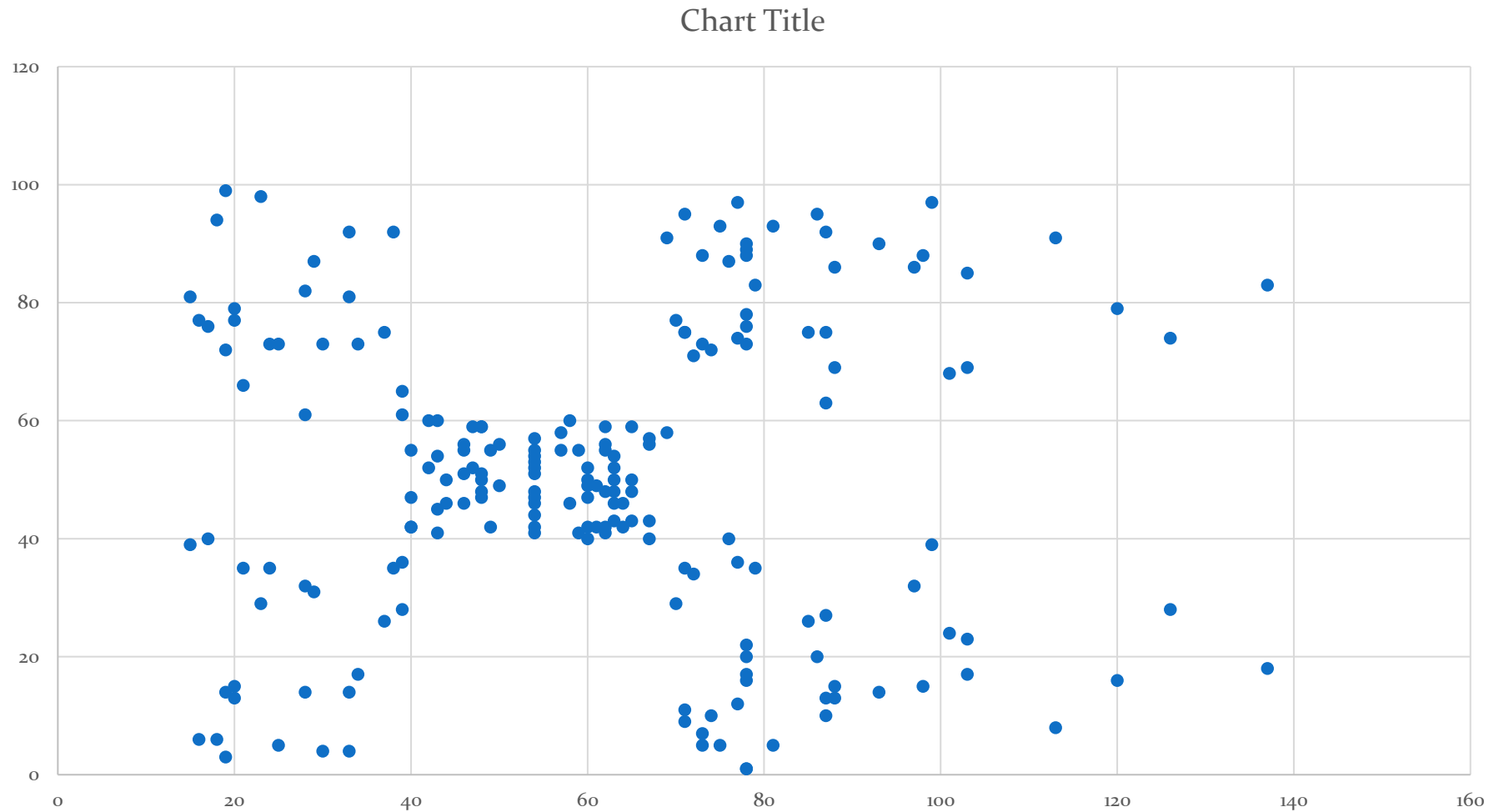
Figure 1



x= y=8.00278

<http://scikit-learn.org/stable/modules/clustering.html>

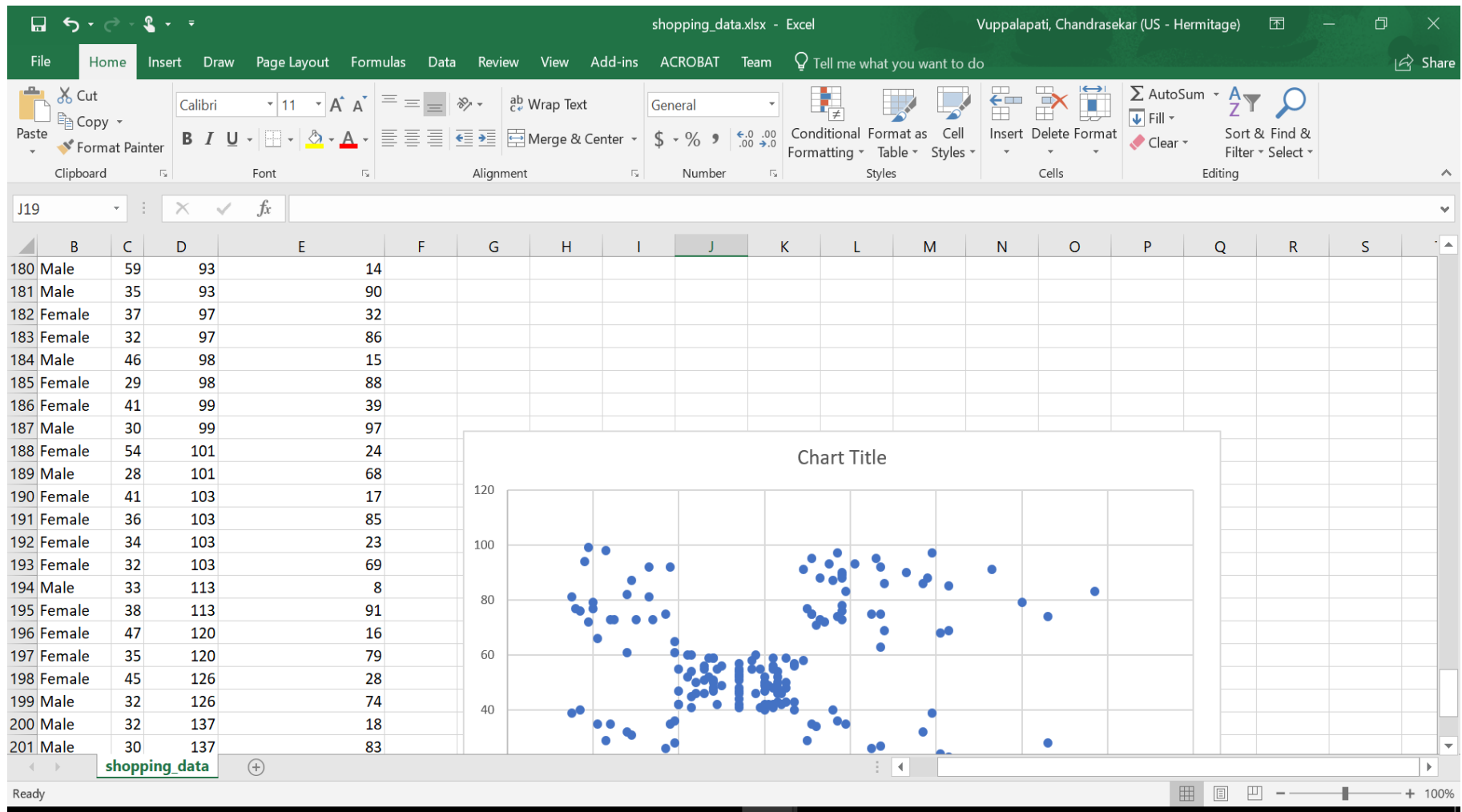
Customer Data



<http://scikit-learn.org/stable/modules/clustering.html>



Customer Data



<http://scikit-learn.org/stable/modules/clustering.html>

Customer Data

Spyder (Python 3.6)

File Edit Search Source Run Debug Consoles Projects Tools View Help

Editor - C:\Hanumayamma\Clustering\agglomeratedustering_plot.py

```
8# -*- coding: utf-8 -*-
9"""
10Created on Wed Oct 3 17:01:24 2018
11
12@author: cvuppapalati
13"""
14# https://stackoverflow.com/questions/35595766/matplotlib-line-magic-causes-syntaxerror-in-python-script
15# https://stackabuse.com/hierarchical-clustering-with-python-and-scikit-learn/
16# https://matplotlib.org/users/installing.html
17
18import matplotlib.pyplot as plt
19import pandas as pd
20##matplotlib inline
21'exec(%matplotlib inline)'
22
23import numpy as np
24customer_data = pd.read_csv('shopping_data.csv')
25customer_data.shape
26print('\ncustomer data\n')
27print(customer_data.head())
28
29data = customer_data.iloc[:, 3:5].values
30import scipy.cluster.hierarchy as shc
31
32plt.figure(figsize=(10, 7))
33plt.title("Customer Dendograms")
34dend = shc.dendrogram(shc.linkage(data, method='complete'))
35print('\npring dend \n')
36print(dend)
37
38plt.show()
39
40from sklearn.cluster import AgglomerativeClustering
41
42cluster = AgglomerativeClustering(n_clusters=5, affinity='manhattan', linkage='complete')
43print(cluster.fit_predict(data))
```

Usage

Here you can get help of any object by pressing **Ctrl+I** in front of it, either on the Editor or the Console.

Help can also be shown automatically.

Variable explorer File explorer Help

IPython console

Console 1/A

Python 3.6.5 [Anaconda, Inc.]
(default, Mar 29 2018, 13:32:41) [MSC v.1900 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 6.4.0 -- An enhanced Interactive Python.

In [1]:

Permissions: RW End-of-lines: CRLF Encoding: UTF-8 Line: 24 Column: 51 Memory: 80 %

<http://scikit-learn.org/stable/modules/clustering.html>

Customer Data

Anaconda Prompt - python agglomerativeclustering_plot.py

```
(base) C:\Users\cvuppapalapati>cd..
(base) C:\Users>cd..
(base) C:\>cd Han*
(base) C:\Hanumayamma>cd clu*
(base) C:\Hanumayamma\Clustering>python agglomerativeclustering_plot.py
```

customer data

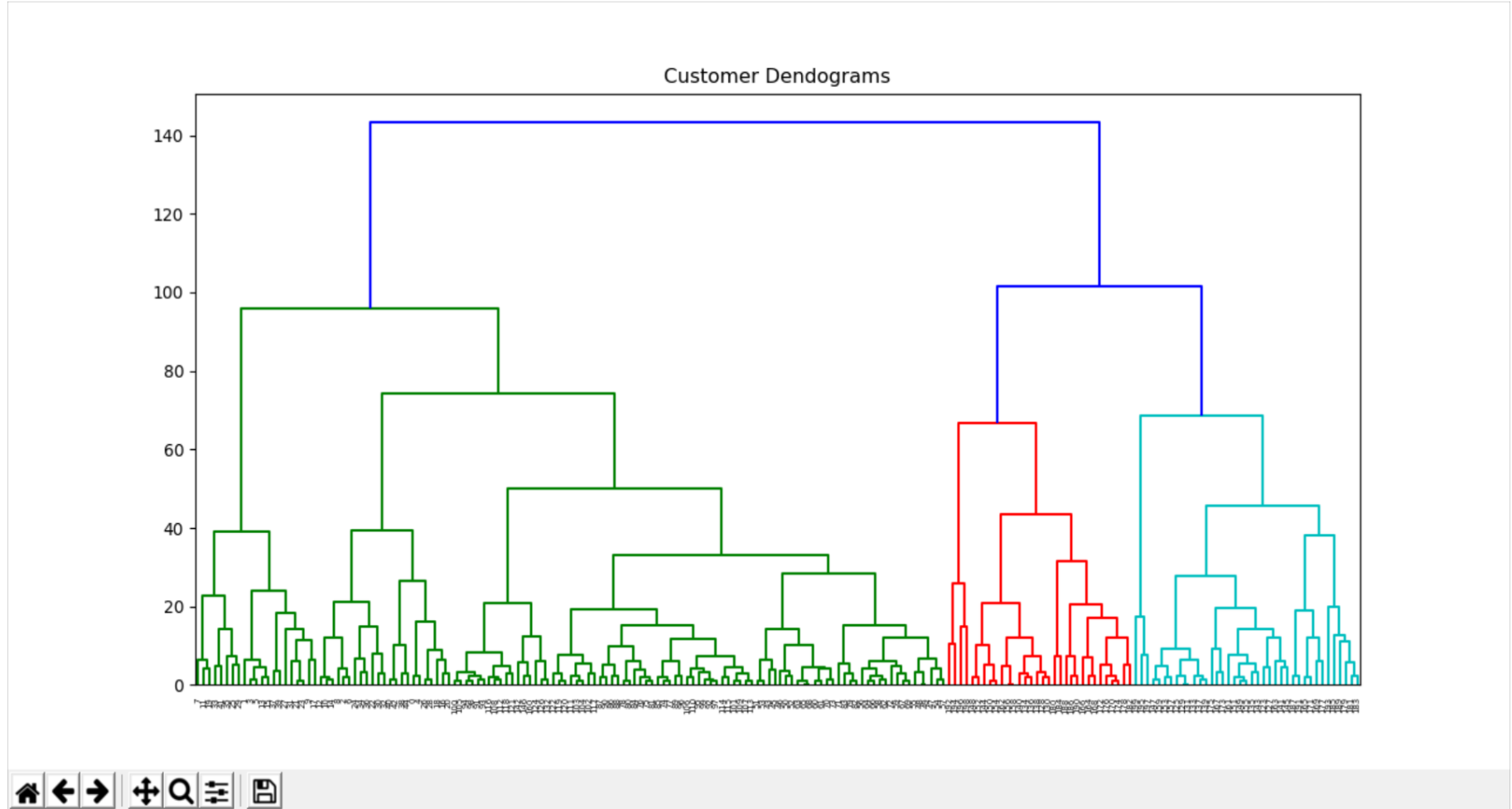
	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

<http://scikit-learn.org/stable/modules/clustering.html>



Customer Data

Figure 1



<http://scikit-learn.org/stable/modules/clustering.html>

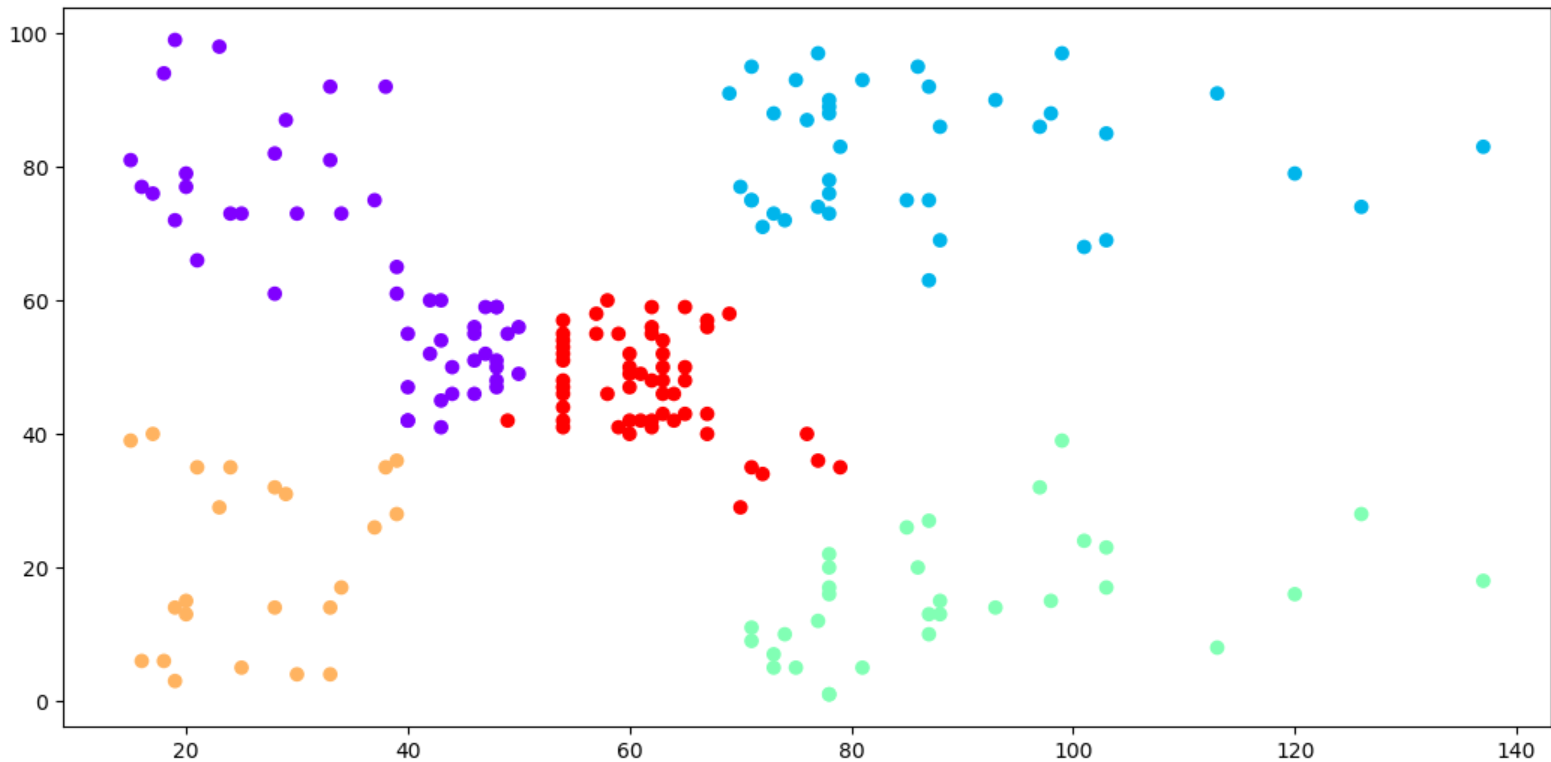
<http://scikit-learn.org/stable/modules/clustering.html>

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CMPE 256 – Largescale Analytics
10/01/2019



Customer Data

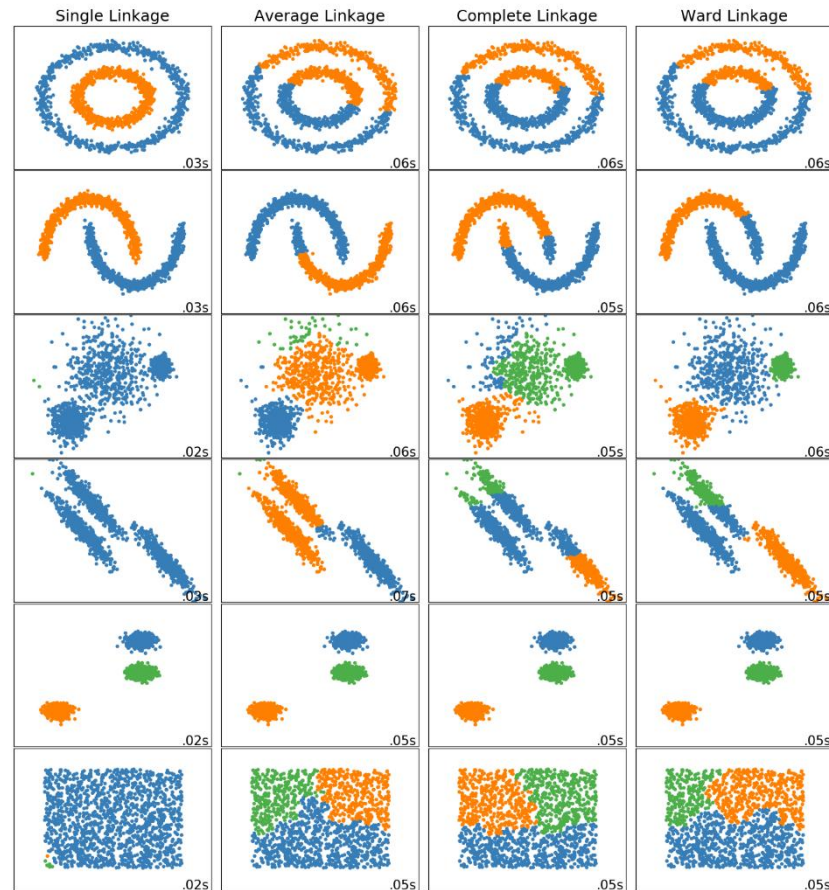
Figure 1



<http://scikit-learn.org/stable/modules/clustering.html>

SCIKIT Reference – Hierarchical Clustering

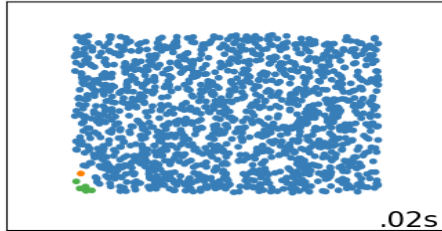
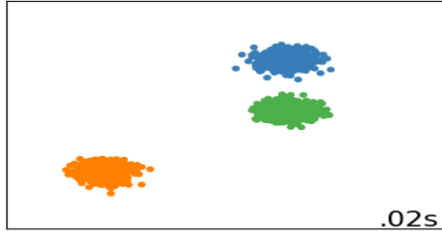
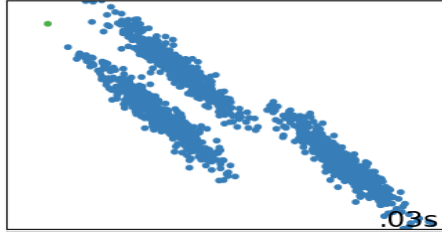
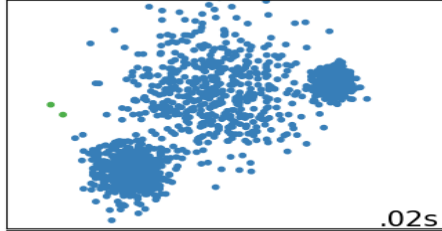
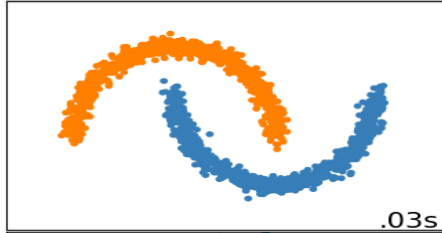
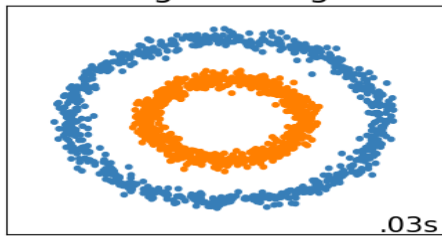
AgglomerativeClustering can also scale to large number of samples when it is used jointly with a connectivity matrix, but is computationally expensive when no connectivity constraints are added between samples: it considers at each step all the possible merges.



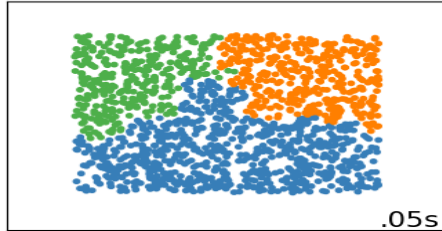
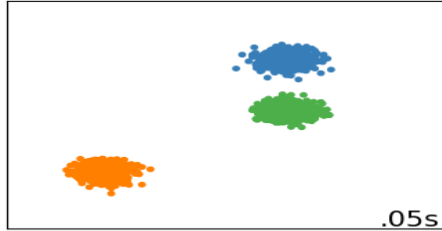
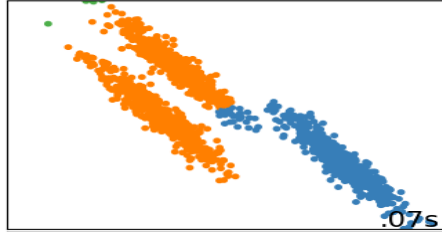
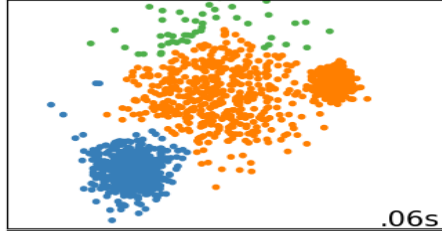
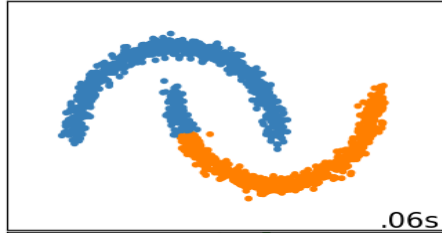
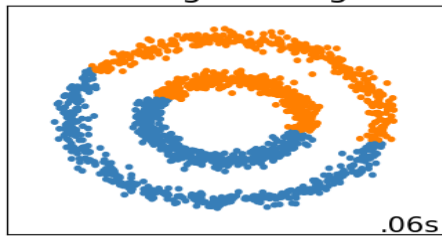
<http://scikit-learn.org/stable/modules/clustering.html>



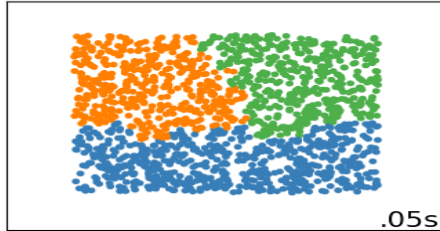
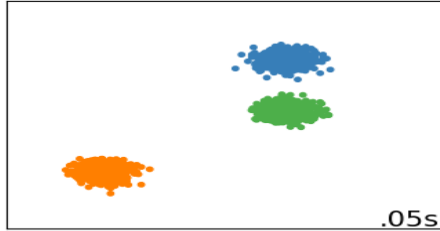
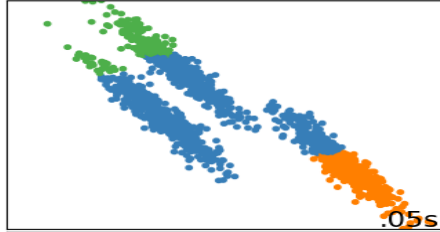
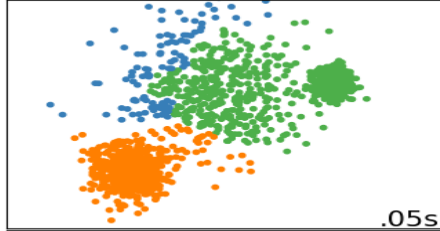
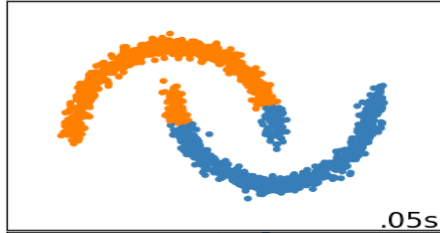
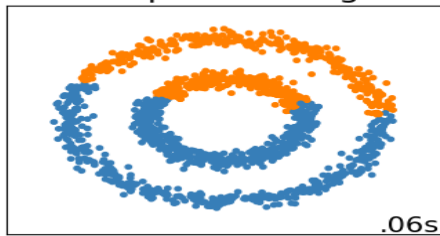
Single Linkage



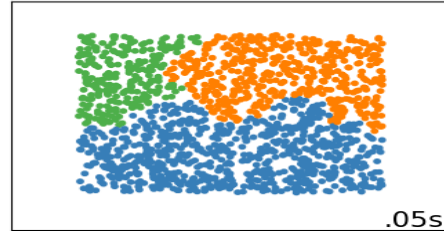
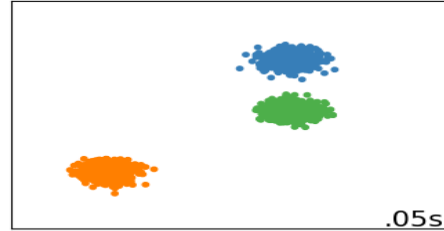
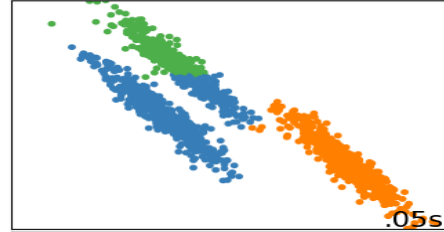
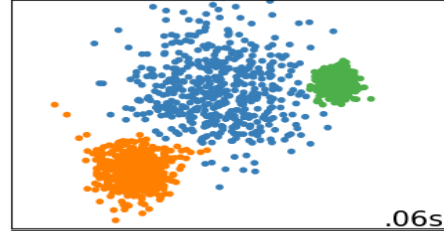
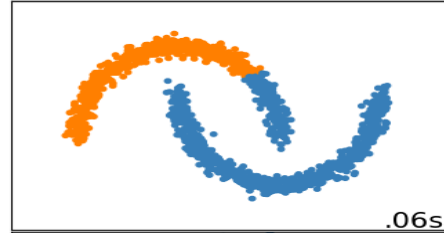
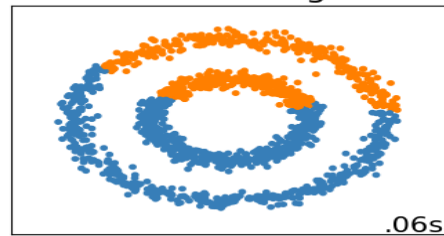
Average Linkage



Complete Linkage



Ward Linkage



SCIKIT Reference – Hierarchical Clustering

Hierarchical clustering is a general family of clustering algorithms that build nested clusters by merging or splitting them successively. This hierarchy of clusters is represented as a tree (or dendrogram). The root of the tree is the unique cluster that gathers all the samples, the leaves being the clusters with only one sample. See the Wikipedia page for more details.

The `AgglomerativeClustering` object performs a hierarchical clustering using a bottom up approach: each observation starts in its own cluster, and clusters are successively merged together. The linkage criteria determines the metric used for the merge strategy:

Ward minimizes the sum of squared differences within all clusters. It is a variance-minimizing approach and in this sense is similar to the k-means objective function but tackled with an agglomerative hierarchical approach.

Maximum or complete linkage minimizes the maximum distance between observations of pairs of clusters.

Average linkage minimizes the average of the distances between all observations of pairs of clusters.

Single linkage minimizes the distance between the closest observations of pairs of clusters.

<http://scikit-learn.org/stable/modules/clustering.html>



Single-Linkage Clustering: The Algorithm

1. Begin with the disjoint clustering having level $L(0) = 0$ and sequence number $m = 0$.
2. Find the least dissimilar pair of clusters in the current clustering, say pair $(r), (s)$, according to

$$d[(r),(s)] = \min d[(i),(j)]$$

where the minimum is over all pairs of clusters in the current clustering.

3. Increment the sequence number : $m = m + 1$. Merge clusters (r) and (s) into a single cluster to form the next clustering m . Set the level of this clustering to

$$L(m) = d[(r),(s)]$$

4. Update the proximity matrix, D , by deleting the rows and columns corresponding to clusters (r) and (s) and adding a row and column corresponding to the newly formed cluster. The proximity between the new cluster, denoted (r,s) and old cluster (k) is defined in this way:

$$d[(k), (r,s)] = \min d[(k),(r)], d[(k),(s)]$$

5. If all objects are in one cluster, stop. Else, go to step 2.



Single-Linkage Clustering: The Algorithm



Single-Linkage Clustering: The Algorithm

Let's now see a simple example: a hierarchical clustering of distances in kilometers between some Italian cities. The method used is single-linkage.

Input distance matrix ($L = 0$ for all the clusters):

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0



Single-Linkage Clustering: The Algorithm

Let's now see a simple example: a hierarchical clustering of distances in kilometers between some Italian cities. The method used is single-linkage.

Input distance matrix ($L = 0$ for all the clusters):

Bari
Florence
Milan
Naples
Rome
Turin

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0



Single-Linkage Clustering: The Algorithm

The nearest pair of cities is MI and TO, at distance 138. These are merged into a single cluster called "MI/TO". The level of the new cluster is $L(\text{MI/TO}) = 138$ and the new sequence number is $m = 1$. Then we compute the distance from this new compound object to all other objects. In single link clustering the rule is that the distance from the compound object to another object is equal to the shortest distance from any member of the cluster to the outside object. So the distance from "MI/TO" to RM is chosen to be 564, which is the distance from MI to RM, and so on. After merging MI with TO we obtain the following matrix:

	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0



Single-Linkage Clustering: The Algorithm

$\min d(i,j) = d(\text{NA}, \text{RM}) = 219 \Rightarrow$ merge NA and RM into a new cluster called NA/RM

$L(\text{NA}/\text{RM}) = 219$

$m = 2$

	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	268	564	0



Single-Linkage Clustering: The Algorithm

$\min d(i,j) = d(\text{BA}, \text{NA/RM}) = 255 \Rightarrow$ merge BA and NA/RM into a new cluster called BA/NA/RM

$L(\text{BA/NA/RM}) = 255$

$m = 3$

	BA/NA/RM	FI	MI/TO
BA/NA/RM	0	268	564
FI	268	0	295
MI/TO	564	295	0



Single-Linkage Clustering: The Algorithm

$\min d(i,j) = d(\text{BA/NA/RM}, \text{FI}) = 268 \Rightarrow$ merge BA/NA/RM and FI into a new cluster called BA/FI/NA/RM
 $L(\text{BA/FI/NA/RM}) = 268$
 $m = 4$

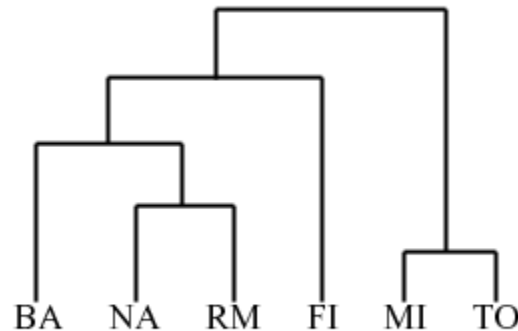
	BA/FI/NA/RM	MI/TO
BA/FI/NA/RM	0	295
MI/TO	295	0



Single-Linkage Clustering: The Algorithm

Finally, we merge the last two clusters at level 295.

The process is summarized by the following hierarchical tree:



Hierarchical Clustering Notes

Given a set of N items to be clustered, and an $N \times N$ distance (or similarity) matrix, the basic process of hierarchical clustering is this:

1. Start by assigning each item to its own cluster, so that if you have N items, you now have N clusters, each containing just one item. Let the distances (similarities) between the clusters equal the distances (similarities) between the items they contain.
2. Find the closest (most similar) pair of clusters and merge them into a single cluster, so that now you have one less cluster.
3. Compute distances (similarities) between the new cluster and each of the old clusters.
4. Repeat steps 2 and 3 until all items are clustered into a single cluster of size N .

Step 3 can be done in different ways, which is what distinguishes single-link from complete-link and average-link clustering. In single-link clustering (also called the connectedness or minimum method), we consider the distance between one cluster and another cluster to be equal to the shortest distance from any member of one cluster to any member of the other cluster. If the data consist of similarities, we consider the similarity between one cluster and another cluster to be equal to the greatest similarity from any member of one cluster to any member of the other cluster.

In complete-link clustering (also called the diameter or maximum method), we consider the distance between one cluster and another cluster to be equal to the longest distance from any member of one cluster to any member of the other cluster.



Hierarchical Clustering Notes

In average-link clustering, we consider the distance between one cluster and another cluster to be equal to the average distance from any member of one cluster to any member of the other cluster. A variation on average-link clustering is the UCLUS method of D'Andrade (1978) which uses the median distance.



US - Cities Single-Linkage Clustering



US - Cities Single-Linkage Clustering

The following pages trace a hierarchical clustering of distances in miles between U.S. cities. The method of clustering is single-link.

	BOS	NY	DC	MIA	CHI	SEA	SF	LA	DEN
BOS	0	206	429	1504	963	2976	3095	2979	1949
NY	206	0	233	1308	802	2815	2934	2786	1771
DC	429	233	0	1075	671	2684	2799	2631	1616
MIA	1504	1308	1075	0	1329	3273	3053	2687	2037
CHI	963	802	671	1329	0	2013	2142	2054	996
SEA	2976	2815	2684	3273	2013	0	808	1131	1307
SF	3095	2934	2799	3053	2142	808	0	379	1235
LA	2979	2786	2631	2687	2054	1131	379	0	1059
DEN	1949	1771	1616	2037	996	1307	1235	1059	0



US - Cities Single-Linkage Clustering

The nearest pair of cities is BOS and NY, at distance 206. These are merged into a single cluster called "BOS/NY".

Then we compute the distance from this new compound object to all other objects. In single link clustering the rule is that the distance from the compound object to another object is equal to the shortest distance from any member of the cluster to the outside object. So the distance from "BOS/NY" to DC is chosen to be 233, which is the distance from NY to DC. Similarly, the distance from "BOS/NY" to DEN is chosen to be 1771.

After merging BOS with NY:

	BOS/NY	DC	MIA	CHI	SEA	SF	LA	DEN
BOS/NY	0	223	1308	802	2815	2934	2786	1771
DC	223	0	1075	671	2684	2799	2631	1616
MIA	1308	1075	0	1329	3273	3053	2687	2037
CHI	802	671	1329	0	2013	2142	2054	996
SEA	2815	2684	3273	2013	0	808	1131	1307
SF	2934	2799	3053	2142	808	0	379	1235
LA	2786	2631	2687	2054	1131	379	0	1059
DEN	1771	1616	2037	996	1307	1235	1059	0



Single-Linkage Clustering: The Algorithm

The nearest pair of objects is BOS/NY and DC, at distance 223. These are merged into a single cluster called "BOS/NY/DC". Then we compute the distance from this new cluster to all other clusters, to get a new distance matrix:

After merging DC with BOS-NY:

	BOS/NY/DC	MIA	CHI	SEA	SF	LA	DEN
BOS/NY/DC	0	1075	671	2684	2799	2631	1616
MIA	1075	0	1329	3273	3053	2687	2037
CHI	671	1329	0	2013	2142	2054	996
SEA	2684	3273	2013	0	808	1131	1307
SF	2799	3053	2142	808	0	379	1235
LA	2631	2687	2054	1131	379	0	1059
DEN	1616	2037	996	1307	1235	1059	0



Single-Linkage Clustering: The Algorithm

Now, the nearest pair of objects is SF and LA, at distance 379. These are merged into a single cluster called "SF/LA". Then we compute the distance from this new cluster to all other objects, to get a new distance matrix:

After merging SF with LA:

	BOS/ NY/DC	MIA	CHI	SEA	SF/LA	DEN
BOS/NY/DC	0	1075	671	2684	2631	1616
MIA	1075	0	1329	3273	2687	2037
CHI	671	1329	0	2013	2054	996
SEA	2684	3273	2013	0	808	1307
SF/LA	2631	2687	2054	808	0	1059
DEN	1616	2037	996	1307	1059	0



Single-Linkage Clustering: The Algorithm

Now, the nearest pair of objects is CHI and BOS/NY/DC, at distance 671. These are merged into a single cluster called "BOS/NY/DC/CHI". Then we compute the distance from this new cluster to all other clusters, to get a new distance matrix:

After merging CHI with BOS/NY/DC:

	BOS/NY/DC/ CHI	MIA	SEA	SF/LA	DEN
BOS/NY/DC/CHI	0	1075	2013	2054	996
MIA	1075	0	3273	2687	2037
SEA	2013	3273	0	808	1307
SF/LA	2054	2687	808	0	1059
DEN	996	2037	1307	1059	0



Single-Linkage Clustering: The Algorithm

Now, the nearest pair of objects is SEA and SF/LA, at distance 808. These are merged into a single cluster called "SF/LA/SEA". Then we compute the distance from this new cluster to all other clusters, to get a new distance matrix:

After merging SEA with SF/LA:

	BOS/NY/DC/CHI	MIA	SF/LA/SEA	DEN
BOS/NY/DC/CHI	0	1075	2013	996
MIA	1075	0	2687	2037
SF/LA/SEA	2054	2687	0	1059
DEN	996	2037	1059	0



Single-Linkage Clustering: The Algorithm

Now, the nearest pair of objects is DEN and BOS/NY/DC/CHI, at distance 996. These are merged into a single cluster called "BOS/NY/DC/CHI/DEN". Then we compute the distance from this new cluster to all other clusters, to get a new distance matrix:

After merging DEN with BOS/NY/DC/CHI:

	BOS/NY/DC/CHI/DEN	MIA	SF/LA/SEA
BOS/NY/DC/CHI/DEN	0	1075	1059
MIA	1075	0	2687
SF/LA/SEA	1059	2687	0



Single-Linkage Clustering: The Algorithm

Now, the nearest pair of objects is BOS/NY/DC/CHI/DEN and SF/LA/SEA, at distance 1059. These are merged into a single cluster called "BOS/NY/DC/CHI/DEN/SF/LA/SEA". Then we compute the distance from this new compound object to all other objects, to get a new distance matrix:

After merging SF/LA/SEA with BOS/NY/DC/CHI/DEN:

	BOS/NY/DC/CHI/DEN/SF/LA/SEA	MIA
BOS/NY/DC/CHI/DEN/SF/LA/SEA	0	1075
MIA	1075	0



Single-Linkage Clustering: The Algorithm

Finally, we merge the last two clusters at level 1075. This process is summarized by the clustering diagram printed by many software packages:

	M	S			B			C	D
	I	E	S	L	O	N	D	H	E
	A	A	F	A	S	Y	C	I	N
Level	4	6	7	8	1	2	3	5	9
-----	-	-	-	-	-	-	-	-	-
206	.	.	.	.	XXX	.	.	.	.
233	.	.	.	.	XXXXX	.	.	.	.
379	.	.	XXX	XXXXX	XXXXX	.	.	.	.
671	.	.	XXX	XXXXXXXX	XXXXXXXX	.	.	.	.
808	.	XXXXX	XXXXX	XXXXXXXX	XXXXXXXX	.	.	.	.
996	.	XXXXX	XXXXX	XXXXXXXXXX	XXXXXXXXXX	.	.	.	.
1059	.	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	.	.	.	.
1075	XXXXXXXXXXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX	XXXXXXXXXX

In the diagram, the columns are associated with the items and the rows are associated with levels (stages) of clustering. An 'X' is placed between two columns in a given row if the corresponding items are merged at that stage in the clustering.



k-Means Clustering

The k-means algorithm searches for a pre-determined number of clusters within an unlabeled multidimensional dataset. It accomplishes this using a simple conception of what the optimal clustering looks like:

- The "cluster center" is the arithmetic mean of all the points belonging to the cluster.
- Each point is closer to its own cluster center than to other cluster centers.

Those two assumptions are the basis of the k-means model. We will soon dive into exactly how the algorithm reaches this solution, but for now let's take a look at a simple dataset and see the k-means result.

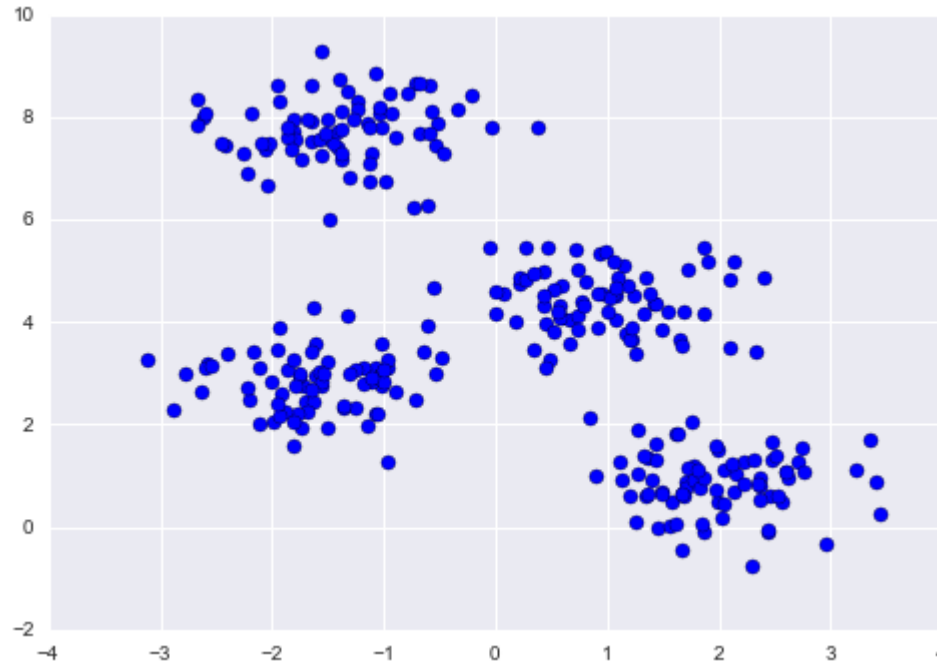


Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering

```
from sklearn.datasets.samples_generator import make_blobs
X, y_true = make_blobs(n_samples=300, centers=4,
                        cluster_std=0.60, random_state=0)
plt.scatter(X[:, 0], X[:, 1], s=50);
```



Source: <https://colab.research.google.com/github/jakevdp/PythonDataScienceHandbook/blob/master/notebooks/05.11-K-Means.ipynb#scrollTo=tui9iaoKX8qV>



k-Means Clustering

Expectation–maximization (E–M) is a powerful algorithm that comes up in a variety of contexts within data science. k-means is a particularly simple and easy-to-understand application of the algorithm, and we will walk through it briefly here. In short, the expectation–maximization approach here consists of the following procedure:

1. Guess some cluster centers
2. Repeat until converged
 1. *E-Step*: assign points to the nearest cluster center
 2. *M-Step*: set the cluster centers to the mean



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means: Step-By-Step Example

As a simple illustration of a k-means algorithm, consider the following data set consisting of the scores of two variables on each of seven individuals:

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

This data set is to be grouped into two clusters. As a first step in finding a sensible initial partition, let the A & B values of the two individuals furthest apart (using the Euclidean distance measure), define the initial cluster means, giving:



k-Means: Step-By-Step Example

This data set is to be grouped into two clusters. As a first step in finding a sensible initial partition, let the A & B values of the two individuals furthest apart (using the Euclidean distance measure), define the initial cluster means, giving:

	Individual	Mean Vector (centroid)
Group 1	1	(1.0, 1.0)
Group 2	4	(5.0, 7.0)

The remaining individuals are now examined in sequence and allocated to the cluster to which they are closest, in terms of Euclidean distance to the cluster mean. The mean vector is recalculated each time a new member is added. This leads to the following series of steps:



k-Means: Step-By-Step Example

The remaining individuals are now examined in sequence and allocated to the cluster to which they are closest, in terms of Euclidean distance to the cluster mean. The mean vector is recalculated each time a new member is added. This leads to the following series of steps:

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Step	Cluster 1		Cluster 2	
	Individual	Mean Vector (centroid)	Individual	Mean Vector (centroid)
1	1	(1.0, 1.0)	4	(5.0, 7.0)
2	1, 2	(1.2, 1.5)	4	(5.0, 7.0)
3	1, 2, 3	(1.8, 2.3)	4	(5.0, 7.0)
4	1, 2, 3	(1.8, 2.3)	4, 5	(4.2, 6.0)
5	1, 2, 3	(1.8, 2.3)	4, 5, 6	(4.3, 5.7)
6	1, 2, 3	(1.8, 2.3)	4, 5, 6, 7	(4.1, 5.4)



k-Means: Step-By-Step Example

Now the initial partition has changed, and the two clusters at this stage having the following characteristics:

	Individual	Mean Vector (centroid)
Cluster 1	1, 2, 3	(1.8, 2.3)
Cluster 2	4, 5, 6, 7	(4.1, 5.4)

But we cannot yet be sure that each individual has been assigned to the right cluster. So, we compare each individual's distance to its own cluster mean and to that of the opposite cluster. And we find:

Individual 1:

Distance between Subject 1 & Mean Vector 1

$$= \text{SQRT}((1.8 - 1.0)^2 + (2.3 - 1)^2)$$

Distance between Subject 1 & Mean Vector 2

$$= \text{SQRT}((4.1 - 1.0)^2 + (5.4 - 1.0)^2)$$

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Individual	Distance to mean (centroid) of Cluster 1	Distance to mean (centroid) of Cluster 2
1	1.5	5.4
2	0.4	4.3
3	2.1	1.8
4	5.7	1.8
5	3.2	0.7
6	3.8	0.6
7	2.8	1.1



Source: <http://mnemstudio.org/clustering-k-means-example-1.htm>

k-Means: Step-By-Step Example

Only individual 3 is nearer to the mean of the opposite cluster (Cluster 2) than its own (Cluster 1). In other words, each individual's distance to its own cluster mean should be smaller than the distance to the other cluster's mean (which is not the case with individual 3). Thus, individual 3 is relocated to Cluster 2 resulting in the new partition:

	Individual	Mean Vector (centroid)
Cluster 1	1, 2	(1.3, 1.5)
Cluster 2	3, 4, 5, 6, 7	(3.9, 5.1)

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5



k-Means: Step-By-Step Example

```
Anaconda Prompt

(base) C:\Hanumayamma\Clustering>python cmpe256_k-means.py

(base) C:\Hanumayamma\Clustering>python cmpe256_k-means.py

(base) C:\Hanumayamma\Clustering>_
```



k-Means: Step-By-Step Example

Spyder (Python 3.6)

File Edit Search Source Run Debug Consoles Projects Tools View Help

Editor - C:\Hanumayamma\Clustering\cmpe256_k-means.py

```
1 """
2 Created on Wed Oct 10 17:33:07 2018
3
4
5 @author: cvuppapalati
6 """
7
8 #- coding: utf-8 -*-
9 """
10 Created on Wed Oct 10 17:31:34 2018
11
12 @author: cvuppapalati
13 """
14
15 # clustering dataset
16 from sklearn.cluster import KMeans
17 from sklearn import metrics
18 import numpy as np
19 import matplotlib.pyplot as plt
20
21 x1 = np.array([1.0, 1.5, 3.0, 5.0, 3.5, 4.5, 3.5])
22 x2 = np.array([1.0, 2.0, 4.0, 7.0, 5.0, 5.0, 4.5])
23
24 plt.plot()
25 plt.xlim([0, 10])
26 plt.ylim([0, 10])
27 plt.title('Dataset')
28 plt.scatter(x1, x2)
29 plt.show()
30
31 # create new plot and data
32 plt.plot()
33 X = np.array(list(zip(x1, x2))).reshape(len(x1), 2)
34 colors = ['b', 'g', 'r']
35 markers = ['o', 'v', 's']
36
37 # KMeans algorithm
```

Usage

Here you can get help of any object by pressing **Ctrl+I** in front of it, either on the Editor or the Console.

Help can also be shown automatically.

Variable explorer File explorer Help

IPython console

Console 1/A

Python 3.6.5 |Anaconda, Inc.| (default, Mar 29 2018, 13:32:41) [MSC v.1900 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 6.4.0 -- An enhanced Interactive Python.

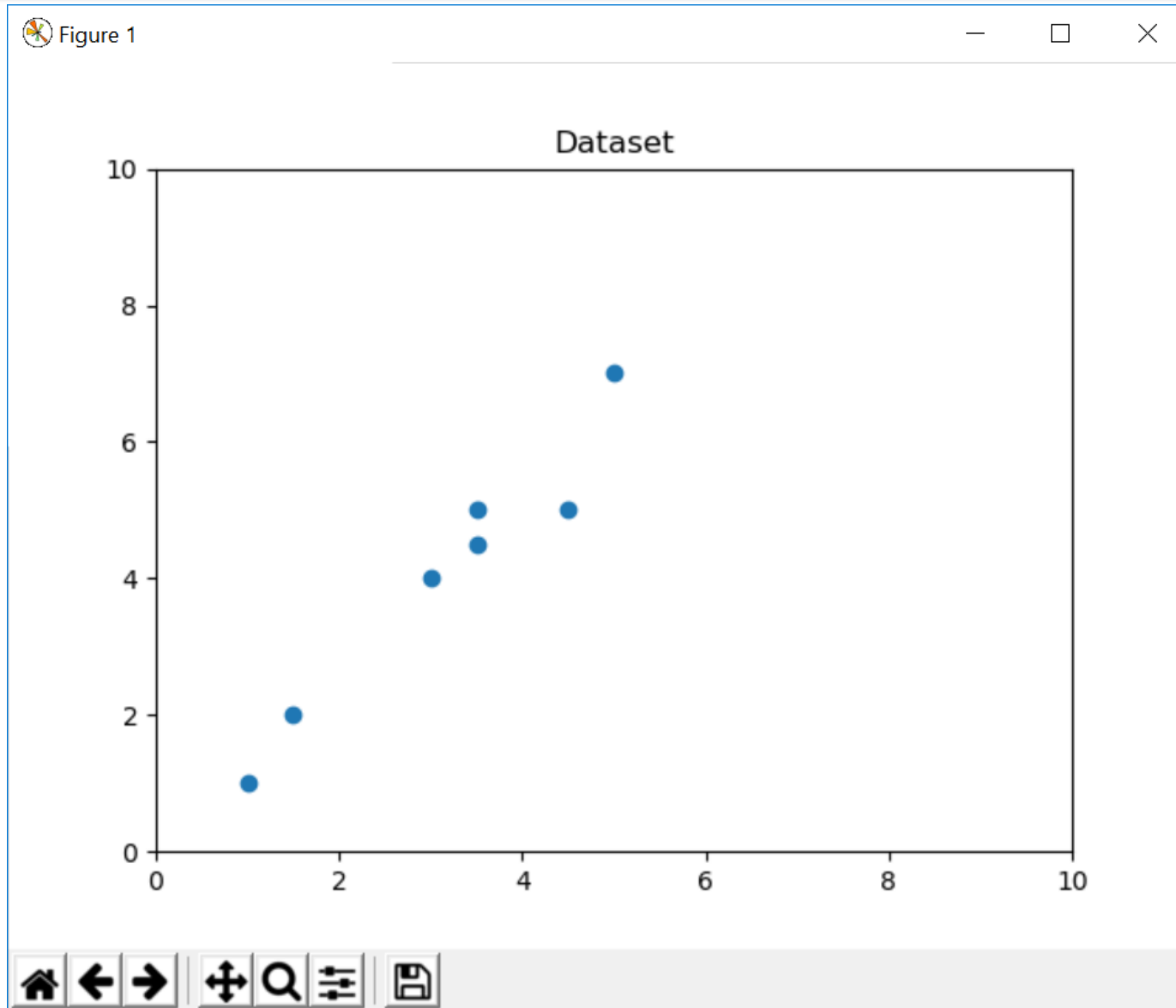
In [1]:

IPython console History log

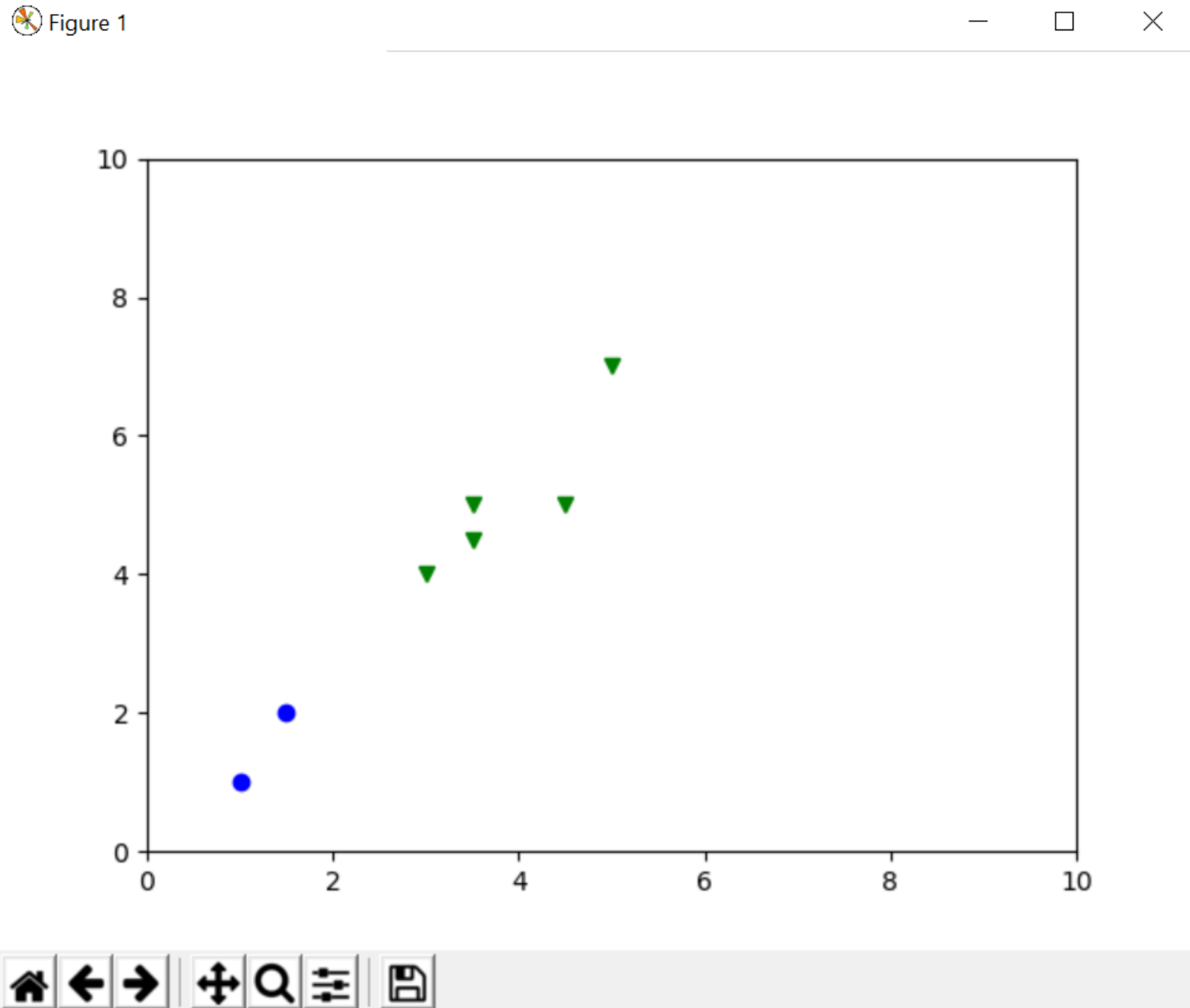
Permissions: RW End-of-lines: CRLF Encoding: UTF-8 Line: 35 Column: 26 Memory: 70 %



k-Means: Step-By-Step Example



k-Means: Step-By-Step Example



Source: <http://mnen>

k-Means: Step-By-Step Example

```
Anaconda Prompt - python k-means2.py

(base) C:\Hanumayamma\Clustering>python k-means2.py
```



k-Means: Step-By-Step Example

Spyder (Python 3.6)

File Edit Search Source Run Debug Consoles Projects Tools View Help

Editor - C:\Hanumayamma\Clustering\k-means2.py

cmpe256_inclass_agglomerative_plot_points.py agglomerativeclustering_plot.py cmpe256_inclass_plot_points.py k-meansclister.py k-means2.py

```
1 #-*- coding: utf-8 -*-
2 """
3 Created on Wed Oct 10 17:31:34 2018
4
5 @author: cvuppapalati
6 """
7
8 # clustering dataset
9 from sklearn.cluster import KMeans
10 from sklearn import metrics
11 import numpy as np
12 import matplotlib.pyplot as plt
13
14 x1 = np.array([3, 1, 1, 2, 1, 6, 6, 6, 5, 6, 7, 8, 9, 8, 9, 9, 9, 8])
15 x2 = np.array([5, 4, 6, 6, 5, 8, 6, 7, 6, 7, 1, 2, 1, 2, 3, 2, 3])
16
17 plt.plot()
18 plt.xlim([0, 10])
19 plt.ylim([0, 10])
20 plt.title('Dataset')
21 plt.scatter(x1, x2)
22 plt.show()
23
24 # create new plot and data
25 plt.plot()
26 X = np.array(list(zip(x1, x2))).reshape(len(x1), 2)
27 colors = ['b', 'g', 'r']
28 markers = ['o', 'v', 's']
29
30 # KMeans algorithm
31 K = 2
32 kmeans_model = KMeans(n_clusters=K).fit(X)
33
34 plt.plot()
35 for i, l in enumerate(kmeans_model.labels_):
36     plt.plot(x1[l], x2[l], color=colors[l], marker=markers[l], ls='None')
```

Usage

Here you can get help of any object by pressing **Ctrl+I** in front of it, either on the Editor or the Console.

Help can also be shown automatically.

Variable explorer File explorer Help

IPython console

Console 1/A

Python 3.6.5 |Anaconda, Inc.| (default, Mar 29 2018, 13:32:41) [MSC v.1900 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 6.4.0 -- An enhanced Interactive Python.

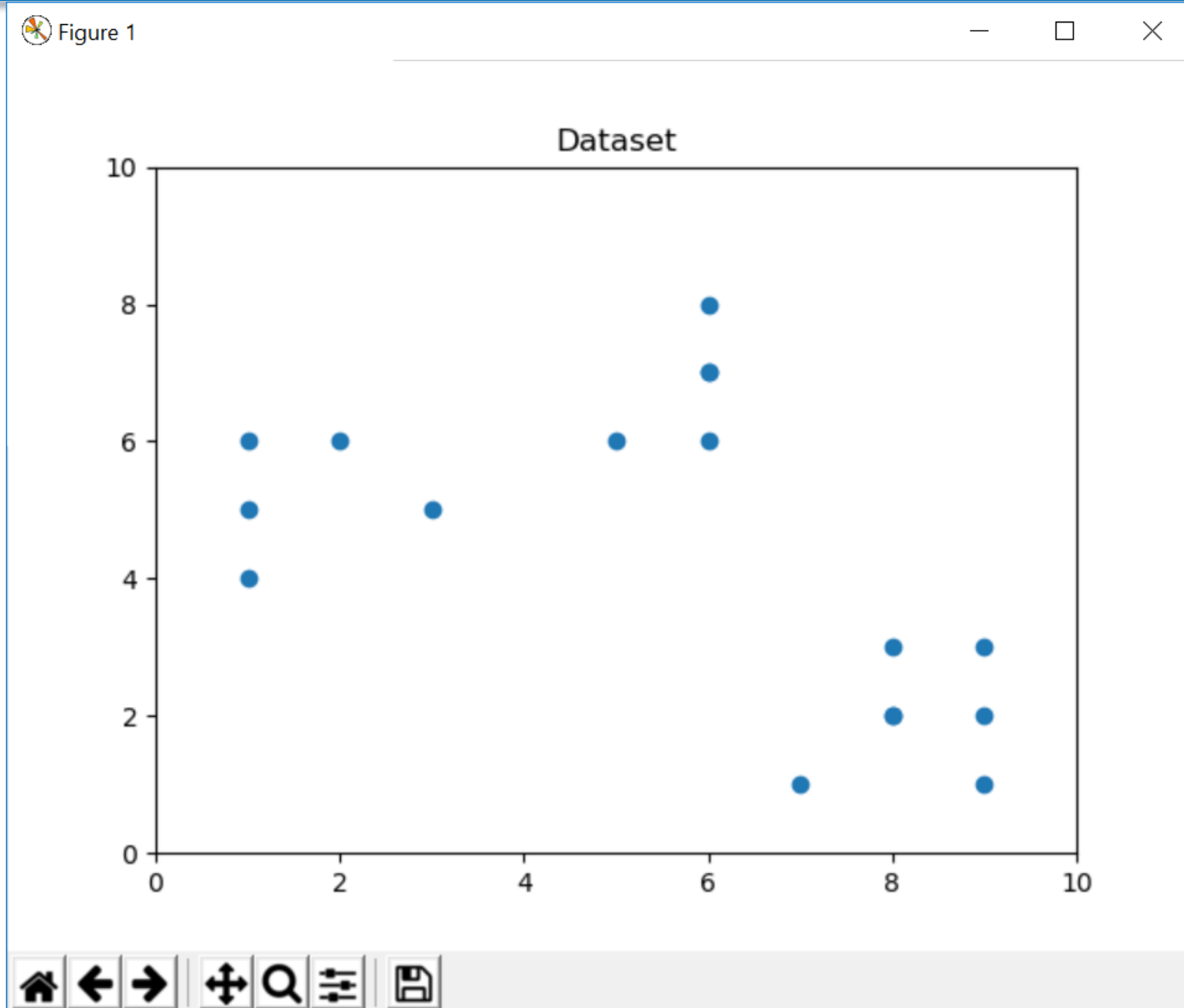
In [1]:

IPython console History log

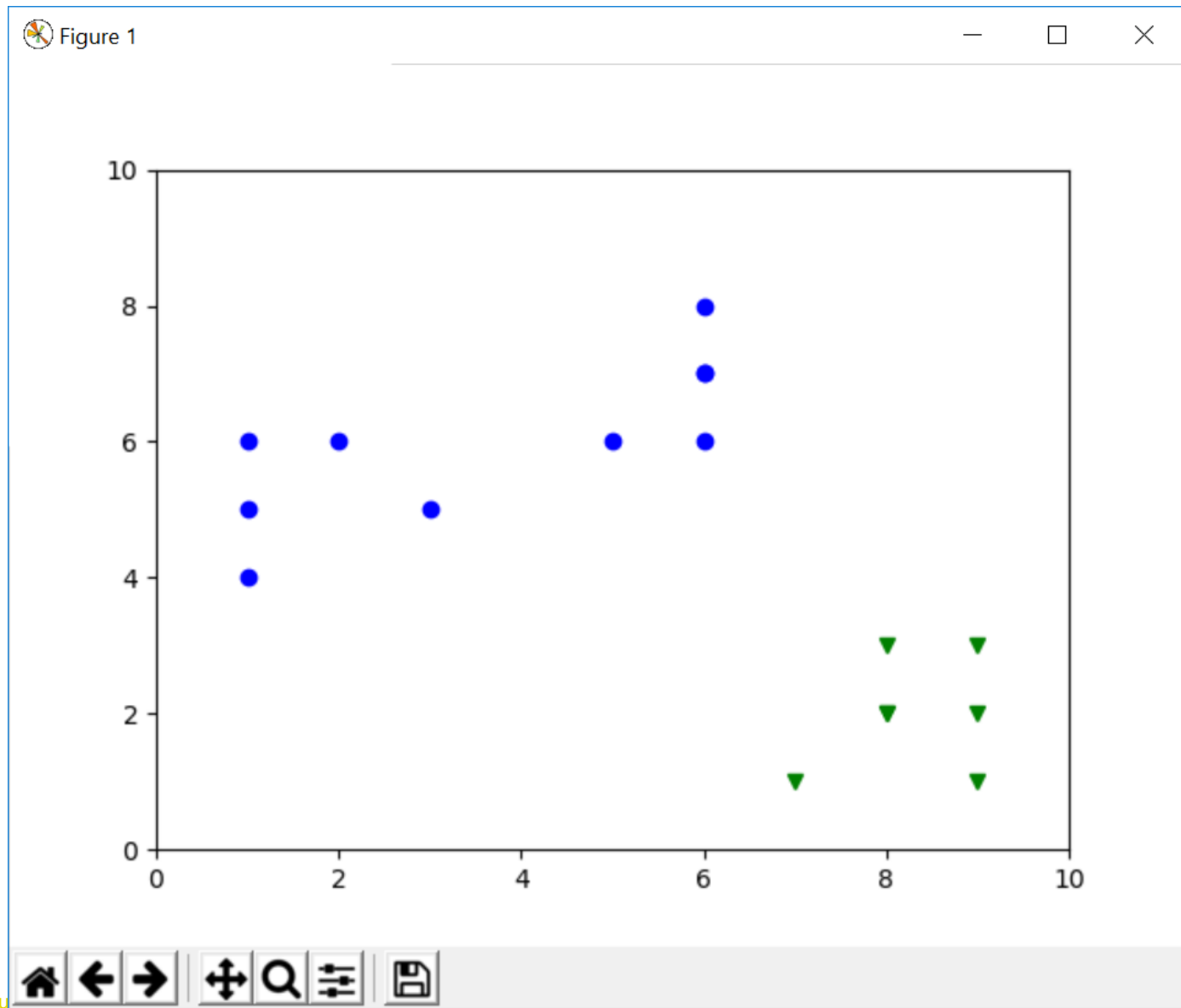
Permissions: RW End-of-lines: CRLF Encoding: UTF-8 Line: 18 Column: 18 Memory: 66 %



k-Means: Step-By-Step Example

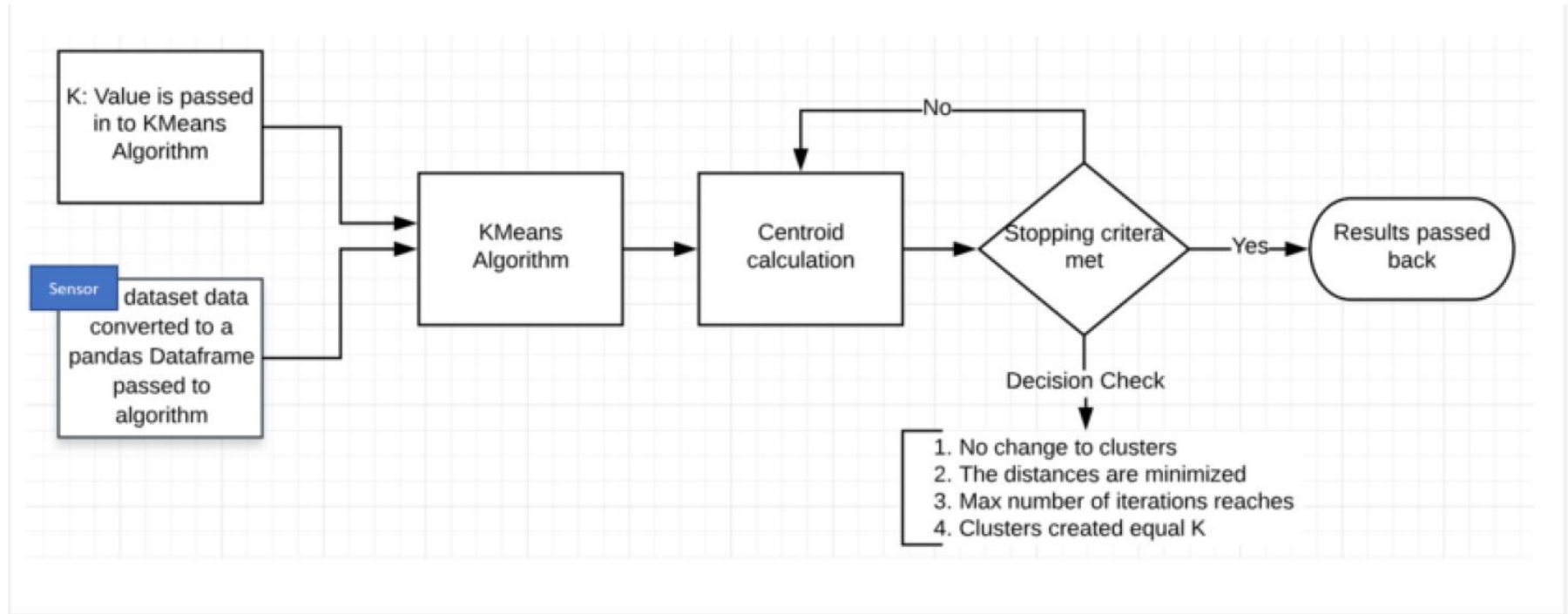


k-Means: Step-By-Step Example



Source: <http://mnemstu>

k-Means At Edge Level (IoT Devices)



Source: https://www.amazon.com/Building-Enterprise-Applications-Chandrasekar-Vuppapapati/dp/0367173859/ref=sr_1_1?keywords=Vuppapapati&qid=1570480052&sr=8-1

k-Means At Edge Level (IoT Devices)

As a simple illustration of a k-means algorithm, consider the following data set consisting of the signal strength (Received Signal Strength Indicator) and Distance from seven sensors:

Sensors	RSSI	Distance
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

The goal at the Edge level is to pick the sensor value that is more accurate and weed out any signal or sensory interferes.



Source: https://www.amazon.com/Building-Enterprise-Applications-Chandrasekar-Vuppapapati/dp/0367173859/ref=sr_1_1?keywords=Vuppapapati&qid=1570480052&sr=8-1



k-Means At Edge Level (IoT Devices)

As a simple illustration of a k-means algorithm, consider the following data set consisting of the signal strength (Received Signal Strength Indicator) and Distance from seven sensors:

Cluster 1 (Closer Sensors)			Cluster 2 (Further Sensors)		
Step	Sensors	Mean Vector (Centroid)	Sensors	Mean Vector (Centroid)	
1	1	(1.0, 1.0)	4	(5.0, 7.0)	
2	1,2	(1.2, 1.5)	4	(5.0, 7.0)	
3	1,2,3	(1.8,2.3)	4	(5.0, 7.0)	
4	1,2,3	(1.8,2.3)	4,5	(4.2,6.0)	
5	1,2,3	(1.8,2.3)	4,5,6	(4.3,5.7)	
6	1,2,3	(1.8,2.3)	4,5,6,7	(4.1,5.4)	



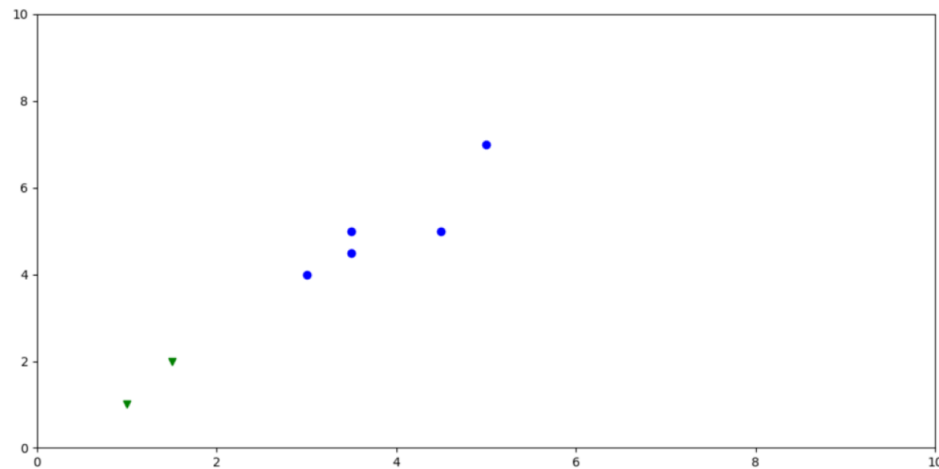
Source: https://www.amazon.com/Building-Enterprise-Applications-Chandrasekar-Vuppapapati/dp/0367173859/ref=sr_1_1?keywords=Vuppapapati&qid=1570480052&sr=8-1



k-Means At Edge Level (IoT Devices)

As a simple illustration of a k-means algorithm, consider the following data set consisting of the signal strength (Received Signal Strength Indicator) and Distance from seven sensors:

	Sensors	Mean Vector (Centroid)
Cluster 1	1,2,3	(1.8, 2.3)
Cluster 2	4,5,6,7	(4.1, 5.4)



Source: https://www.amazon.com/Building-Enterprise-Applications-Chandrasekar-Vuppapapati/dp/0367173859/ref=sr_1_1?keywords=Vuppapapati&qid=1570480052&sr=8-1



k-Means Clustering

Here the "E-step" or "Expectation step" is so-named because it involves updating our expectation of which cluster each **point belongs to**. The "M-step" or "Maximization step" is so-named because it involves **maximizing some fitness function that defines the location of the cluster centers—in this case, that maximization is accomplished by taking a simple mean of the data in each cluster.**

Limitations of E-M:

The globally optimal result may not be achieved

- First, although the E–M procedure is guaranteed to improve the result in each step, there is no assurance that it will lead to the global best solution.



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering – Limitation of E-M

The globally optimal result may not be achieved

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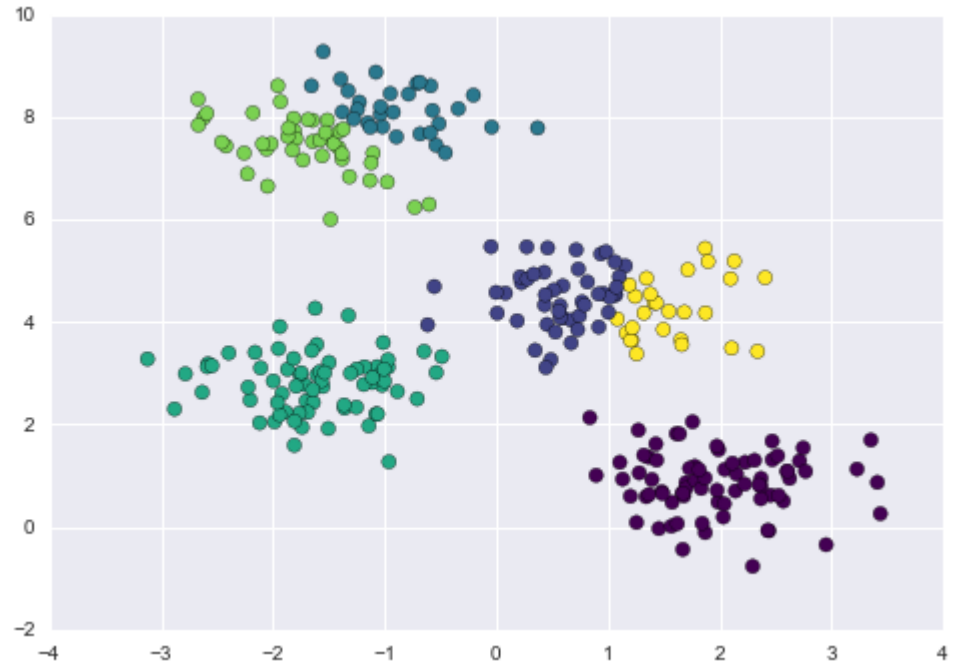
Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

k-Means Clustering – Limitation of E-M

The number of clusters must be selected beforehand

Another common challenge with k-means is that you must tell it how many clusters you expect: it cannot learn the number of clusters from the data. For example, if we ask the algorithm to identify six clusters, it will happily proceed and find the best six clusters.

```
labels = KMeans(6,  
random_state=0).fit_predict(X)  
plt.scatter(X[:, 0], X[:, 1], c=labels,  
s=50, cmap='viridis');
```



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering – Limitation of E-M

The number of clusters must be selected beforehand

Whether the result is meaningful is a question that is difficult to answer definitively; one approach that is rather intuitive, but that we won't discuss further here, is called **silhouette analysis**.

Alternatively, you might use a more complicated clustering algorithm which has a better quantitative measure of **the fitness per number of clusters** (e.g., Gaussian mixture models) or which can choose a suitable number of clusters (e.g., DBSCAN, mean-shift, or affinity propagation, all available in the sklearn.cluster submodule)



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering – Limitation of E-M

k-means is limited to linear cluster boundaries

The fundamental model assumptions of k-means (points will be closer to their own cluster center than to others) means that the algorithm will often be ineffective if **the clusters have complicated geometries.**

In particular, the boundaries between k-means clusters **will always be linear**, which means that it will fail for more complicated boundaries. Consider the following data, along with the cluster labels found by the typical k-means approach:



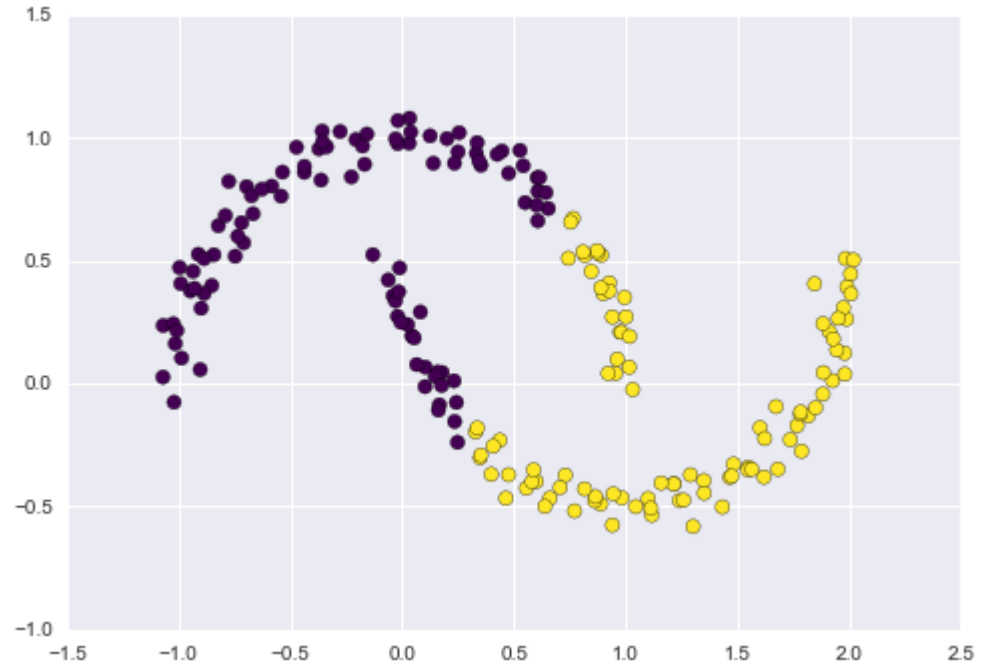
Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering – Limitation of E-M

```
from sklearn.datasets import  
make_moons  
X, y = make_moons(200, noise=.05,  
random_state=0)
```

```
labels = KMeans(2,  
random_state=0).fit_predict(X)  
plt.scatter(X[:, 0], X[:, 1], c=labels,  
s=50, cmap='viridis');
```



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



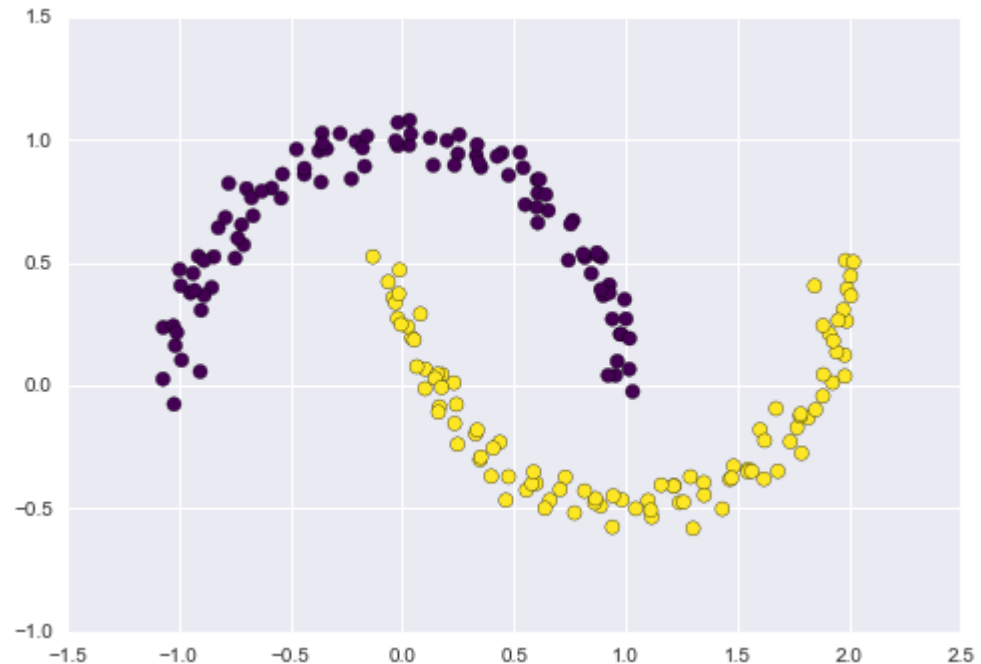
k-Means Clustering – Limitation of E-M

Important

k-means is limited to linear cluster boundaries

One version of this kernelized k-means is implemented in Scikit-Learn within **the SpectralClustering** estimator. *It uses the graph of nearest neighbors to compute a higher-dimensional representation of the data, and then assigns labels using a k-means algorithm:*

```
from sklearn.cluster import  
SpectralClustering  
model =  
SpectralClustering(n_clusters=2,  
affinity='nearest_neighbors',  
assign_labels='kmeans')  
labels = model.fit_predict(X)  
plt.scatter(X[:, 0], X[:, 1], c=labels,  
s=50, cmap='viridis');
```



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means Clustering – Limitation of E-M

k-means can be slow for large numbers of samples

Because each iteration of k-means must access every point in the dataset, the algorithm can be relatively slow as the number of samples grows. You might wonder if this requirement to **use all data at each iteration can be relaxed; for example, you might just use a subset of the data to update the cluster centers at each step. This is the idea behind batch-based k-means algorithms, one form of which is implemented in sklearn.cluster.MiniBatchKMeans**. The interface for this is the same as for standard KMeans; we will see an example of its use as we continue our discussion.



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

MiniBatch Cluster

One interesting application of clustering is in color compression within images. For example, imagine you have an image with millions of colors. In most images, a large number of the colors will be unused, and many of the pixels in the image will have similar or even identical colors.

For example, consider the image shown in the following figure, which is from the Scikit-Learn datasets module (for this to work, you'll have to have the pillow Python package installed).



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

```
# Note: this requires the ``pillow``  
package to be installed  
from sklearn.datasets import  
load_sample_image  
china =  
load_sample_image("china.jpg")  
ax = plt.axes(xticks=[], yticks=[])  
ax.imshow(china);
```



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

k-Means : Color Compression

Assignments: FA19: x | Files x | Annual Author and x | In Depth: k-Means x | 05.11-K-Means.ipynb x

colab.research.google.com/github/jakevdp/PythonDataScienceHandbook/blob/master/notebooks/05.11-K-Means.ipynb#sc...

05.11-K-Means.ipynb

File Edit View Insert Runtime Tools Help

+ Code + Text | Copy to Drive

RAM Disk Editing

For example, consider the image shown in the following figure, which is from the Scikit-Learn datasets module (for this to work, you'll have to have the pillow Python package installed).

```
# Note: this requires the `pillow` package to be installed
from sklearn.datasets import load_sample_image
china = load_sample_image('china.jpg')
ax = plt.axes(xticks=[], yticks=[])
ax.imshow(china);
```



<https://colab.research.google.com/github/jakevdp/PythonDataScienceHandbook/blob/master/notebooks/05.11-K-Means.ipynb#scrollTo=OaPe23QX8r>

MyCo_CompanyLo....pdf ^ | WS2.png ^ | WS1.png ^ | notes.js ^ | Show all x

Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

k-Means : Color Compression

Tensor or Size of Image:

The image itself is stored in a three-dimensional array of size (height, width, RGB), containing red/blue/green contributions as integers from 0 to 255:

`China.shape()`

`(427, 640, 3)`



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

k-Means : Color Compression

Tensor or Size of Image:

One way we can view this set of pixels is as a cloud of points in a three-dimensional color space. We will reshape the data to [n_samples x n_features], and rescale the colors so that they lie between 0 and 1:

```
data = china / 255.0 # use  
0...1 scale  
data = data.reshape(427 *  
640, 3)  
data.shape
```

(273280, 3)

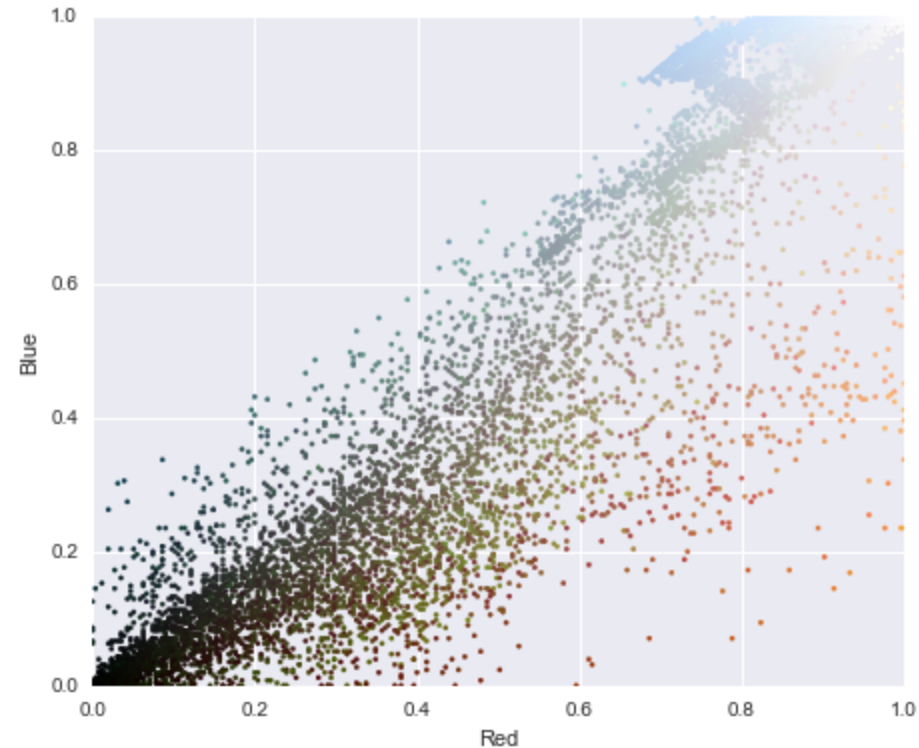
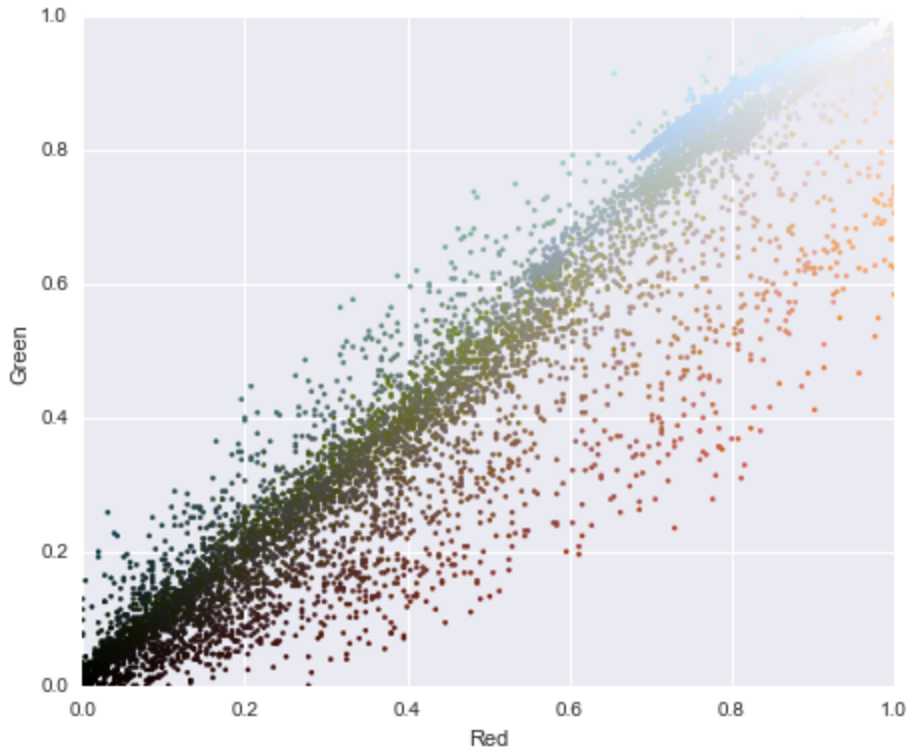


Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

Input color space: 16 million possible colors



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>

k-Means : Color Compression

Viewing of the Image:

We can visualize these pixels in this color space, using a subset of 10,000 pixels for efficiency::

```
def plot_pixels(data, title, colors=None, N=10000):
    if colors is None:
        colors = data

    # choose a random subset
    rng = np.random.RandomState(0)
    i = rng.permutation(data.shape[0])[:N]
    colors = colors[i]
    R, G, B = data[i].T

    fig, ax = plt.subplots(1, 2, figsize=(16, 6))
    ax[0].scatter(R, G, color=colors, marker='.')
    ax[0].set(xlabel='Red', ylabel='Green', xlim=(0, 1), ylim=(0, 1))

    ax[1].scatter(R, B, color=colors, marker='.')
    ax[1].set(xlabel='Red', ylabel='Blue', xlim=(0, 1), ylim=(0, 1))

    fig.suptitle(title, size=20);
```

plot_pixels(data,
title='Input color
space: 16 million
possible colors')



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

K Means in action:

Now let's reduce these 16 million colors to just 16 colors, using a k-means clustering across the pixel space. Because we are dealing with a very large dataset, we will use the mini batch k-means, which operates on subsets of the data to compute the result much more quickly than the standard k-means algorithm:

```
import warnings; warnings.simplefilter('ignore') # Fix NumPy issues.
```

```
from sklearn.cluster import MiniBatchKMeans
```

```
kmeans = MiniBatchKMeans(16)
```

```
kmeans.fit(data)
```

```
new_colors = kmeans.cluster_centers_[kmeans.predict(data)]
```

```
plot_pixels(data, colors=new_colors,  
            title="Reduced color space: 16 colors")
```

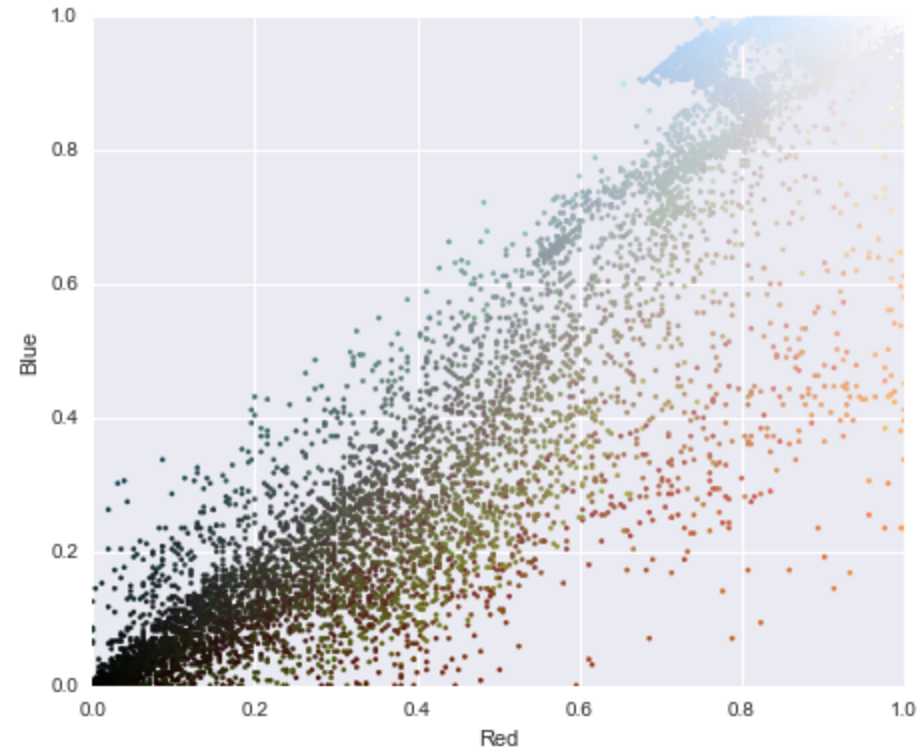
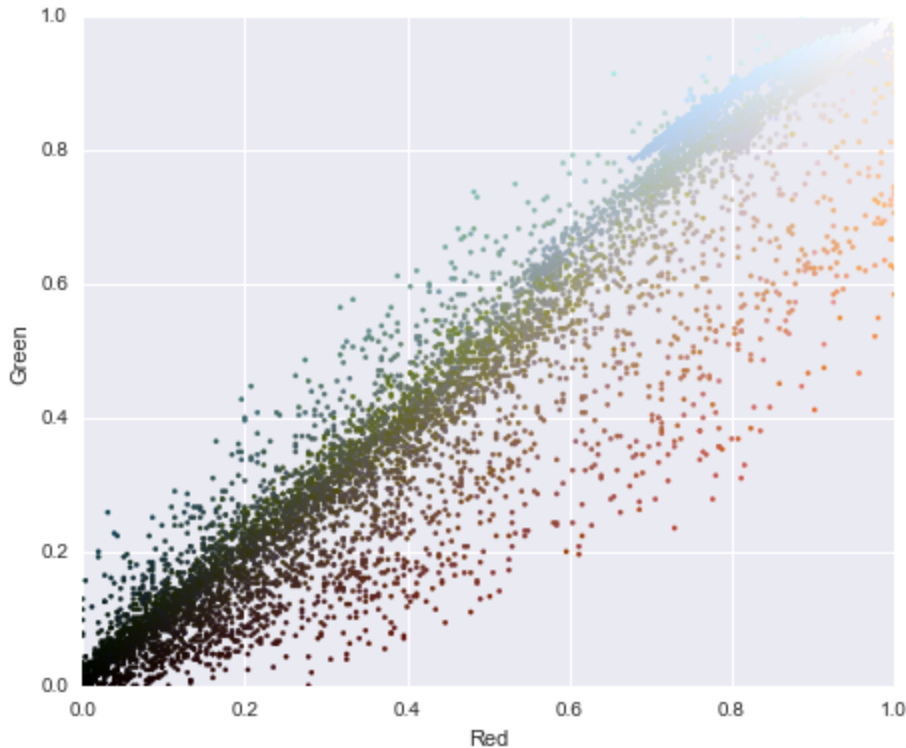


Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

Input color space: 16 million possible colors



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

```
china_recolored = new_colors.reshape(china.shape)

fig, ax = plt.subplots(1, 2, figsize=(16, 6),
                       subplot_kw=dict(xticks=[], yticks=[]))
fig.subplots_adjust(wspace=0.05)
ax[0].imshow(china)
ax[0].set_title('Original Image', size=16)
ax[1].imshow(china_recolored)
ax[1].set_title('16-color Image', size=16);
```



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



k-Means : Color Compression

The result is a re-coloring of the original pixels, where each pixel is assigned the color of its closest cluster center. Plotting these new colors in the image space rather than the pixel space shows us the effect of this:

Original Image



16-color Image



Source: <https://jakevdp.github.io/PythonDataScienceHandbook/05.11-k-means.html>



Texture of Image

The purpose of Local Binary Pattern (LBP) operator is to transform an image into an array or image of integer labels describing small scale appearance of image. In other words, LBP translates an image into histogram of integer labels and used to image texture analysis purposes.

The local binary pattern assumes that the image texture has two complimentary features: pattern and its strength[LBP]. The original version of LBP is based on 3x3 pixel block of an image.

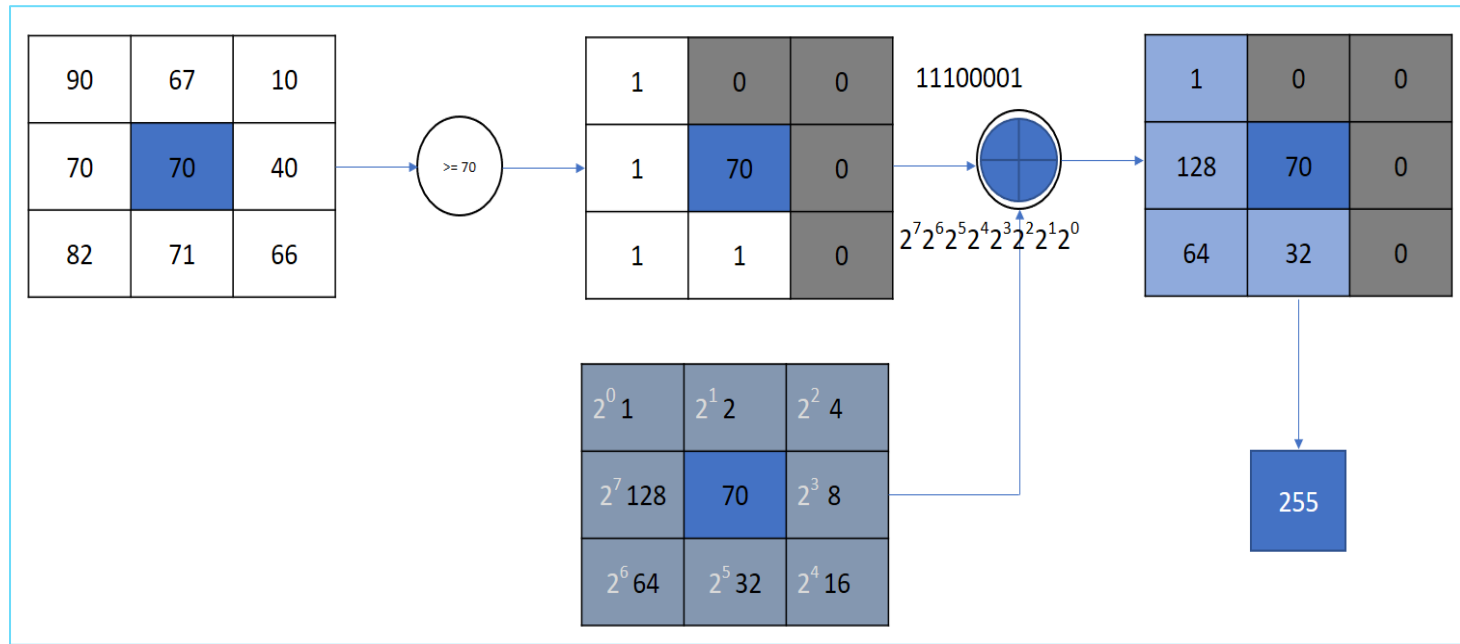
The LBP performs texture classification based on principle looking points surrounding a central point and tests whether surrounding points are greater than or less than the central point. In essence, performs binary transition.

In the below figure, the central pixel value [70] is compared with surrounding pixel values [90][67][10][40][66][71][82][70] and produced bit pattern [11100001] and translated into binary bit values [2726252423222120] and resulted into $[(1*27)+(1*26)+(1*25)+(0*24)+(0*23)+(0*22)+(0*21)+(1*20)]$ and thus producing value of 255.



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Texture of Image



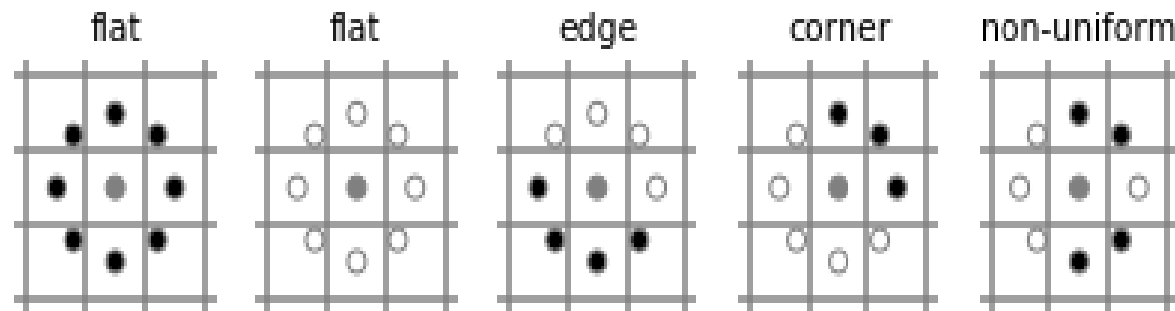
Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>



Texture of Image

As shown in below figure, based on color patterns of surrounding pixels with that of the central one, following types of LBP is possible [LBP Plot] [L3]:

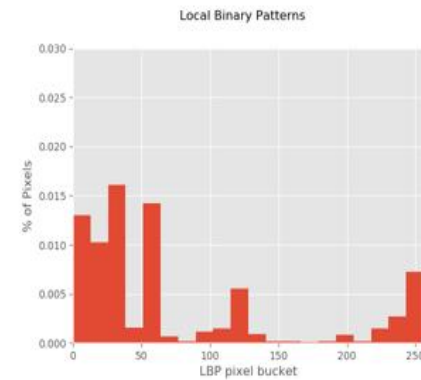
- When surrounding pixels are all white ([11111111]) or all black ([00000000]), then the image is flat or featureless. This is due to the values being produced either 255 or 0 respectively.
- Group of continuous black or white pixels are considered “uniform” patterns that can be interpreted as corners ([00001110]) or edges ([01110000]).
- If surrounding pixels switch back-and-forth between black and white pixels, the pattern is considered non-uniform.



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>



Texture of Image

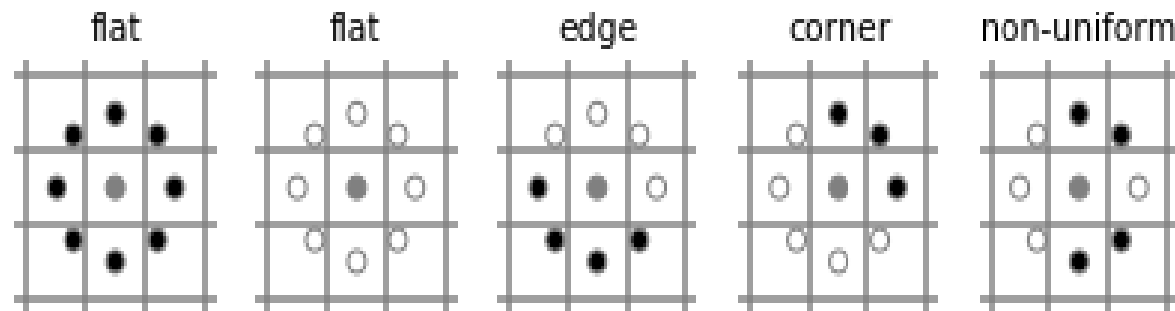


Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

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Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>



Texture of Image

```
Administrator: Anaconda Prompt - pip install --upgrade scikit-image

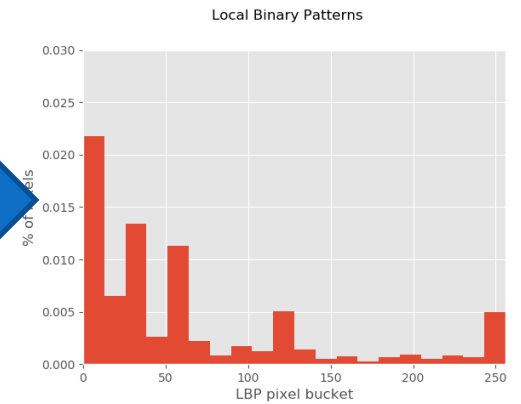
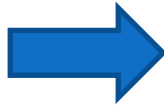
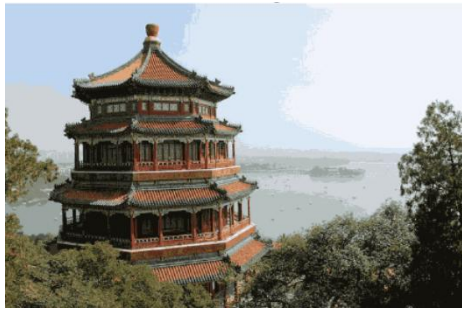
(base) C:\Hanumayamma\Clustering>pip install --upgrade scikit-image
Collecting scikit-image
  Downloading https://files.pythonhosted.org/packages/24/d5/a016b686ba4030c7e6039033bef899ff8a790819479016b2365eb929eacc/scikit_image-0.15.0-cp36-none-win_amd64.whl (25.6MB)
    | 25.6MB 467kB/s
Requirement already satisfied, skipping upgrade: PyWavelets>=0.4.0 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from scikit-image) (0.5.2)
Requirement already satisfied, skipping upgrade: pillow>=4.3.0 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from scikit-image) (5.1.0)
Requirement already satisfied, skipping upgrade: imageio>=2.0.1 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from scikit-image) (2.3.0)
Requirement already satisfied, skipping upgrade: networkx>=2.0 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from scikit-image) (2.1)
Requirement already satisfied, skipping upgrade: numpy>=1.9.1 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from PyWavelets>=0.4.0->scikit-image) (1.17.2)
Requirement already satisfied, skipping upgrade: decorator>=4.1.0 in c:\program files (x86)\microsoft visual studio\shared\anaconda3_64\lib\site-packages (from networkx>=2.0->scikit-image) (4.3.0)
Installing collected packages: scikit-image
  Found existing installation: scikit-image 0.13.1
  Uninstalling scikit-image-0.13.1:
    Successfully uninstalled scikit-image-0.13.1
```

<https://stackoverflow.com/questions/54241226/import-error-cannot-import-name-validate-lengths>



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Texture of Image



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Texture of Image

```
# plot a histogram of the LBP features and show it

# displaying default to make cool image
features = feature.local_binary_pattern(gray, 10, 5, method="default") # method="uniform")
# imShow cv2.imshow("LBP", features.astype("uint8"))
cv2.waitKey(10)

# Save figure of lbp_image
cv2.imwrite("lbp_lbpSource_16_color.png", features.astype("uint8"))

ax.hist(features.ravel(), normed=True, bins=20, range=(0, 256))
ax.set_xlim([0, 256])
ax.set_ylim([0, 0.030])
# save figure
fig.savefig('lbp_lbpSource_16_color_histogram.png') # save the figure to file
# imShow plt.show()

cv2.destroyAllWindows()
```



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Texture of Image

```
# for the lbp
from skimage import feature

import matplotlib.pyplot as plt
import numpy as np
import argparse
import imutils
import cv2

def frange(x, y, jump):
    while x < y:
        yield x
        x += jump

def resizeImage(image):
    (h, w) = image.shape[:2]

    width = 360 # This "width" is the width of the resize`ed image
    # calculate the ratio of the width and construct the
    # dimensions
    ratio = width / float(w)
    dim = (width, int(h * ratio))
    resized = cv2.resize(image, dim, interpolation=cv2.INTER_AREA)
    #resized = cv2.resize(image, dim, interpolation=cv2.INTER_CUBIC)
    return resized

# 1 load the image
```



Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Texture of Image

```
imagepath = r"lbpSource_16_color.png"
# , double it in size, and grab the cell size
image = cv2.imread(imagepath)
#image = imutils.resize(image, width=image.shape[1] * 2, inter=cv2.INTER_CUBIC)

# 2 resize the image
image = resizeImage(image)
(h, w) = image.shape[:2]
#cellSize = 16 * 2
cellSize = h/10

# 3 convert the image to grayscale and show it
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# imShow cv2.imshow("Image", gray)
cv2.waitKey(10)

# save the image
cv2.imwrite("gray_lbpSource_16_color.png", gray)

# Construct the figures
# construct the figure
plt.style.use("ggplot")
(fig, ax) = plt.subplots()
fig.suptitle("Local Binary Patterns")
plt.ylabel("% of Pixels")
plt.xlabel("LBP pixel bucket")
```



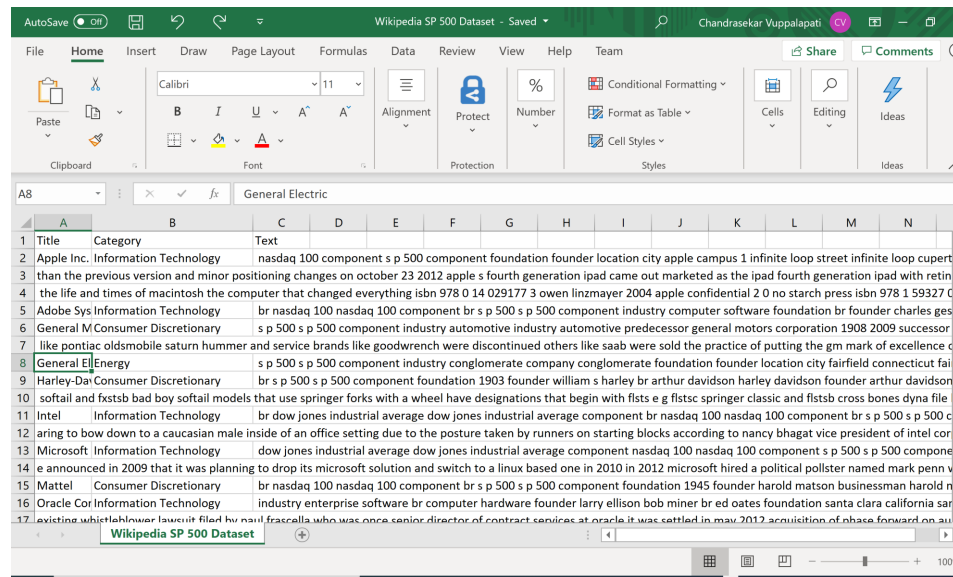
Source: <http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8622458&isnumber=8621858>

Clustering (K-Means) basic - Wikipedia

Clustering: Find similar companies

Largescale Analytics

This experiment demonstrates how to use the K-Means clustering algorithm to perform segmentation on companies from the Standard & Poor (S&P) 500 index, based on the text of Wikipedia articles about each company.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1		Title	Category	Text										
2		Apple Inc.	Information Technology	nasdaq 100 component s p 500 component foundation founder location city apple campus 1 infinite loop street infinite loop cupert										
3		than the previous version and minor positioning changes on october 23 2012 apple s fourth generation ipad came out marketed as the ipad fourth generation ipad with retin												
4		the life and times of macintosh the computer that changed everything isbn 978 0 14 029177 3 owen linzmayer 2004 apple confidential 2 0 no starch press isbn 978 1 59327 0												
5		Adobe Sys	Information Technology	br nasdaq 100 nasdaq 100 component br s p 500 s p 500 component industry computer software foundation br founder charles ges										
6		General M	Consumer Discretionary	s p 500 s p 500 component industry automotive predecessor general motors corporation 1908 2009 successor c										
7		like pontiac	oldsmobile saturn hummer and service brands like goodwrench were discontinued others like saab were sold the practice of putting the gm mark of excellence c											
8		General El	Energy	s p 500 s p 500 component industry conglomerate company conglomerate foundation founder location city fairfield connecticut fai										
9		Harley-Da	Consumer Discretionary	br s p 500 s p 500 component foundation 1903 founder william s harley br arthur davidson harley davidson founder arthur davidson										
10		softail and fxtsb	bad boy softail models that use springer forks with a wheel have designations that begin with flsts e g flstsc springer classic and flstsb cross bones dyna file											
11		Intel	Information Technology	br dow jones industrial average dow jones industrial average component br nasdaq 100 nasdaq 100 component br s p 500 s p 500 c										
12		aring to bow down to a caucasian male inside of an office setting due to the posture taken by runners on starting blocks according to nancy bhagat vice president of intel cor												
13		Microsoft	Information Technology	dow jones industrial average dow jones industrial average component nasdaq 100 nasdaq 100 component s p 500 s p 500 compone										
14		e announced in 2009 that it was planning to drop its microsoft solution and switch to a linux based one in 2010 in 2012 microsoft hired a political pollster named mark penn v												
15		Mattel	Consumer Discretionary	br nasdaq 100 nasdaq 100 component br s p 500 s p 500 component foundation 1945 founder harold matson businessman harold n										
16		Oracle Cor	Information Technology	industry enterprise software br computer hardware founder larry ellison bob miner br ed oates foundation santa clara california sar										
17		avicting whilelabourer lawsuit filed by nort fraccella who was once senior director of contract services at oracle.it was settled in may 2012 acquisition of nbase forward on au												

Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

Clustering (K-Means) basic - Wikipedia

AutoSave ☐ Off Wikipdia SP 500 Dataset - Saved Chandrasekar Vuppalapati CV

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Calibri 11 B I U A A Alignment Protect Number Conditional Formatting Format as Table Cell Styles

A8 General Electric

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
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17			existing whistleblower lawsuit filed by paul frascella who was once senior director of contract services at oracle it was settled in may 2012 acquisition of phase forward on au											

Wikipedia SP 500 Dataset

100%

Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

Chandrasekar Vuppalapati, Professor, MS Software Engineering,
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Clustering (K-Means) basic - Wikipedia

The screenshot displays the Microsoft Azure Machine Learning Studio interface. The top navigation bar includes tabs for 'Azure Diagnostic Logs supp', 'Clustering (K-Means) basic', and 'Experiments - Microsoft'. The browser address bar shows the URL: <https://studio.azureml.net/Home/ViewWorkspaceCached/8ab75f7fa3864b279ed7c5b263f4e69c#Workspaces>. The main workspace area is titled 'Clustering (K-Means) bas' and is in 'In draft' status. The draft was saved at 1:15:44 PM. The workflow consists of the following steps: 1. 'Wikipedia SP 500 Dataset' (input), 2. 'Feature Hashing', 3. 'Execute R Script', 4. 'Select Columns in Dataset', 5. 'K-Means Clustering', 6. 'Train Clustering Model', and 7. 'Edit Metadata'. The left sidebar contains a search bar and a list of experiment items: Saved Datasets, Data Format Conversions, Data Input and Output, Data Transformation, Feature Selection, Machine Learning, OpenCV Library Modules, Python Language Modules, R Language Modules, Statistical Functions, and Text Analytics. The right sidebar shows the 'Properties' tab for the 'Wikipedia SP 500 Dataset', with details: SUBMITTED BY: Microsoft C..., SIZE: 5.30 MB, FORMAT: GenericCSV, and CREATED ON: 4/8/2015 3:... A 'View dataset' link is also present. Below the properties is a 'Quick Help' section for the 'Wikipedia SP 500 Dataset'. The bottom toolbar includes icons for 'NEW', 'RUN HISTORY', 'SAVE', 'SAVE AS', 'DISCARD CHANGES', 'RUN', 'SET UP WEB SERVICE', and 'PUBLISH TO GALLERY'.

Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

Clustering (K-Means) basic - Wikipedia

Largescale Analytics

Clustering: Find similar companies

Data

The articles from **Wikipedia were pre-processed outside** Azure ML Studio to extract and partially clean text content related to each company. The processing included:

- Removing wiki formatting
- Removing non-alphanumeric characters
- Converting all text to lowercase
- Adding company categories, where known

For some companies, articles could not be found; therefore the number of records is less than 500.



Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

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Clustering (K-Means) basic - Wikipedia

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Model

First, the contents of each Wiki article were passed to the Feature Hashing module, which tokenizes the text string and then transforms the data into a series of numbers, based on the hash value of each token.

Even with this transformation, the dimensionality of the data is too high and sparse to be used by the K-Means clustering algorithm directly. Therefore, Principal Component Analysis (PCA) was applied using a custom R script in the Execute R Script module to reduce the dimensionality to 10 variables. You can review the result of PCA by double-clicking the right-hand output of the Execute R Script R module.



Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

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Clustering (K-Means) basic - Wikipedia

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Clustering: Find similar companies

Model

From trial and error, we learned that the first variable in the PCA transformed data had the highest variance and appears to have had a detrimental effect on clustering. Therefore, we removed it from the feature set using Project Columns. Once the data was prepared, we created K-Means Clustering module and trained models on the text data. Finally, we used Metadata Editor to change the cluster labels into categorical values.



Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

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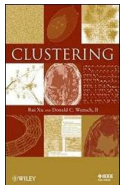
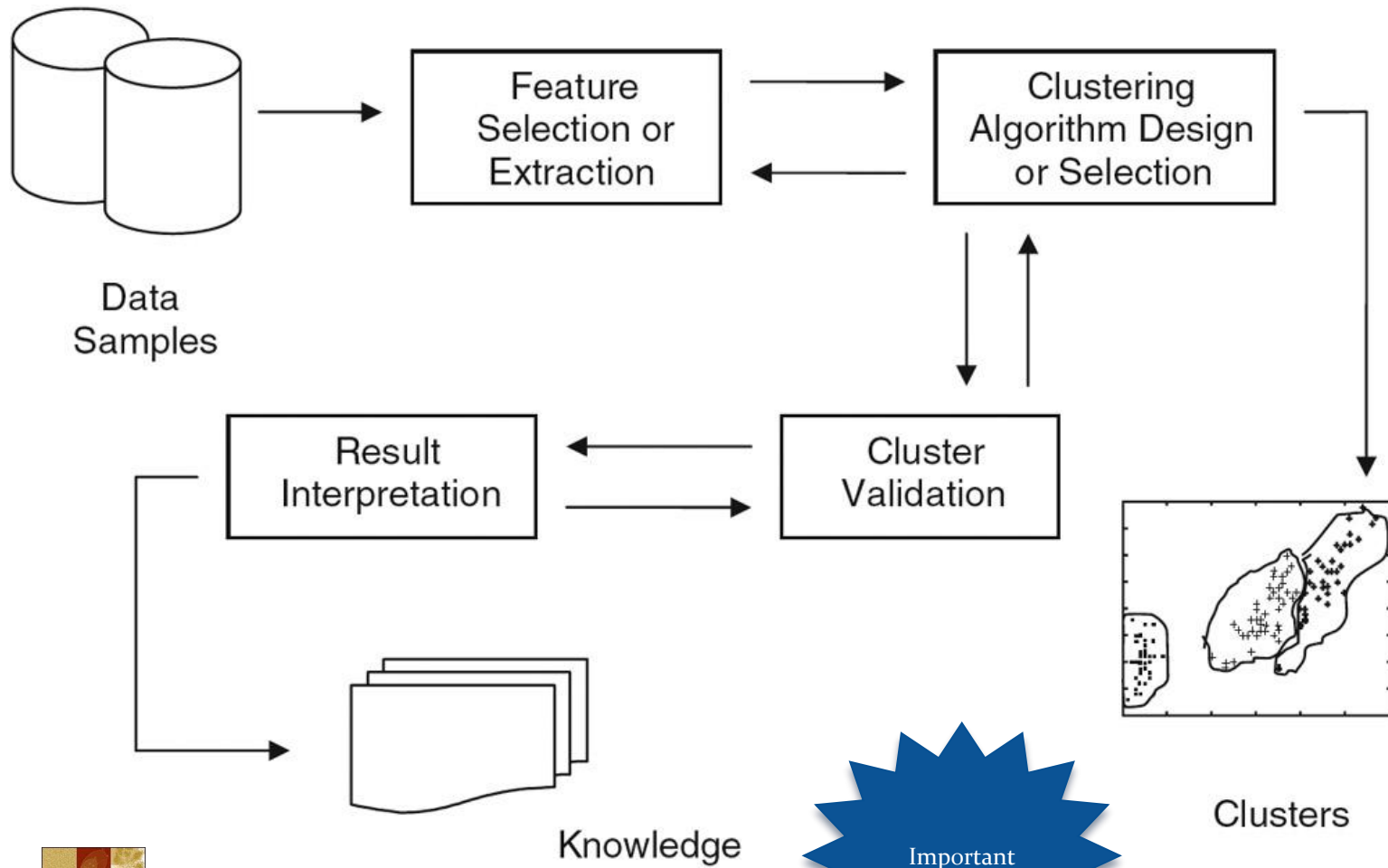
 Principal Component Analysis (PCA) was applied

Source: <https://gallery.azure.ai/Experiment/Clustering-K-Means-basic>

using a custom R script in the Execute R Script



Clustering procedure

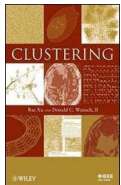


Clustering procedure

1. Feature selection or extraction:

Feature selection chooses distinguishing features from a set of candidates, while feature extraction utilizes some transformations to generate useful and novel features from the original ones. Clearly, **feature extraction is potentially capable of producing features that could be of better use in uncovering the data structure**. However, feature extraction may generate features that are not physically interpretable, while feature selection assures the retention of the original physical meaning of the selected features. In the literature, these two terms sometimes are used interchangeably without further identifying the difference.

Both feature selection and feature extraction are very important to the effectiveness of clustering applications. **Elegant selection or generation of salient features can greatly decrease the storage requirement and measurement cost**, simplify the subsequent design process, and facilitate the understanding of the data. Generally, ideal features should be of use in distinguishing patterns belonging to different clusters, immune to noise, and easy to obtain and interpret.

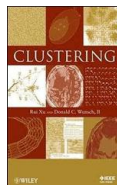


Clustering procedure

2. Cluster algorithm design or selection

This step usually consists of determining an appropriate proximity measure and constructing a criterion function. Intuitively, data objects are grouped into different clusters according to whether they resemble one another or not. Almost all clustering algorithms are explicitly or implicitly connected to some particular **definition of proximity measure**. Some algorithms even work directly on the proximity matrix. Once a proximity measure is determined, clustering could be construed as an optimization problem with a specific criterion function. Again, the obtained clusters are dependent on the selection of the criterion function. The subjectivity of cluster analysis is thus inescapable.

Clustering is ubiquitous, and a wealth of clustering algorithms has been developed to solve different problems from a wide variety of fields. However, there is no universal clustering algorithm to solve all problems. “It has been very difficult to develop a unified framework for reasoning about it (clustering) at a technical level, and profoundly diverse approaches to clustering” (Kleinberg, 2002). Therefore, it is important to carefully investigate the characteristics of a problem in order to select or design an appropriate clustering strategy. Clustering algorithms that are developed to solve a particular problem in a specialized field usually make assumptions in favor of the application of interest. For example, **the K-means algorithm is based on the Euclidean measure and hence tends to generate hyperspherical clusters**. However, if the real clusters are in other geometric forms, K-means may no longer be effective, and we need to resort to other schemes. Similar considerations must be kept in mind for mixture-model clustering, in which data are assumed to come from some specific models that are already known in advance.

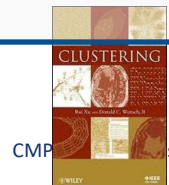


Clustering procedure

3. Cluster Validation

Given a data set, each clustering algorithm can always produce a partition whether or not there really exists a particular structure in the data. Moreover, different clustering approaches usually lead to different clusters of data, and even for the same algorithm, the selection of a parameter or the presentation order of input patterns may affect the final results. Therefore, effective evaluation standards and criteria are critically important to provide users with a degree of confidence for the clustering results. These assessments should be objective and have no preferences to any algorithm. Also, they should be able to provide meaningful insights in answering questions like how many clusters are hidden in the data, whether the clusters obtained are meaningful from a practical point of view or just artifacts of the algorithms, or why we choose one algorithm instead of another. Generally, there are three categories of

testing criteria: external indices, internal indices, and relative indices. They are defined on three types of clustering structures, known as partitional clustering, hierarchical clustering, and individual clusters. Tests for situations in which no clustering structure exists in the data are also considered but seldom used because users are usually confident of the presence of clusters in the data of interest. External indices are based on some pre-specified structure, which is the reflection of prior information on the data and is used as a standard to validate the clustering solutions. Internal tests are not dependent on external information (prior knowledge). Instead, they examine the clustering structure directly from the original data. Relative criteria emphasize the comparison of different clustering structures in order to provide a reference to decide which one may best reveal the characteristics of the objects.



Different Aspects of Cluster Validation

1. Determining the **clustering tendency** of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.
2. Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
3. Evaluating how well the results of a cluster analysis fit the data *without* reference to external information.
 - Use only the data
4. Comparing the results of two different sets of cluster analyses to determine which is better.
5. Determining the 'correct' number of clusters.

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.



Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
 - **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - Entropy
 - **Internal Index:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - Sum of Squared Error (SSE)
 - **Relative Index:** Used to compare two different clusterings or clusters.
 - Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as **criteria** instead of **indices**
 - However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.



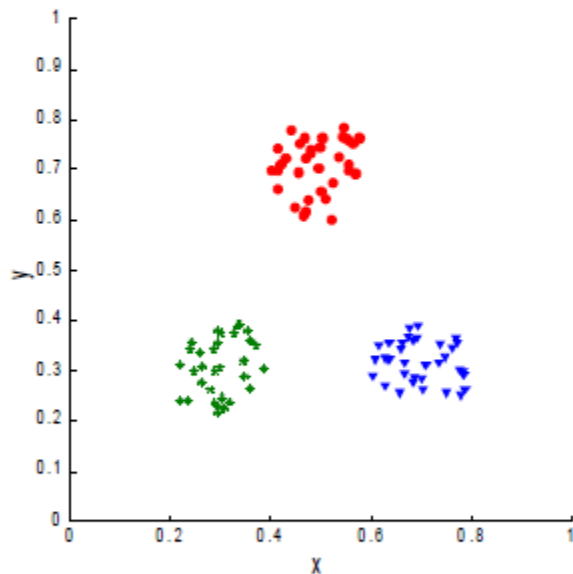
Measuring Cluster Validity Via Correlation

- Two matrices
 - Proximity Matrix
 - “Incidence” Matrix
 - One row and one column for each data point
 - An entry is 1 if the associated pair of points belong to the same cluster
 - An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
 - Since the matrices are symmetric, only the correlation between $n(n-1) / 2$ entries needs to be calculated.
- High correlation indicates that points that belong to the same cluster are close to each other.
- Not a good measure for some density or contiguity based clusters.

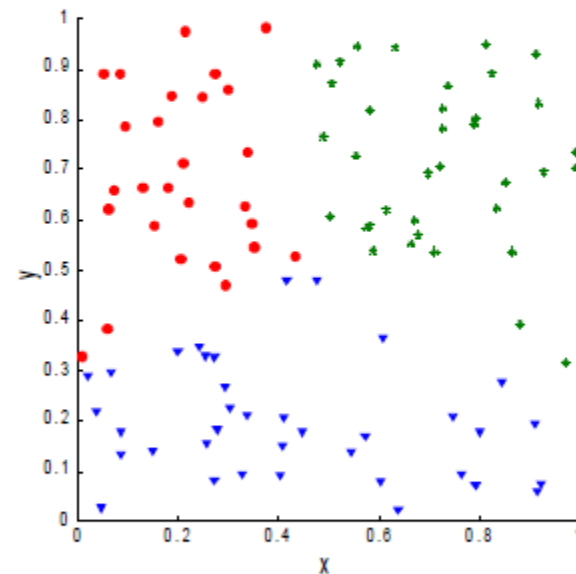


Measuring Cluster Validity Via Correlation

- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



Corr = -0.9235

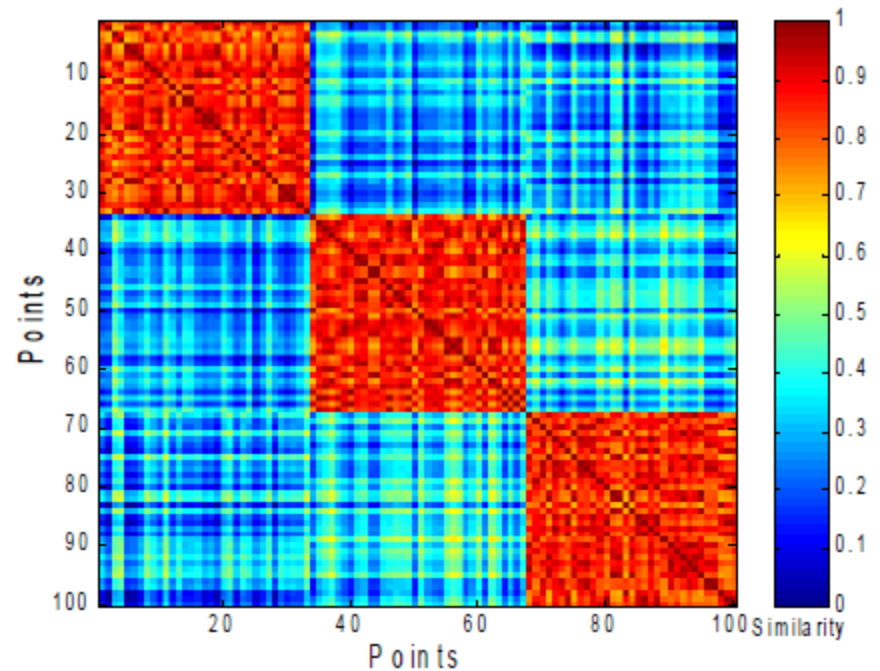
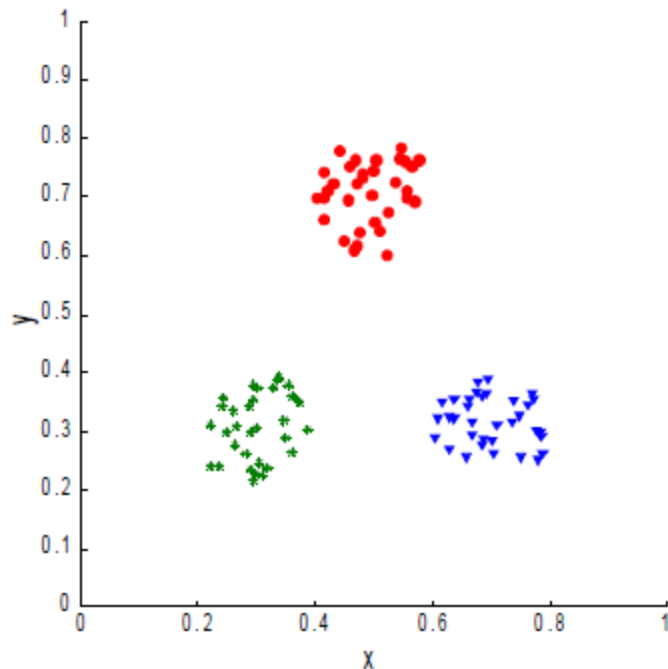


Corr = -0.5810



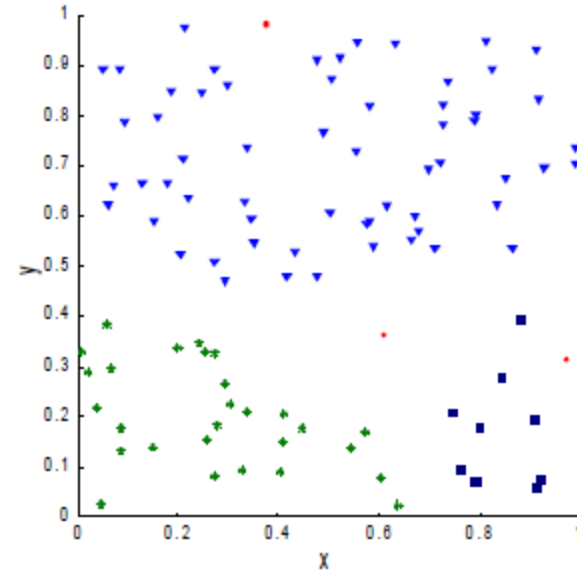
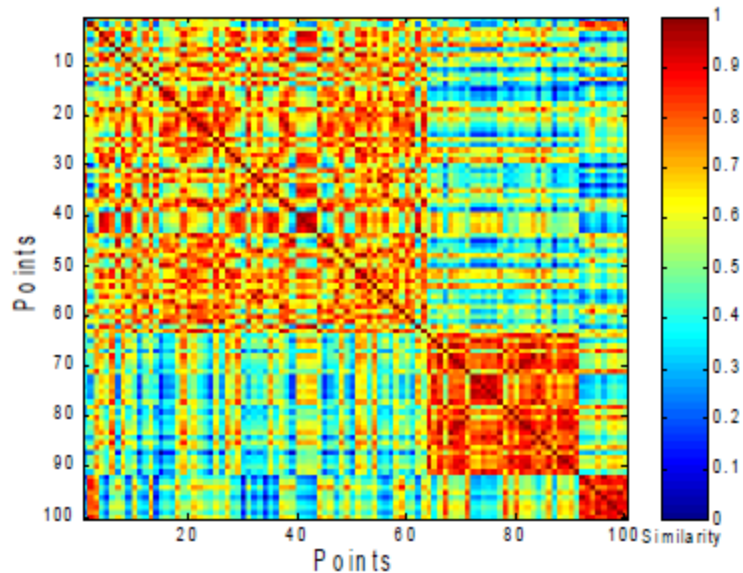
Using Similarity Matrix for Cluster Validation

- Order the similarity matrix with respect to cluster labels and inspect visually.



Using Similarity Matrix for Cluster Validation

- Clusters in random data are not so crisp

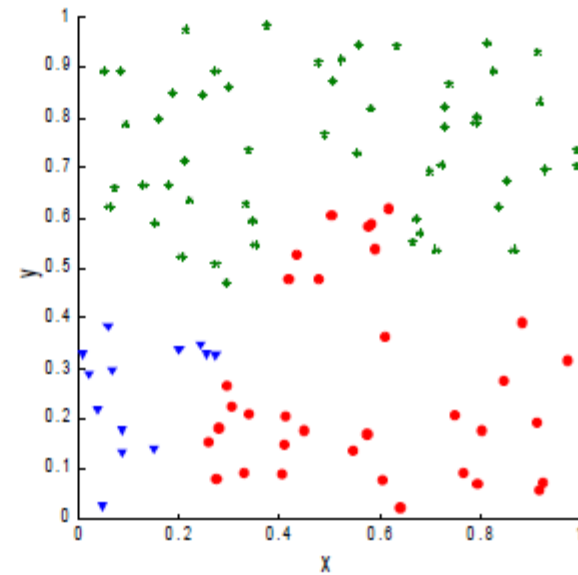
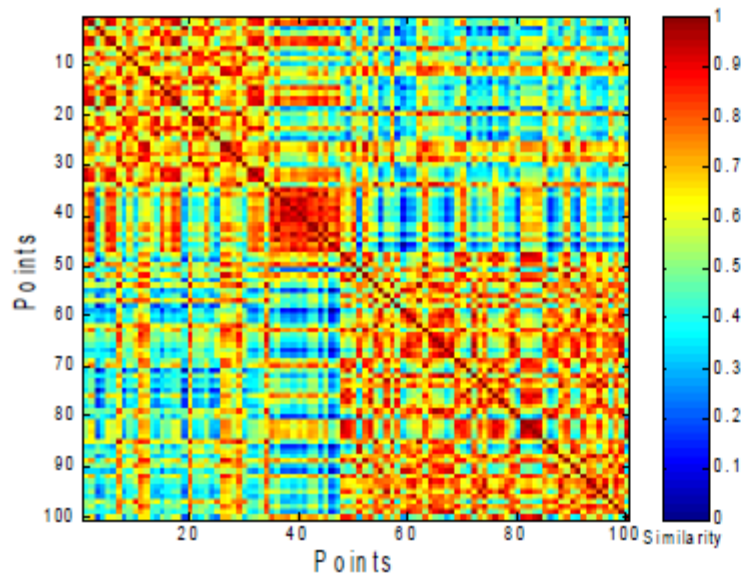


DBSCAN



Using Similarity Matrix for Cluster Validation

- Clusters in random data are not so crisp



Complete Link

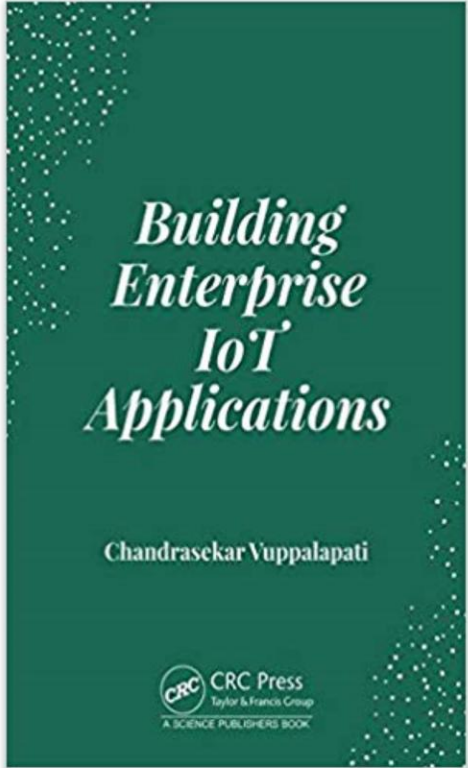


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Source: <https://gallery.azure.ai/Experiment/Recommendation-Engines-Using-Apriori-Algorithm>

Chandrasekar Vuppapapati, Professor, MS Software Engineering,
Computer Engineering, San Jose State University.

Framework for Cluster Validity

- Need a framework to interpret any measure.
 - For example, if our measure of evaluation has the value, 10, is that good, fair, or poor?
- Statistics provide a framework for cluster validity
 - The more “atypical” a clustering result is, the more likely it represents valid structure in the data
 - Can compare the values of an index that result from random data or clusterings to those of a clustering result.
 - If the value of the index is unlikely, then the cluster results are valid
 - These approaches are more complicated and harder to understand.
- For comparing the results of two different sets of cluster analyses, a framework is less necessary.
 - However, there is the question of whether the difference between two index values is significant



Internal Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

- Separation is measured by the between cluster sum of squares

$$BSS = \sum_i |C_i| (m - m_i)^2$$

- Where $|C_i|$ is the size of cluster i

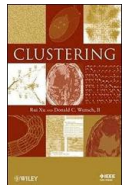


Clustering procedure

4. Result Interpretation

The ultimate goal of clustering is to provide users with meaningful insights from the original data so that they can develop a clear understanding of the data and therefore effectively solve the problems encountered. Experts in the relevant fields are encouraged to interpret the data partition, integrating other experimental evidence and domain information without restricting their observations and analyses to any specific clustering result. Consequently, further analyses and experiments may be required.

Cluster analysis is not a one-shot process. In many circumstances, clustering requires a series of trials and repetitions. Moreover, there are no universally effective criteria to guide the selection of features and clustering schemes. Validation criteria provide some insights into the quality of clustering solutions, but even choosing an appropriate criterion is a demanding problem.

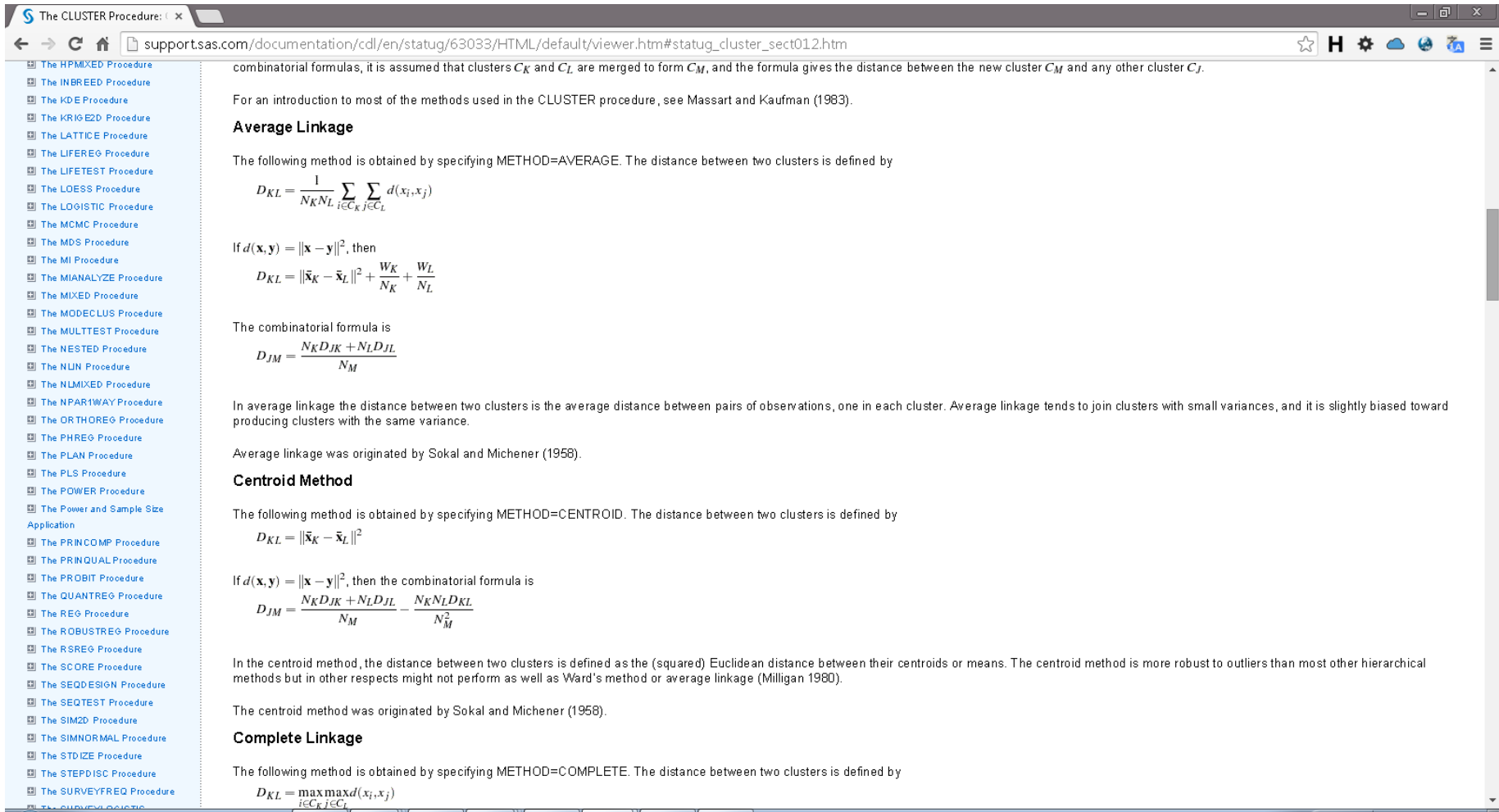


Proximity measures and their applications.

Measure	Metric	Examples and Applications
Minkowski distance	Yes	Fuzzy c-means with measures based on Minkowski family (Hathaway et al., 2000)
City block distance	Yes	FuzzyART (Carpenter et al., 1991)
Euclidean distance	Yes	K-means with its variants (Ball and Hall, 1967; Forgy, 1965; MacQueen, 1967)
Sup distance	Yes	Fuzzy c-means with sup norm (Bobrowski and Bezdek, 1991)
Mahalanobis	Yes	Ellipsoidal ART (Anagnostopoulos and M. Georgiopoulos, 2001); Hyperellipsoidal clustering algorithm (Mao and Jain, 1996)
Point symmetry distance	No	Symmetry-based K-means (Su and Chou, 2001)
Pearson correlation	No	Widely used as the measure for microarray gene expression data analysis (Eisen et al., 1998)
Cosine similarity	No	The most commonly used measure in document clustering (Steinbach et al., 2000)



SAS Reference - Clustering



The CLUSTER Procedure: support.sas.com/documentation/cdl/en/statug/63033/HTML/default/viewer.htm#statug_cluster_sect012.htm

combinatorial formulas, it is assumed that clusters C_K and C_L are merged to form C_M , and the formula gives the distance between the new cluster C_M and any other cluster C_J .

For an introduction to most of the methods used in the CLUSTER procedure, see Massart and Kaufman (1983).

Average Linkage

The following method is obtained by specifying METHOD=AVERAGE. The distance between two clusters is defined by

$$D_{KL} = \frac{1}{N_K N_L} \sum_{i \in C_K} \sum_{j \in C_L} d(x_i, x_j)$$

If $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$, then

$$D_{KL} = \|\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L\|^2 + \frac{W_K}{N_K} + \frac{W_L}{N_L}$$

The combinatorial formula is

$$D_{JM} = \frac{N_K D_{JK} + N_L D_{JL}}{N_M}$$

In average linkage the distance between two clusters is the average distance between pairs of observations, one in each cluster. Average linkage tends to join clusters with small variances, and it is slightly biased toward producing clusters with the same variance.

Average linkage was originated by Sokal and Michener (1958).

Centroid Method

The following method is obtained by specifying METHOD=CENTROID. The distance between two clusters is defined by

$$D_{KL} = \|\bar{\mathbf{x}}_K - \bar{\mathbf{x}}_L\|^2$$

If $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$, then the combinatorial formula is

$$D_{JM} = \frac{N_K D_{JK} + N_L D_{JL}}{N_M} - \frac{N_K N_L D_{KL}}{N_M^2}$$

In the centroid method, the distance between two clusters is defined as the (squared) Euclidean distance between their centroids or means. The centroid method is more robust to outliers than most other hierarchical methods but in other respects might not perform as well as Ward's method or average linkage (Milligan 1980).

The centroid method was originated by Sokal and Michener (1958).

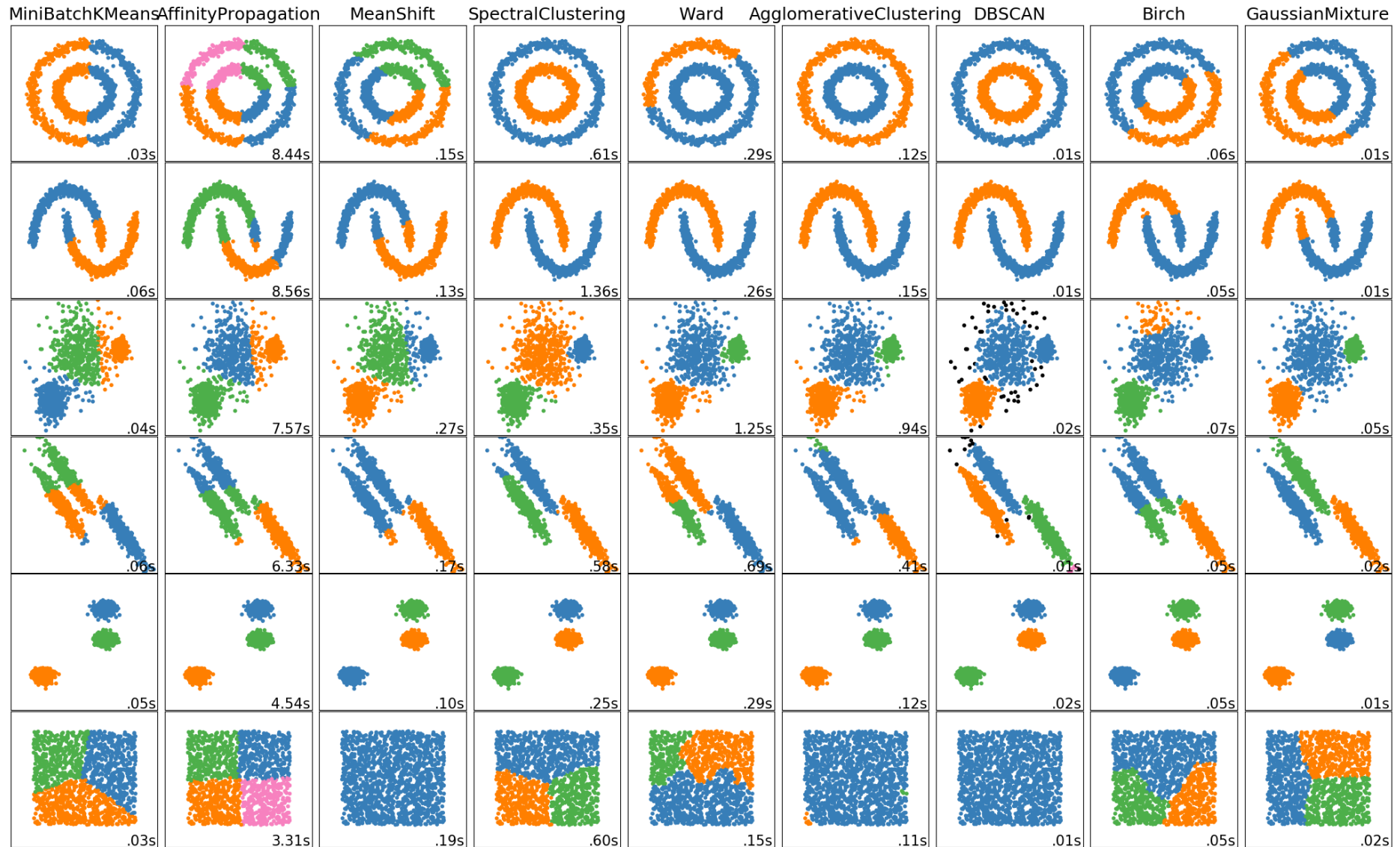
Complete Linkage

The following method is obtained by specifying METHOD=COMPLETE. The distance between two clusters is defined by

$$D_{KL} = \max_{i \in C_K} \max_{j \in C_L} d(x_i, x_j)$$


http://support.sas.com/documentation/cdl/en/statug/63033/HTML/default/viewer.htm#statug_cluster_sect012.htm

SCIKIT Reference - Clustering



<http://scikit-learn.org/stable/modules/clustering.html>



SCIKIT Reference - Clustering

Method name	Parameters	Scalability	Usecase	Geometry (metric used)
K-Means	number of clusters	Very large n_samples, medium n_clusters with MiniBatch code	General-purpose, even cluster size, flat geometry, not too many clusters	Distances between points
Affinity propagation	damping, sample preference	Not scalable with n_samples	Many clusters, uneven cluster size, non-flat geometry	Graph distance (e.g. nearest-neighbor graph)
Mean-shift	bandwidth	Not scalable with n_samples	Many clusters, uneven cluster size, non-flat geometry	Distances between points
Spectral clustering	number of clusters	Medium n_samples, small n_clusters	Few clusters, even cluster size, non-flat geometry	Graph distance (e.g. nearest-neighbor graph)
Ward hierarchical clustering	number of clusters	Large n_samples and n_clusters	Many clusters, possibly connectivity constraints	Distances between points
Agglomerative clustering	number of clusters, linkage type, distance	Large n_samples and n_clusters	Many clusters, possibly connectivity constraints, non Euclidean distances	Any pairwise distance
DBSCAN	neighborhood size	Very large n_samples, medium n_clusters	Non-flat geometry, uneven cluster sizes	Distances between nearest points
Gaussian mixtures	many	Not scalable	Flat geometry, good for density estimation	Mahalanobis distances to centers
Birch	branching factor, threshold, optional global clusterer.	Large n_clusters and n_samples	Large dataset, outlier removal, data reduction.	Euclidean distance between points

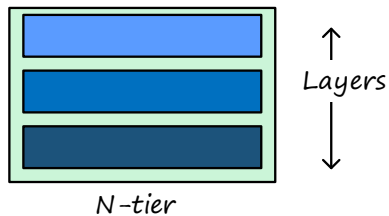
<http://scikit-learn.org/stable/modules/clustering.html>



Architecture styles

An architecture style is a family of architectures that share certain characteristics. For example, N-tier is a common architecture style. More recently, microservice architectures have started to gain favor. Architecture styles don't require the use of particular technologies, but some technologies are well-suited for certain architectures. For example, containers are a natural fit for microservices.

N Tier



N-tier is a traditional architecture for enterprise applications.

Dependencies are managed by dividing the application into layers that perform logical functions, such as presentation, business logic, and data access. A layer can only call into layers that sit below it. However, this horizontal layering can be a liability. It can be hard to introduce changes in one part of the application without touching the rest of the application. That makes frequent updates a challenge, limiting how quickly new features can be added.

N-tier is a natural fit for migrating existing applications that already use a layered architecture. For that reason, N-tier is most often seen in infrastructure as a service (IaaS) solutions, or application that use a mix of IaaS and managed services.

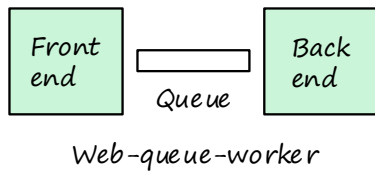
<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/index>



Architecture styles

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Web-Queue-Worker For a purely PaaS solution, consider a Web-Queue-Worker architecture. In this style, the application has a web front end that handles HTTP requests and a back-end worker that performs CPU-intensive tasks or long-running operations. The front end communicates to the worker through an asynchronous message queue.



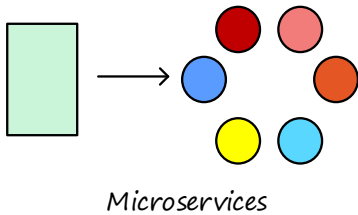
Web-queue-worker is suitable for relatively simple domains with some resource-intensive tasks. Like N-tier, the architecture is easy to understand. The use of managed services simplifies deployment and operations. But with a complex domains, it can be hard to manage dependencies. The front end and the worker can easily become large, monolithic components that are hard to maintain and update. As with N-tier, this can reduce the frequency of updates and limit innovation.

<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/index>



Architecture styles

Microservices



If your application has a more complex domain, consider moving to a Microservices architecture. A microservices application is composed of many small, independent services. Each service implements a single business capability. Services are loosely coupled, communicating through API contracts.

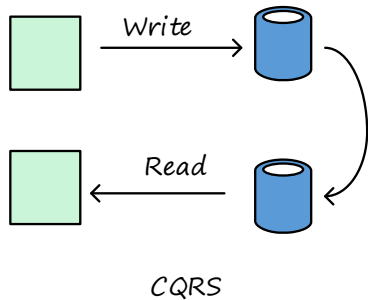
Each service can be built by a small, focused development team. Individual services can be deployed without a lot of coordination between teams, which encourages frequent updates. A microservice architecture is more complex to build and manage than either N-tier or web-queue-worker. It requires a mature development and DevOps culture. But done right, this style can lead to higher release velocity, faster innovation, and a more resilient architecture.

<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/index>



Architecture styles

CQRS



The CQRS (Command and Query Responsibility Segregation) style separates read and write operations into separate models. This isolates the parts of the system that update data from the parts that read the data. Moreover, reads can be executed against a materialized view that is physically separate from the write database. That lets you scale the read and write workloads independently, and optimize the materialized view for queries.

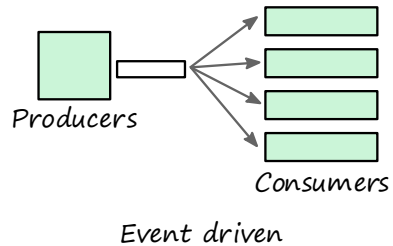
CQRS makes the most sense when it's applied to a subsystem of a larger architecture. Generally, you shouldn't impose it across the entire application, as that will just create unneeded complexity. Consider it for collaborative domains where many users access the same data.

<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/index>



Architecture styles

Event-Driven Architecture



Event-Driven Architectures use a publish-subscribe (pub-sub) model, where producers publish events, and consumers subscribe to them. The producers are independent from the consumers, and consumers are independent from each other.

Consider an event-driven architecture for applications that ingest and process a large volume of data with very low latency, such as IoT solutions. The style is also useful when different subsystems must perform different types of processing on the same event data.

<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/index>



Architecture styles

Architecture style	Dependency management	Domain type
N-tier	Horizontal tiers divided by subnet	Traditional business domain. Frequency of updates is low.
Web-Queue-Worker	Front and backend jobs, decoupled by async messaging.	Relatively simple domain with some resource intensive tasks.
Microservices	Vertically (functionally) decomposed services that call each other through APIs.	Complicated domain. Frequent updates.
CQRS	Read/write segregation. Schema and scale are optimized separately.	Collaborative domain where lots of users access the same data.
Event-driven architecture.	Producer/consumer. Independent view per sub-system.	IoT and real-time systems
Big data	Divide a huge dataset into small chunks. Parallel processing on local datasets.	Batch and real-time data analysis. Predictive analysis using ML.
Big compute	Data allocation to thousands of cores.	Compute intensive domains such as simulation.

<https://docs.microsoft.com/en-us/azure/architecture/guide/architecture-styles/event-driven>



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- SAS Reference - http://support.sas.com/documentation/cdl/en/statug/63033/HTML/default/viewer.htm#statug_cluster_sect012.htm
- How to Explain Hierarchical - Clustering <http://www.analytictech.com/networks/hiclus.htm>
- Cluster Validation, <http://www.cs.kent.edu/~jin/DM08/ClusterValidation.pdf>



SCIKIT Clustering Methods



Methods and systems for constructing a taxonomy based on hierarchical clustering

US 9110984 B1

ABSTRACT

Methods and systems for constructing a taxonomy based on hierarchical clustering are provided. The taxonomy is generated by first constructing a hierarchy of clusters using a clustering algorithm. A first level of the hierarchy of clusters is generated by providing a plurality of content files to a clustering algorithm. Subsequent levels of the hierarchy are generated by providing the clusters of the preceding levels to the clustering algorithm. Labels that characterize each cluster within the hierarchy are assigned to corresponding clusters. Labels and clusters are combined to form the taxonomy.

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Publication type Grant

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Priority date Dec 27, 2011

Inventors Glenn M. Lewis, Kirill Buryak, Nadav Benbarak, Aner Ben-Artzi, Jun Peng

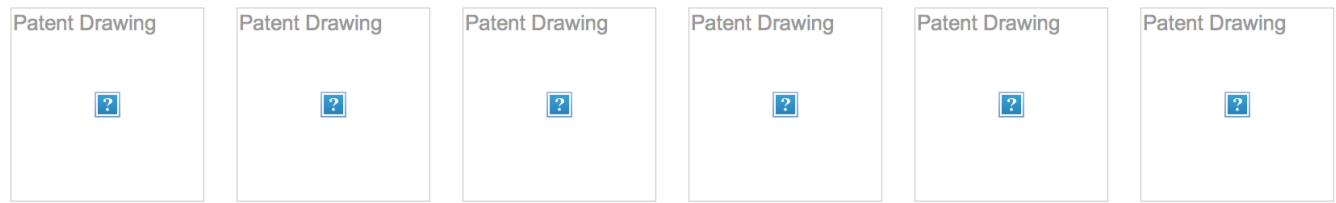
Original Assignee Google Inc.

Export Citation BiBTeX, EndNote, RefMan

Patent Citations (37), **Classifications** (3), **Legal Events** (1)

External Links: [USPTO](#), [USPTO Assignment](#), [Espacenet](#)

IMAGES (6)



<https://www.google.com/patents/US9110984?dq=US+9110984&hl=en&sa=X&ved=0CB0Q6AEwAGoVChMIxp3SxpWdyAIVBlssCh1oaA9b>

DESCRIPTION



CLAIMS (20)

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to provisional patent application Ser. No. 61/580,503 filed on Dec. 27, 2011, and provisional patent application Ser. No. 61/594,615 filed on Feb. 3, 2012, which are both hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

This invention relates generally to providing hierarchical clusters of data and, more particularly, to network-based methods and systems for constructing a taxonomy of product issues based on hierarchical clustering.

Organizations and businesses can receive a large number of messages from customers, potential customers, users and/or other people. For example, a business and/or organization can receive messages from its customers and potential customers, such as email messages, messages from online forums, e.g., support forums or message boards, and other types of messages. These messages can be related to a variety of different topics or issues. For example, the messages can be related to problems experienced by a user and can include a request for assistance to solve the problem. Oftentimes, these request messages are directed to a support center at the organization/business.

In addition, the Internet provides these organizations and businesses with access to a wide variety of resources, including web pages for particular topics, reviews of products and/or services, news articles, editorials and blogs. The authors of these resources can express their opinions and/or views related to a myriad of topics such a product and/or service, politics, political candidates, fashion, design, etc. For example, an author can create a blog entry supporting a political candidate and express their praise in the candidate's position regarding fiscal matters or social issues. As another example, authors can create a restaurant review on a blog or on an online review website and provide their opinions of the restaurant using a numerical rating (e.g., three out of five stars), a letter grade (e.g., A+) and/or a description of their dining experience to indicate their satisfaction with the restaurant.

Such a large volume of documents (i.e., different types of electronic documents,

What is claimed is:

1. A computer system including instructions stored on one or more computer-readable medium and executable by one or more processors, the computer system comprising:

a statistic calculator configured to receive a plurality of content files and to determine at least one statistical measure of the content of the plurality of content files, said statistic calculator configured to store the at least one statistical measure within a statistics repository;

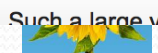
a cluster controller configured to cause the one or more processors to generate a hierarchy of clusters based on the at least one statistical measure stored in the statistics repository, wherein the hierarchy of clusters comprises at least two levels, each cluster within a first level of the hierarchy of clusters comprising at least one content file, each cluster within a second level of the hierarchy of clusters comprising at least one cluster of the first level of the hierarchy of clusters;

an aggregator configured to aggregate the content files of each cluster, and transmit the aggregated content file to said cluster controller to form a third level of the hierarchy of clusters;

a label manager configured to cause the one or more processors to determine a label for each cluster within the hierarchy of clusters based on the at least one statistical measure, the label identifying a topic of information contained within each cluster, the topic related to at least one of a problem experienced by a user and a request for assistance in solving the problem; and

a taxonomy manager configured to cause the one or more processors to output a taxonomy based on the hierarchy of clusters and the determined labels.

2. The computer system of claim 1, wherein said cluster controller further comprises at least one cluster algorithm module, said at least one cluster algorithm module configured to assign the at least one content



Clustering images using an image region graph

US 8243988 B1

ABSTRACT

Methods for grouping images from image corpora using graph clustering are presented. In one embodiment, a method is presented where grouping of images from a collection of digital images is done by: representing regions of images as vertices in a graph; connecting each pair of matching-vertices with a matching-edge; connecting each pair of overlap-vertices with an overlap-edge; assigning weights to each said matching-edge and to each said overlap-edge; clustering the graph, wherein clustering generates one or more vertex-clusters; and grouping the digital images into visual-clusters based on the vertex-clusters. Corresponding systems and computer program products are also presented.

IMAGES (7)



DESCRIPTION

BACKGROUND

This invention relates generally to the grouping of digital images and more specifically to using image region graphs in clustering of images.

CLAIMS (23)

1. A computer-implemented method for grouping images from a collection of digital images, comprising:

(a) representing, using a processor, regions of images as vertices in a graph, wherein the images are from the collection of digital images;

Publication number US8243988 B1
Publication type Grant
Application number US 12/183,613
Publication date Aug 14, 2012
Filing date Jul 31, 2008
Priority date ? Jul 31, 2008

Inventors Ulrich Buddemeier

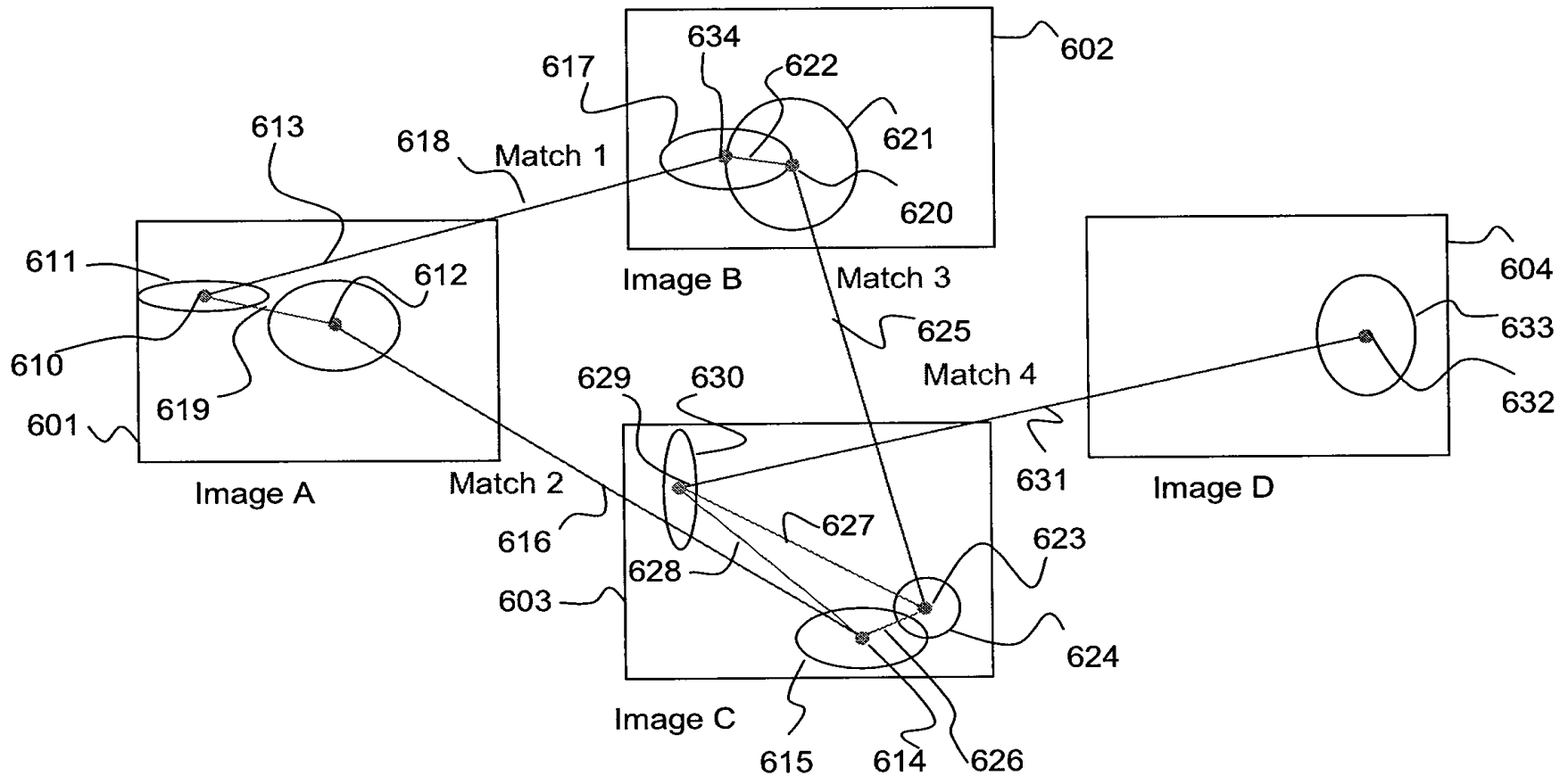
Original Assignee Google Inc.

Export Citation BiBTeX, EndNote, RefMan

Patent Citations (4), **Non-Patent Citations** (7), **Referenced by** (4),
Classifications (9), **Legal Events** (1)

External Links: USPTO, USPTO Assignment, Espacenet

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100

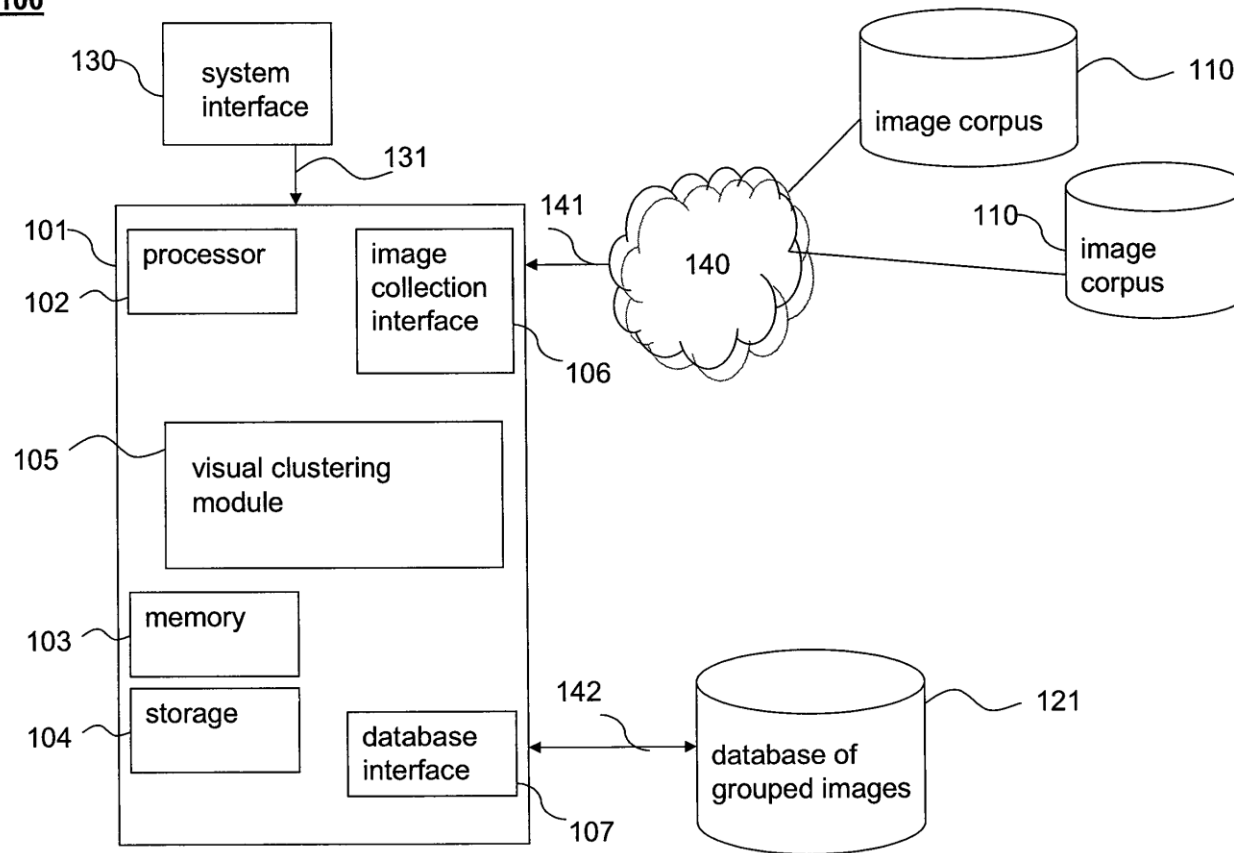


FIG. 1



Big Data Industry Insights

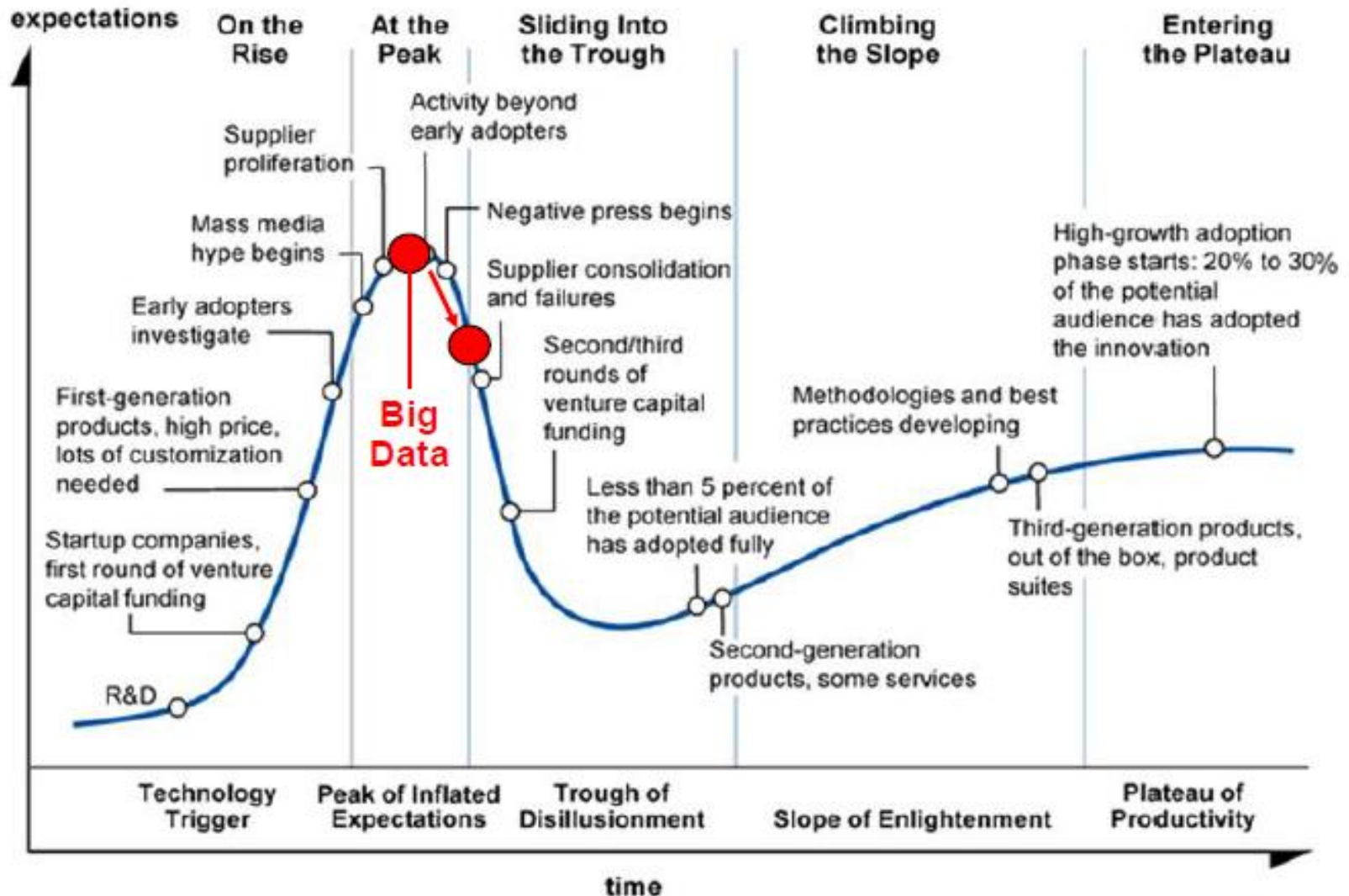
Lisa Kart
Twitter: @Kart_Lisa

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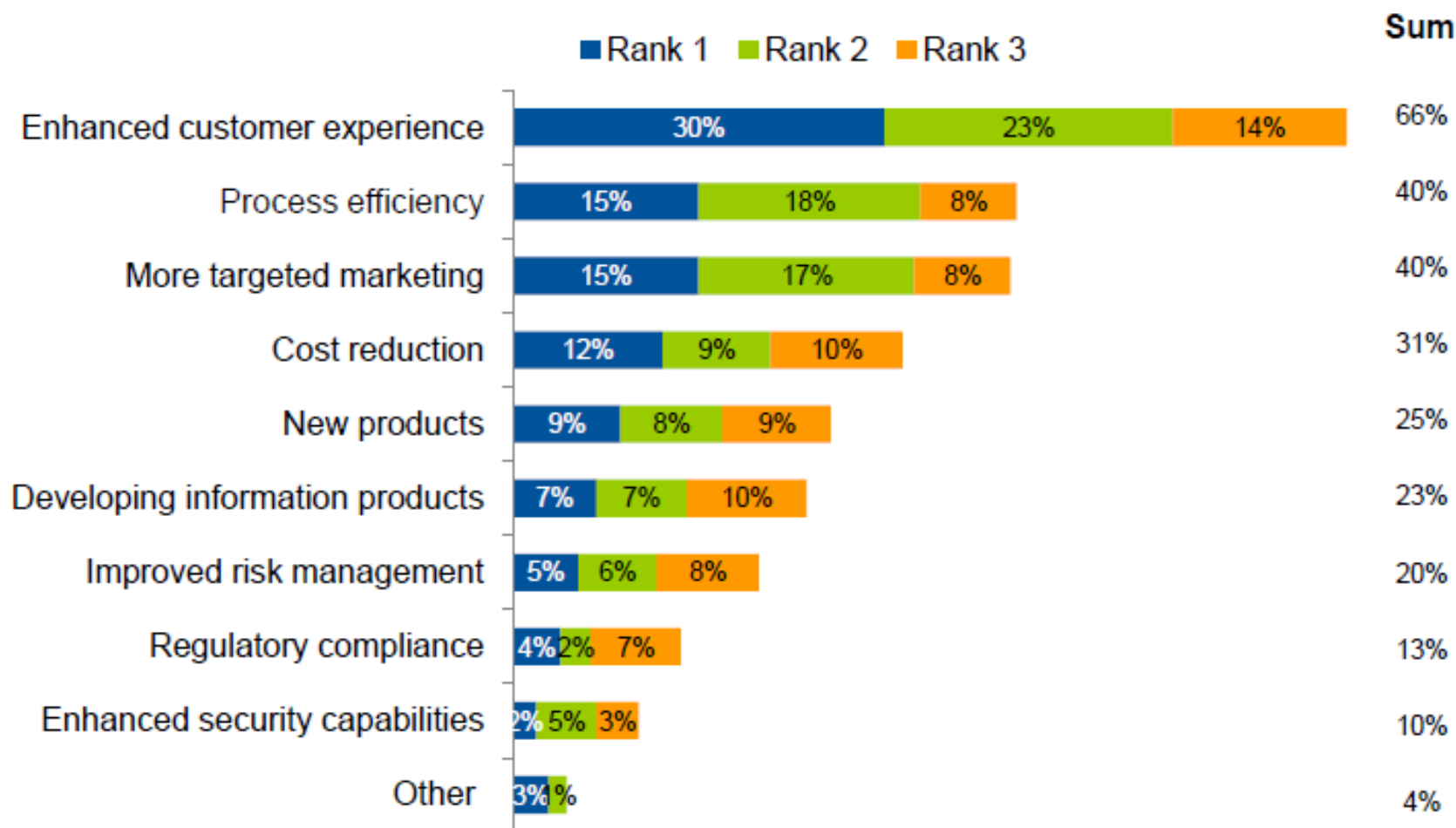
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Gartner

What's Beyond the Big Data Hype?



Customers Top the List of Priorities



Banking on Knowing and Engaging Your Customers Better

- Opportunity:
 - Actively engaging a greater number of customer to ensure their satisfaction and retention
- Data and Analytics:
 - Capturing and centralizing customer activity of 12M customers, including bank website browsing history, ATMs usage, call center, and so on
 - Matching behavior versus dozens of offerings and potential acceptance using SAS
- Results:
 - Launched "KnowMe" program to engage customers
 - Grew customer engagement (offers presented) from less than 1% to 25% of customers
 - Recommends nearby Westpac ATMs to save money when customer regularly uses another ATM



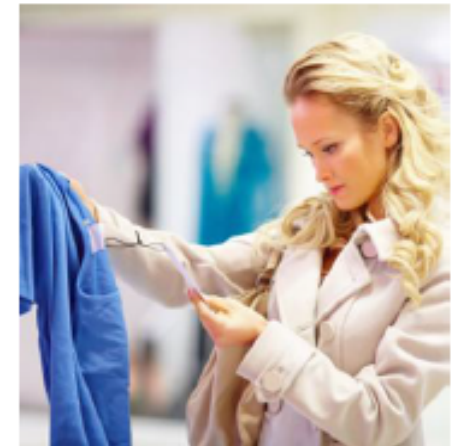
Unleashing Big Data and Analytics

- Opportunity:
 - Improve revenue at this small Washington zoo
- Data and Analytics:
 - Monitors, integrates, and analyzes data collected at exhibits and event sites, social channels, and ticket sales and historical data from 600,000 annual visitors
 - Created simpler and more targeted marketing campaigns with the help of IBM and BrightStar Partners
- Results:
 - Increased online ticket sales by 700%
 - Doubled annual member renewals from 3% to 6%



"Sixth Floor: Big Data-based Pricing"

- Opportunity:
 - Better react to changing market conditions with near-real-time product pricing
- Data and Analytics:
 - Sales, inventory data on 73M items; 50% annual data growth
 - Hadoop, R, Impala, SAS, Vertica, Tableau
 - Data science organization generates and tests hundreds of thousands of analytic models on granular data versus the dozens previously possible
- Results:
 - Reduced time to price items from 27 hours to about one hour
 - 70% hardware cost reduction



Squeezing Every Drop of Data to Ensure Product Consistency

- Opportunity:
 - Inconsistencies in orange juice due to variations in orange crop, sourcing, and seasonality
- Data and Analytics:
 - "Black Book" model algorithm developed by Revenue Analytics crunches data up to *one quintillion* variables including satellite images, weather, expected crop yields, cost pressures, regional preferences and detailed data about the 600 flavors that comprise an orange, plus variables such as acidity and sweetness
- Results:
 - Precise dynamic formula for how to blend orange juice for consistent taste, including pulp content, for its \$2B orange juice business
 - After a hurricane or freeze, it can re-plan the business in 5-10 minutes

Coca-Cola



Blowing Away Previous Wind Turbine Modeling Capabilities

- Opportunity:
 - Precise placement of wind turbines affects performance and useful life and energy costs
- Data and Analytics:
 - 10x increase in breadth (178 parameters) and history (18-24 petabytes) of weather data, including temperature, barometric pressure, humidity, precipitation, wind direction and velocity from 0-300 feet altitude
 - Supercomputing-based analytics from IBM enables 10 sq. meter grids versus 27 sq. km grids previously modeled
- Results:
 - Reduced wind forecast modeling by 97% (3 weeks to 15 minutes) to pinpoint optimal placement of each turbine
 - One month faster turbine site development (one turbine installed every three hours)
 - Reduced energy costs to utilities and consumers

Vestas



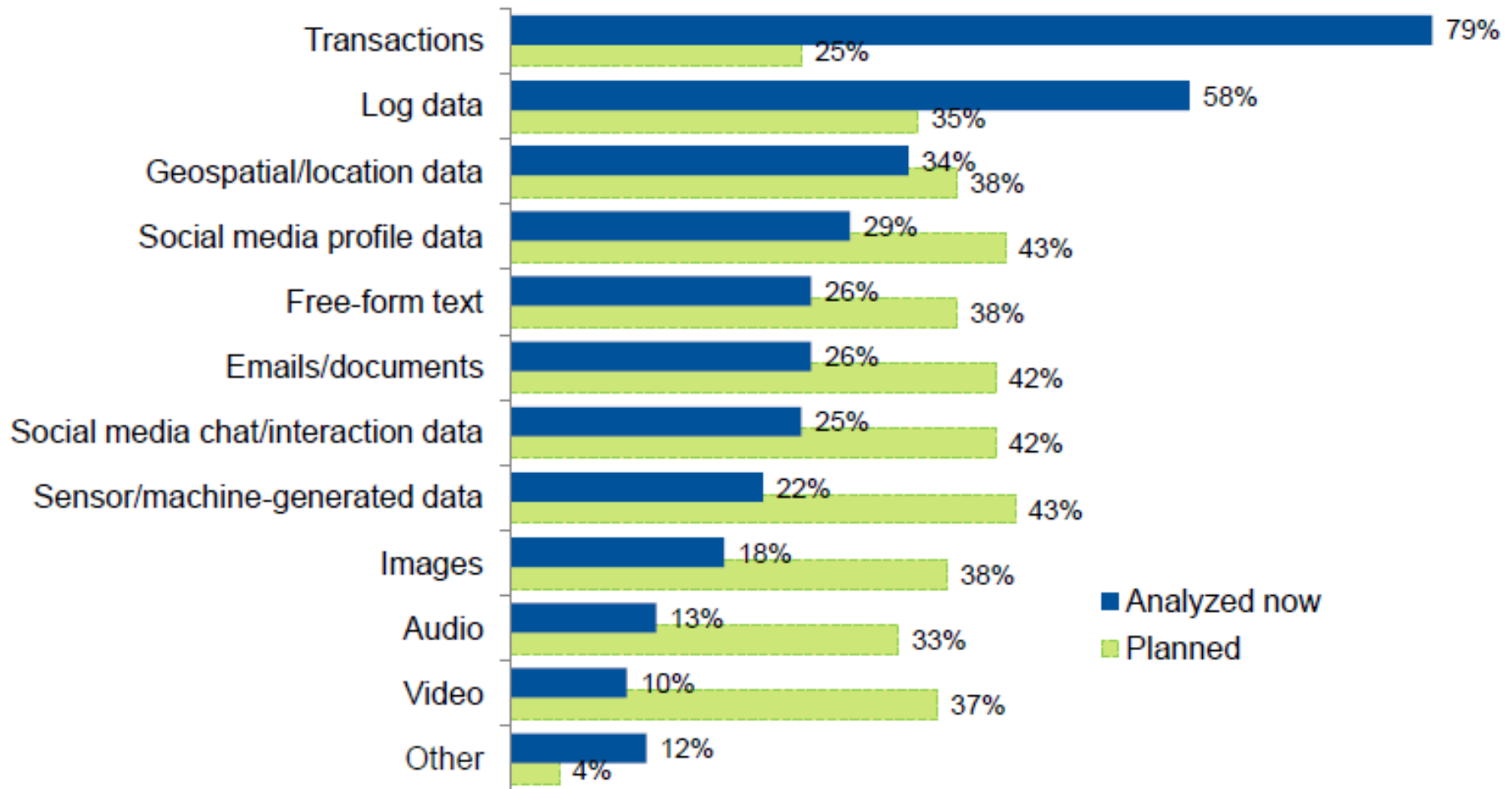
Gartner

Predictive Policing

- Opportunity:
 - Increase Los Angeles police presence where it's most likely needed
- Data and Analytics:
 - PredPol applies models for predicting earthquake aftershocks to historical crime data and other factors
- Results:
 - Predicted twice as many crimes as experienced crime analysts in controlled trials
 - 33% reduction in burglaries and 21% reduction in violent crimes in test region of Los Angeles, compared to a slight increase in the rest of the city



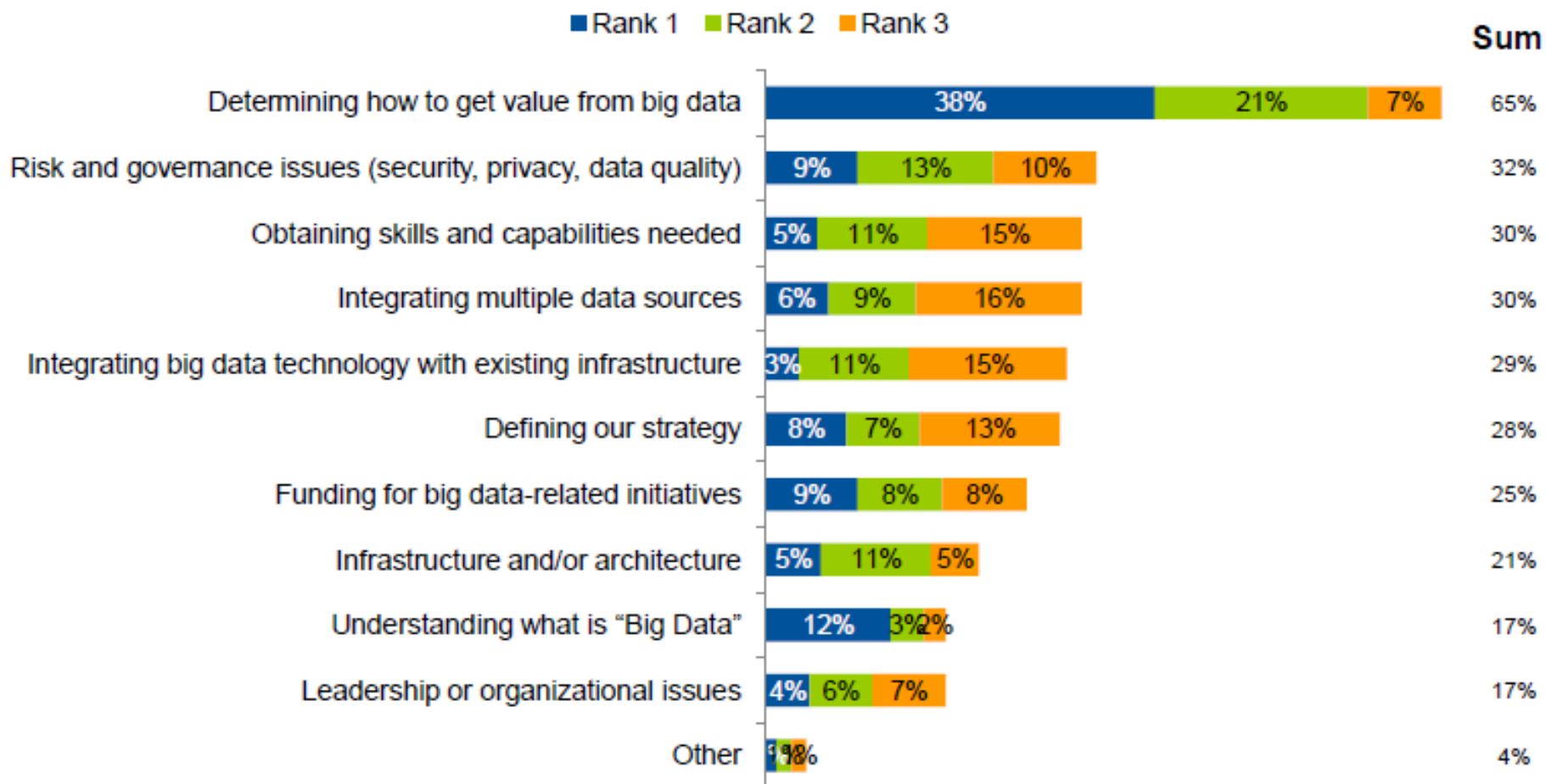
Traditional Data Sources Dominate, But Many New Sources Are Planned



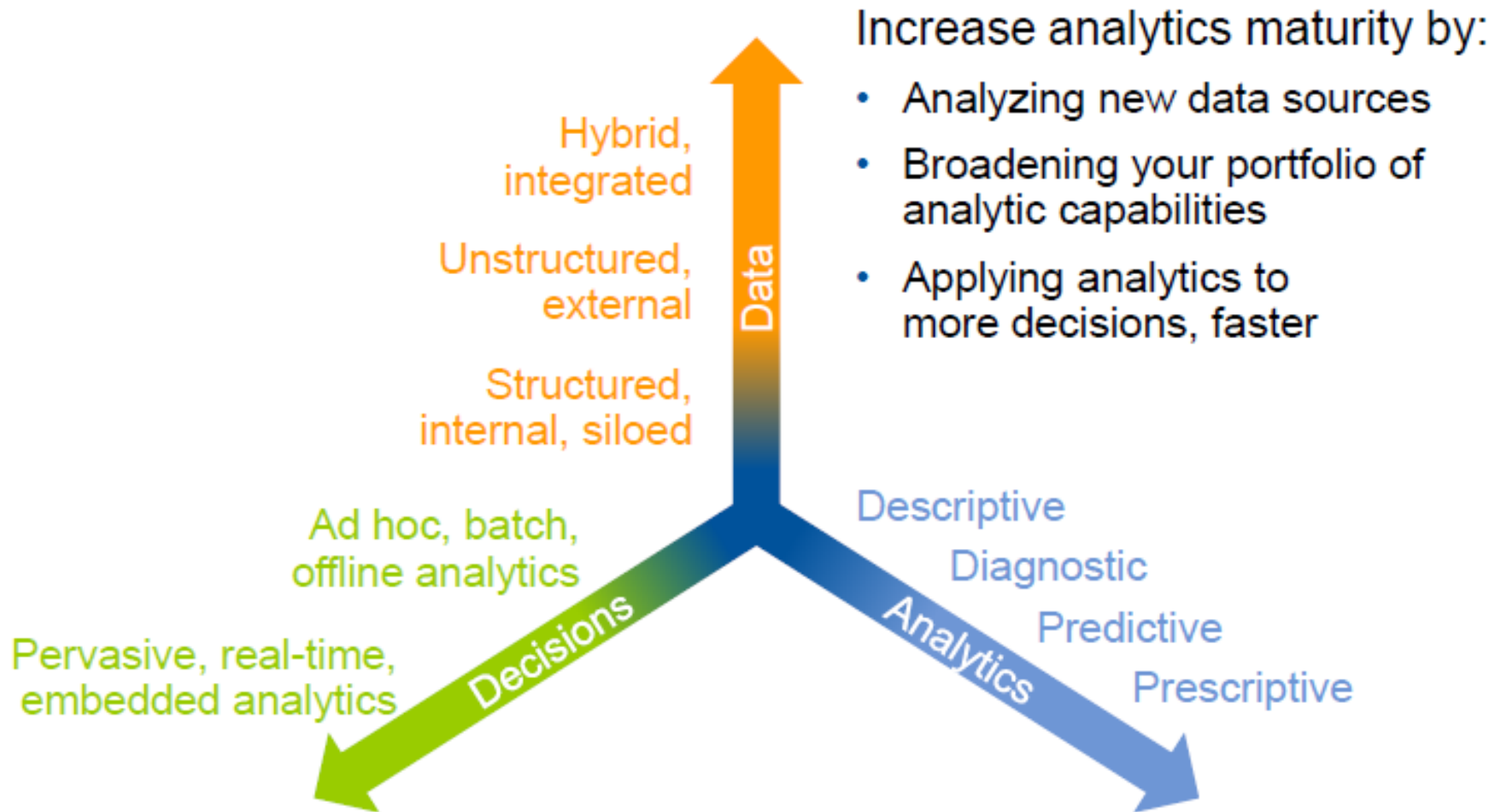
Multiple responses allowed

Organizations Are Struggling with How to Get Value from Big Data

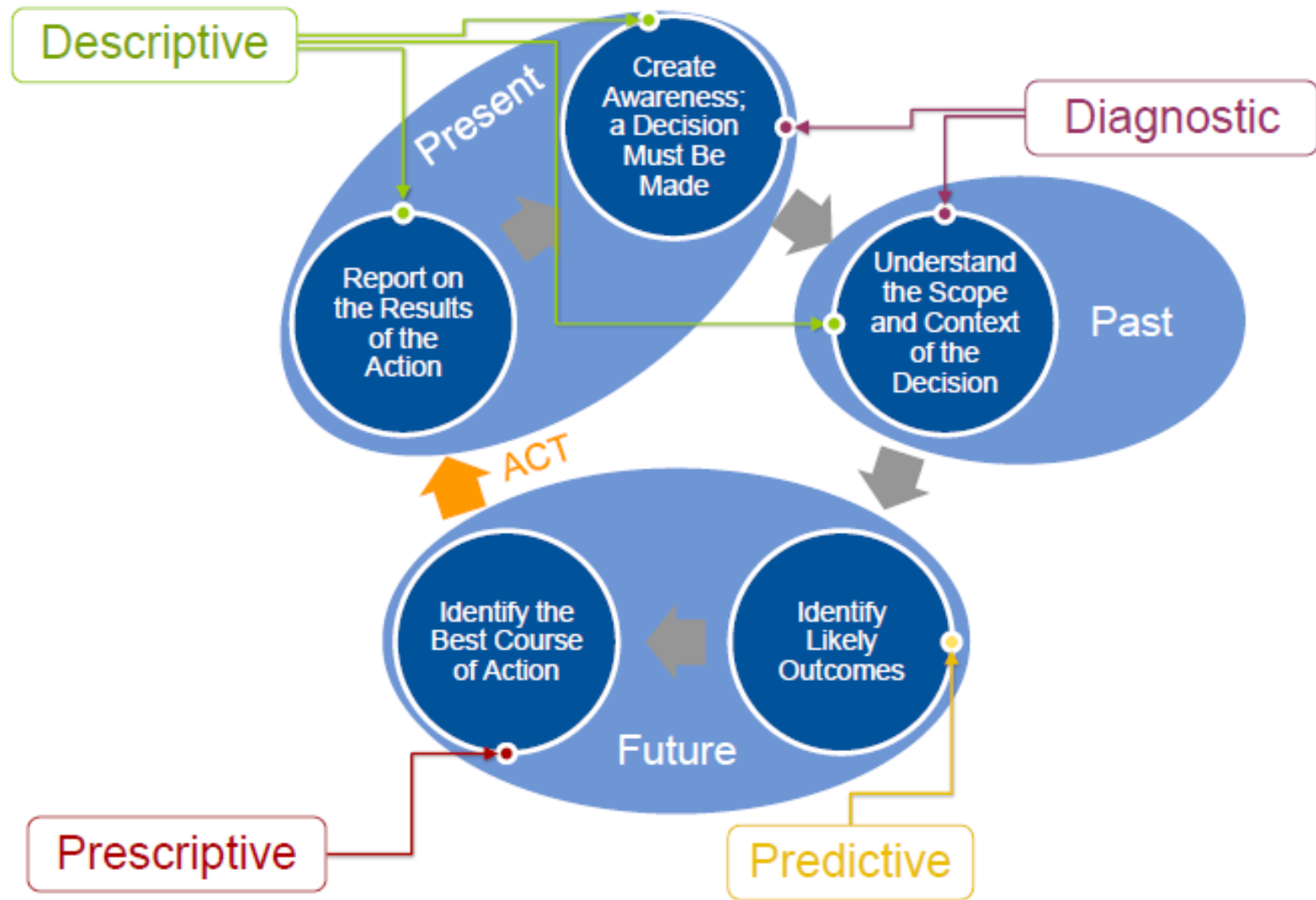
What are your organization's top 3 hurdles or challenges with big data?



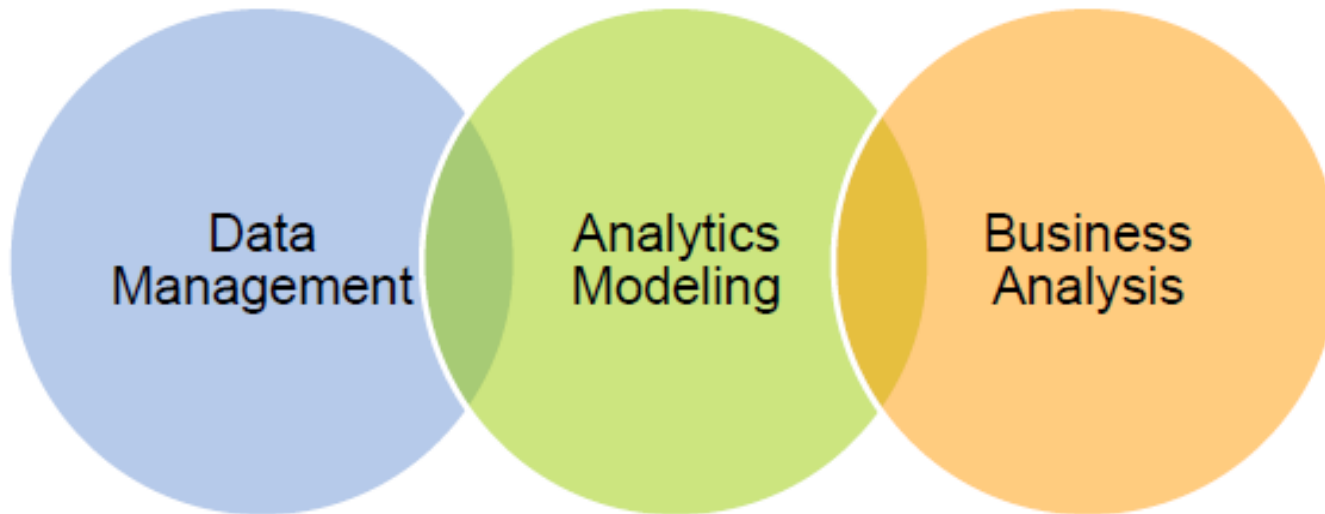
It's All About Analytics Maturity



Apply Relevant Data and Analytics to Decision Making



What Skills Do I Need?



Understand Data:

- Integrate and Manipulate
- Structured, Content, and Hybrid Data
- Quality Assurance
- Big Data

Know Analytics:

- Appropriate Techniques and Platforms
- Interpret Data and Diagnose Models
- Meet Business Requirements

Focus on the Business:

- Goals
- Constraints
- Decisions
- Change Management
- Communication of Results